“The cosmos is full beyond measure of elegant truths; of exquisite interrelationships; of the awesome machinery of nature.”

—Carl Sagan

Taxon-Specific Ecological Drivers of Diatom-Dominated Microphytobenthos: Experimental and Spatio-Temporal Insights from a Temperate Hypertidal Mudflat

by

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ABSTRACT

Photosynthetic microorganisms form the productive base of many coastal ecosystems. Yet, the relative importance of ecological drivers underlying variability in their abundance, diversity, and species composition remains poorly resolved. To address this, I examined variability in microalgal assemblages in relation to infaunal and epifaunal grazers, heterotrophic microorganisms, and sediment physico-chemical properties in a Bay of Fundy intertidal mudflat. These variables were quantified using environmental DNA metabarcoding, chlorophyll a fluorescence, microscopy, and field measurements. Data were obtained from a field experiment that manipulated epifaunal grazer abundance with exclosures and nutrient availability with fertilizers. My findings suggest differing sensitivities of the biomass and species composition of the diatom-dominated microphytobenthos to ecological drivers. Species composition was more closely linked to biotic interactions, whereas biomass varied under resource-driven control within the broader environmental context. My thesis contributes to ongoing efforts to clarify ecological controls on the productive, carbon-sequestering base of coastal ecosystems.

Keywords: Bay of Fundy; tidal flats; intertidal biofilm; microphytobenthos; microbe-fauna interactions; polar centric diatoms; raphid pennate diatoms; migratory shorebirds; mud snails; nutrient enrichment; annual variability; seasonal transitions

DEDICATION

I dedicate this thesis first and foremost to my mother, Edna Kalu. You nurtured my curiosity, cultivated my passions, and always encouraged me. Your unwavering support has been my foundation; your strength, a guiding light. Thank you for being my rock, my compass, and my inspiration. To my father, Ikechi Kalu—thank you for the many invaluable life lessons that have shaped who I am today.

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I respectfully acknowledge that my fieldwork was conducted in Mi’kma’ki, the ancestral and unceded territory of the Mi’kmaq people. I also acknowledge that the University of New Brunswick's Fredericton campus, where I studied, is situated on the unsurrendered and unceded traditional Wolastoqey land of the Wolastoqiyik people.

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STATEMENT OF CONTRIBUTIONS TO RESEARCH

I developed the objectives of this research with guidance from Dr. Myriam Barbeau and Dr. Adrian Reyes-Prieto. I performed all analyses presented in this thesis with guidance and assistance from Dr. Myriam Barbeau and Dr. Adrian Reyes-Prieto. These analyses were conducted specifically for this thesis, reflecting original work and contributing novel insights. I wrote all sections of this thesis with feedback from Dr. Myriam Barbeau, Dr. Adrian Reyes-Prieto and Dr. Diana Hamilton.

The research presented in this thesis is based on data I collected during fieldwork conducted between 2019 and 2021 as part of my graduate and undergraduate studies (Kalu 2020; Kalu *et al.* 2023), as well as data collected by Leandra Vanbaelinghem and Madeleine Robitaille during their respective Honours research projects in the Hamilton Lab at Mount Allison University (Robitaille 2021; Vanbaelinghem 2021). As a master's student, I collected sediment samples for environmental DNA metabarcoding in 2020 and 2021 to characterize bacterial, microbial eukaryotic, and meiofaunal communities, and carried out the associated laboratory work, including DNA extraction. As part of my Honours research project in the Barbeau and Reyes-Prieto Labs at the University of New Brunswick, I collected sediment samples for environmental DNA metabarcoding and performed measurements of mudflat sediment physico-chemical properties in 2019. The 2019 dataset was used in my analyses of microalgal depth and inter-seasonal variability, as well as correlations with environmental variables. It also contributed to my analyses of microalgal intra-seasonal and inter-annual variability.

Leandra Vanbaelinghem and Madeleine Robitaille designed and led the setup of the field experiment conducted in July and August 2020 to manipulate epifaunal

predator abundance and nutrient availability at the Grande Anse mudflat in the upper Bay of Fundy. This work is described in Section 2.3.1. They collected sediment samples for chlorophyll *a* extraction and quantification, and carried out the associated laboratory work. Additionally, they collected sediment samples for macro-infauna quantification, performing taxonomic identification, size classification, and enumeration using a microscope. Their work also included field observations of mud snails and shorebirds to assess population density and habitat usage, respectively. These activities are described in Section 2.3.2. I incorporated the chlorophyll *a* data they generated as one of five microalgal response variables in my analyses. Furthermore, the shorebird, mud snail, and macro-infauna datasets contributed 10 of the 20 biotic covariates used in my analyses of the variability of benthic microalgal assemblages.

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1 Introduction

1.1 The microphytobenthos forms an ecologically important "secret garden" in mudflats

Oxygenic photosynthetic microorganisms, both bacterial (i.e., cyanobacteria) (Hoffman 1999; Thiel *et al.* 2018) and eukaryotic (i.e., microalgae like dinoflagellates, diatoms, haptophytes, and green algae) (Karlusich et al. 2020; Singer et al. 2021), constitute the productive base of many marine and coastal food webs. Murray et al. (2019) estimated global tidal flats, a class of unvegetated coastal wetlands, to cover 127,921 km², with Davidson & Finlayson (2019) estimating the area to be 2.3 times larger than the global extent of salt marshes. In intertidal mudflats, microphytobenthos—which refers to the collection of photosynthetic microorganisms that live in, on, or near the sedimentary seabed (i.e., the benthos)—typically dominate autotrophic assemblages, surpassing the contributions of their planktonic counterparts in often turbid overlying waters (Cahoon 2002; Hargrave *et al.* 1983; MacIntyre *et al.* 1996; Underwood & Kromkamp 1999). The microphytobenthos has been metaphorically likened to a "secret garden" in so-called unvegetated, silty-to-sandy tidal flats, nestled within the broader "black box" of the marine sedimentary microbiome. This representation of the microphytobenthos underscores its ecologically important, yet often overlooked and understudied, trophodynamics, biogeochemical contributions, and other ecosystem services (Forster *et al.* 2016; Gerwing *et al.* 2015a; Hope *et al.* 2020; Miller *et al.* 1996). It also emphasizes the implicit complexity and environmental heterogeneity of their benthic habitats.

The microphytobenthos in unvegetated coastal marine ecosystems serves analogous ecological roles to those of vascular plants in terrestrial habitats. They play

crucial roles in the health and functioning of intertidal mudflats, a highly dynamic and productive coastal marine ecosystem, where the microphytobenthos often organizes into patches of brownish to greenish biofilm at the sediment surface (MacIntyre *et al.* 1996; Miller *et al.* 1996). These photosynthetic microorganisms drive the gross productivity of intertidal mudflats, which can exceed 300 g of fixed carbon per square metre annually (MacIntyre *et al.* 1996; Underwood & Kromkamp 1999). Hence, they facilitate the sequestration of carbon in both their biomass and extracellularly in sediments, thereby aiding in the mitigation of climate change (Chen & Lee 2022; Lin *et al.* 2020). It has been estimated that the microphytobenthos contributes 500 million tons of carbon to global annual productivity across all benthic environments (Cahoon 2002), thus playing a crucial role in the carbon cycle, both within coastal sediments (Middelburg *et al.* 2000; van Oevelen *et al.* 2006), and globally, in addition to their involvement in nitrogen cycling (Cook *et al.* 2004). They also nutritionally sustain biodiversity at higher trophic levels in mudflats, like invertebrate animals, benthic fish, and migratory shorebirds (Prouse *et al.* 1984). Critically, they contribute to sediment stability and shoreline resilience by secreting extracellular polymeric substances (Black *et al.* 2002; Van Colen *et al.* 2014; Decho 2000; Stal 2003; Underwood & Paterson 2003) that help preserve the structural and functional integrity of surficial biofilms; with 80% of biostabilization attributable to diatoms (like *Nitzschia*, *Gyrosigma,* and *Pennales*) and 12% to bacteria acting synergistically with diatoms (Lundkvist *et al.* 2007). The microphytobenthos in Bay of Fundy mudflats is especially important for migratory shorebirds, such as the declining eastern Canadian Arctic population of Semipalmated Sandpipers (*Calidris pusilla*) (Andres *et al.* 2012; Smith *et al.* 2023). Recent estimates indicate a 37% decline in the abundance of North American shorebirds from 1970 to 2017 (Rosenberg *et al.*

2019). Benthic assemblages of photosynthetic microorganisms constitute a highly variable, energy-dense food resource and a significant source of essential fatty acids for these migratory shorebirds, fueling their annual southward migration (Huggins *et al.* 2004; Kuwae *et al.* 2012; Maillet & Weber 2006, 2007; Mogle 2021; Schnurr *et al.* 2019, 2020). Alongside the mud snail *Tritia obsoleta*, shorebirds function as top epibenthic predators in these mudflats (Booty *et al.* 2020; Cheverie *et al.* 2014; Hamilton *et al.* 2006; Hicklin & Smith 1979; Mathot *et al.* 2018; Vanbaelinghem 2021), likely influencing microphytobenthic dynamics in turn.

1.2 Intrinsic and extrinsic factors shaping microphytobenthic dynamics

In coastal habitats, the general nature of feeding relationships between local consumers and the microphytobenthos is well known, based on linkages inferred from stable isotope ratios of nitrogen and carbon (Post 2002), as well as on gut content analysis using traditional microscopy or DNA metabarcoding. Previous studies have examined microphytobenthic relationships with migratory shorebirds (Gerwing *et al.* 2016; Kuwae *et al.* 2008, 2012; Mathot *et al.* 2010; Quinn & Hamilton 2012), benthic fish (Carpentier *et al.* 2014; Kostecki *et al.* 2012; Schwinghamer *et al.* 1983; Shahraki *et al.* 2014; Takai *et al.* 2002), mud snails (Blanchard *et al.* 2000; Cheverie *et al.* 2014; Hamilton *et al.* 2006), *Corophium* amphipods (Gerdol & Hughes 1994b; Hagerthey *et al.* 2002; Smith *et al.* 1996), meiofauna (i.e., minute invertebrates) like nematodes and ostracods (Moens *et al.* 2002; Montagna *et al.* 1995; Pinckney *et al.* 2003; Riera *et al.* 1996), and other invertebrates including polychaete worms (Christianen *et al.* 2017; Daggers *et al.* 2020b; Galván *et al.* 2008; Herman *et al.* 2000; Kanaya *et al.* 2008; Kang *et al.* 2003). Microalgal interactions with bacteria (Koedooder *et al.* 2019; Yao *et al.* 2019), microbial

consumers (Berney *et al.* 2013; Epstein *et al.* 1992; Leander 2020; Sherr & Sherr 2002; Tillmann 2004), parasites (Anderson & Harvey 2020; Yan *et al.* 2023), and other microorganisms (Adl *et al.* 2019; Caron *et al.* 2012; Crump & Bowen 2024) have also been examined. Environmental control of microalgal diversity and productivity has also been studied across a range of conditions and lineages (Juneja *et al.* 2013; Kholssi *et al.* 2023; Scholz & Liebezeit 2012a, b, c; Thornton *et al.* 2002; van der Wal *et al.* 2010), likely more so than biotic variables, often to aid efficient biofuel production. Microbial functional ecology has become an increasingly important research priority, particularly in the contexts of climate change mitigation, microbial biogeochemistry, harmful algal blooms, and emerging biotechnological applications (Caron *et al.* 2017; Cavicchioli *et al.* 2019; Escalas *et al.* 2019; Hutchins & Fu 2017; Madsen 2011; Salazar & Sunagawa 2017; Wells *et al.* 2020). This growing emphasis has been supported by the advancement of meta-omics techniques, such as metagenomics for exploring genetic repertoires and metatranscriptomics for assessing differential gene expression at community scales (Knight *et al.* 2018; Prosser 2015; Taş *et al.* 2021). These tools complement environmental DNA (eDNA) metabarcoding (Deiner *et al.* 2017; Ruppert *et al.* 2019), which is often used to survey microbial and other biodiversity under variable environmental and experimental conditions. While the microphytobenthos is widely recognized as a dynamic and vital component of coastal marine ecosystems, much remains unknown about the relative importance of the biotic, environmental, and stochastic factors shaping its spatial and temporal diversity and productivity. Furthermore, although the importance of these factors across different scales—such as inter-seasonal versus interannual and local versus regional—remains under investigation, a substantial body of knowledge exists.

Photosynthetic microbial production varies vertically between highly turbid shallow coastal waters and exposed surficial intertidal sediments (Hargrave *et al.* 1983). Differences in taxonomic composition also exist between planktonic and benthic communities, with dinoflagellates, green algae, and haptophytes typically exhibiting higher abundance and diversity in the plankton, while diatoms tend to be more dominant in the benthos (Forster *et al.* 2016; Massana *et al.* 2015). Within sediments, microalgal biomass is typically concentrated in the upper 2 mm (MacIntyre *et al.* 1996; Yin *et al.* 2016). However, the formation of microalgal resting stages in deep marine sediments that can remain viable for extremely long periods (Anderson *et al.* 2012; Pawlowski *et al.* 2011; Sanyal *et al.* 2022; Seckbach & Oren 2007) often drives similarities in microbial community composition across sediment depth layers (Kalu *et al.* 2023). This occurs even as decreasing light and oxygen availability with sediment depth (Gerwing *et al.* 2015b; Glud 2008; Kühl *et al.* 1994) create steep vertical gradients in microalgal abundance, productivity, and metabolic activity (Bourke *et al.* 2017; Broman *et al.* 2017; Fenchel & Straarup 1971; Kamp *et al.* 2011; MacIntyre *et al.* 1996). Spatial variability in microalgal assemblages can also occur both within individual mudflats (i.e., locally), where the patchy distribution of organisms prevails (Daggers *et al.* 2020a; Guarini *et al.* 1998; Virta *et al.* 2019), as well as among mudflats in a region (e.g., the Bay of Fundy) (Gerwing *et al.* 2015a; Kalu *et al.* 2023). Differences in microalgal assemblages between mudflats are often driven by site-specific factors, including sediment grain size, water content, nutrient availability, and infaunal densities (Cahoon *et al.* 1999; Gerwing *et al.* 2015a; Mitbavkar & Anil 2002). At the same time, hydrodynamic processes—including sediment reworking by tidal currents (Bazin *et al.* 2014; de Jorge & van Beusekom 1995; Koh *et al.* 2006; Tweedley *et al.* 2016)—and the influence of regional species

pools (Cornell & Harrison 2014; Zhou & Ning 2017) can drive spatial homogeneity in microalgal diversity. Additionally, the abundance and activity of nutrient-cycling and organic matter-degrading microorganisms, including certain fungal (Amend *et al.* 2019; Hyde *et al.* 1998; Richards *et al.* 2012) and bacterial (Cavicchioli *et al.* 2019; Falkowski *et al.* 2008; Kuypers *et al.* 2018; Middelburg 2018) lineages, contribute to the availability of nutrients for benthic microalgal growth and metabolism. Lastly, latitudinal gradients in the abundance and diversity of microbial eukaryotes, including microalgae, result from multiple interacting factors (Burki *et al.* 2021; Cordier *et al.* 2022; Malviya *et al.* 2016).

From a temporal perspective, variability can occur within a diurnal or semi-diurnal tidal cycle, where diel (vertical) migration of microalgae ensues in response to the rhythmicity of ebb (outgoing) and flood (incoming) tides (Consalvey *et al.* 2004; Pinckney *et al.* 1994). Daily variability in photosynthetic pigment content, or the distribution of microalgal cells within surficial intertidal sediments, can arise from variable light conditions and the associated behavioral and physiological photoacclimation and photoregulation processes that optimize light harvesting while minimizing photodamage (Barnett *et al.* 2020; Blommaert *et al.* 2018; Cartaxana *et al.* 2011, 2016; Prins *et al.* 2020). Similar photoprotection mechanisms have been documented in cyanobacteria (Lichtenberg *et al.* 2020). Seasonal shifts in temperature and photoperiod, alongside interannual environmental variation, further influence microphytobenthic variability (Barranguet *et al.* 1998; Blanchard *et al.* 1997; Jacobs *et al.* 2021; de Jonge *et al.* 2012; Savelli *et al.* 2018; Thornton *et al.* 2002; van der Wal *et al.* 2010). Additionally, seasonality in the presence and foraging intensity of intertidal

consumers (Cranford 1988; Hicklin 1987; Peer *et al.* 1986; Sheahan *et al.* 2009) also contributes to microphytobenthic variability. Across spatial and temporal scales, the availability and specificity of parasites also play a role in shaping microalgal bloom dynamics and species composition (Anderson & Harvey 2020; Berdjeb *et al.* 2018; Clarke *et al.* 2019).

1.3 A multi-faceted approach to studying microphytobenthic drivers and dynamics

The upper reaches of the Bay of Fundy in eastern Canada form a hyper-tidal (tidal range >6 m) system where coastal sites are semi-diurnally inundated by highly turbid waters (Archer 2013; Desplanque & Mossman 2004). In such highly energetic systems, cyclical tidal action can obscure natural gradients, and stochastic processes may play an outsized role in shaping the spatial and temporal distributions of benthic organisms (Demers *et al.* 1987; de Jorge & van Beusekom 1995; Petro *et al.* 2017; Ubertini *et al.* 2012). This can make it challenging to characterize the scales at which the microphytobenthos exhibits its greatest variability. Notably, in some parts of the Bay of Fundy, microphytobenthos inhabiting extensive mudflats accounts for approximately half of the gross primary production, exceeding the individual contributions of phytoplankton and saltmarsh-based production (Prouse *et al.* 1984). Despite this ecological significance, the relative importance of key drivers of microphytobenthic variability across spatial and temporal scales remains quantitatively uncertain in these highly dynamic ecosystems.

My thesis aimed to address this gap by examining the spatial and temporal scales of, and contributors to, microphytobenthic variability in intertidal mudflat sediments of the Bay of Fundy. To this end, I explored five complementary questions: How diverse and abundant are benthic cyanobacteria and microalgae during the summer in surficial

sediments? How do microalgal assemblages vary with depth within surficial sediments? How variable are microalgal assemblages intra-seasonally during the summer, inter-seasonally across the spring–summer and summer–fall transitions, and interannually over three consecutive summers? Which sediment environmental variables are most associated with, and likely influence, variability in microalgal assemblages? Lastly, what are the relative contributions of ecological variables—including infaunal and epifaunal grazers, heterotrophic microbial eukaryotes, bacteria, and nutrient availability—to intra-seasonal variability in microalgal assemblages during the summer?

To address these questions, I examined five complementary microalgal response variables (i.e., microalgal metrics). First, I examined chlorophyll *a* content as a proxy for microalgal biomass (MacIntyre *et al.* 1996), consistent with previous studies validating the reliability of this pigment (Mitbavkar & Anil 2002), which is considered universal among oxygenic photosynthetic organisms, with the exception of certain cyanobacteria (Björn *et al.* 2009). Second, I examined microalgal relative abundance—defined as the percent of 18S rRNA gene reads attributable to microalgal taxa among all microbial eukaryotic taxa—to provide insights into the dynamics of photosynthetic microbial eukaryotes relative to their heterotrophic counterparts. For instance, an increase in microalgal relative abundance may indicate that microalgal biomass has increased while heterotrophic microbial eukaryotic biomass has decreased, remained constant, or increased to a lesser extent. It may also suggest that microalgal biomass decreased less than the heterotrophic microbial eukaryotic biomass. Third, I examined ASV richness—measured as the number of amplicon sequence variants attributable to microalgal taxa—to provide insights into variability in alpha-diversity (Willis 2019). Fourth, I examined

ASV presence-absence, indicating the presence or absence of microalgal amplicon sequence variants. Finally, I examined taxonomic composition, primarily at the genus- or species-level—based on the percentages of 18S rRNA gene reads for microalgal taxa—to identify potential species- or lineage-specific relationships with other microbes and benthic fauna. Together, these metrics synergistically provided a nuanced understanding of the patterns and processes underlying microalgal dynamics at the Grande Anse mudflat in the upper Bay of Fundy.

I applied a multi-faceted approach that integrated traditional ecological methods with high-throughput molecular techniques. I synthesized multi-year (2019–2021) local-scale observations of the microphytobenthos, combining environmental DNA metabarcoding of bacterial, microbial eukaryotic, and meiofaunal taxa; in-situ profiling of sediment physico-chemical properties; chlorophyll fluorescence analysis; microscopy-based identification and enumeration of macro-infauna; and experimental field manipulations of epifauna through exclosures and nutrient availability using fertilizers. Although intra-seasonal summer observations span relatively short timescales compared to the lifespan of a mudflat, they are critical for understanding ecosystem function, particularly when paired with inter-seasonal and interannual observations. Ultimately, by helping to unravel the complexities and clarify the relationships that govern microphytobenthic variability, my thesis provides new insights that inform conservation strategies aimed at protecting biodiversity and sustaining ecosystem services in coastal sedimentary environments vital to climate change mitigation.

2 Methods

2.1 Study site

My study was conducted at Grande Anse (45.807° N, 64.495° W), a large, silt-dominated intertidal mudflat in the Chignecto Bay arm of the upper Bay of Fundy. Typical tidal amplitudes in this region range from 10 to 14 m (Desplanque & Mossman 2004). This mudflat extends approximately 9–10 km alongshore and 2–2.5 km from shore at low tide, and is situated downstream of the Petitcodiac River, a notable freshwater influence. This site was selected for its representativeness of extensive mudflats in the upper Bay of Fundy that are visited annually in summer by large flocks of migratory shorebirds. These Arctic-breeding shorebirds use these mudflats as staging areas and foraging grounds during their southward migration to wintering grounds that are mostly along the northern coast of South America. Semipalmated Sandpipers (*Calidris pusilla*)—with an estimated global breeding population of 2.26 million individuals and eastern Canadian Arctic breeding population of 405,000 (Andres *et al.* 2012)—are the most numerous shorebird species using these mudflats (Hicklin 1987). As much as 30–50% of the global population of Semipalmated Sandpipers has been estimated to use these mudflats (Neima 2016). Grande Anse mudflat sediments generally have higher water content (peaking at about 60%), organic matter content (~6% peak), penetrability, and ostracod density (~120,000 individuals/m^2 peak), but lower particle size (~30 μm peak), and *Corophium* amphipod density (~18,000 individuals/m^2 peak) compared to other mudflats in the upper Bay of Fundy (Gerwing *et al.* 2015a). Peak densities of other infauna are in the hundreds to tens of thousands of individuals per m^2 in Grande Anse mudflat sediments and are generally more comparable to other mudflats

(Gerwing *et al.* 2015a). Mud snail density and chlorophyll *a* content in Grande Anse sediments are also comparable to those of some mudflats but far from the peaks of others in the upper Bay of Fundy (Gerwing *et al.* 2015a). Cercozoans, ciliates, and apusomonads dominate assemblages of microbial consumers in Grande Anse sediments, while syndineans are the dominant parasites of microbial eukaryotes, particularly microalgae (Kalu *et al.* 2023).

2.2 Natural dynamics sampling of the microphytobenthos

Sediment core samples were collected 300–500 m from the shore, between late spring and early fall, from 2019 to 2021. I collected three replicate cores on each of the eight sampling dates in 2019 and 2020, and two replicate cores on each of the three dates in 2021, for a total of 30 cores across three years to investigate the natural dynamics of the microphytobenthos (Table S1). Replicate samples were collected at least 5 m apart, and the same location was never resampled. Sediment cores were collected using tip-less plastic syringes (2–4 cm diameter, 10–13 cm long) and stored at −80 °C until processing. I measured temperature, oxidation-reduction (redox) potential, and pH just below the sediment surface and at a depth of 1 cm using a portable meter and probe (models HI991003 and HI12973, respectively; Hanna Instruments, USA). Sediment penetrability, an integrated assessment of sediment environmental conditions, was also measured as described in Gerwing et al. (2020).

Sediment cores were subsampled at the top 2–5 mm and the 1 ± 0.25 cm depth in 2019 and 2020, and over the top 1 cm in 2021, to obtain 54 samples for environmental DNA isolation (Table S1). The 4 ± 0.25 cm and 7 ± 0.25 cm depths were also examined for cyanobacteria in 2020. I extracted total environmental DNA from each 250–350 mg

sediment sample using the DNeasy PowerSoil (Qiagen, Germany) and NucleoSpin Soil (Takara Bio, Japan) kits, according to the manufacturers' instructions. I used gel electrophoresis (0.9% agarose in TAE buffer) to qualitatively validate the isolation protocols and assess the quality of the isolated DNA. I measured DNA concentrations using a Qubit 4 Fluorometer (Thermo Fisher Scientific, USA) and a NanoVue Plus Spectrophotometer (GE Healthcare, USA). For each environmental DNA sample, PCR amplicons of the SSU rRNA gene (gene regions encoding the small subunit ribosomal RNA) were generated and sequenced using Illumina technology (MiSeq platform) at the Integrated Microbiome Resource facility (Dalhousie University, Halifax, Canada), following reported protocols (Comeau *et al.* 2011), thus generating paired-end reads, each 300 bases long. To investigate microalgal natural dynamics, amplicons of the eukaryote-specific hypervariable V4 region of the 18S rRNA gene were generated using primers E572F and E1009R (Comeau *et al.* 2011) for all samples. To investigate cyanobacterial natural dynamics, amplicons of the bacteria-specific hypervariable V6–V8 region of the 16S rRNA gene were generated using primers B969F and BA1406R (Comeau *et al.* 2011) for samples collected in summer 2020. To further investigate cyanobacterial natural dynamics, metagenomes (i.e., the collective genomic material of all organisms present) of sediment samples collected in summer 2021 were sequenced on the Illumina NextSeq platform at the Integrated Microbiome Resource facility.

2.3 Predator exclosure and fertilizer application field experiment

2.3.1 Experimental design

A field experiment was conducted in July and August 2020 to manipulate the presence of epifaunal predators and the availability of nutrients in mudflat sediments. A study area

approximately 320 m long (alongshore) and 50 m wide (across-shore) was established 300–350 m from the shoreline (Figure 1A). Within this study area, twenty 5 m × 5 m plots were established in two parallel rows of ten, ensuring that each plot was submerged and exposed for approximately equal durations during each tidal cycle. Rows were spaced approximately 30 m apart, with plots within each row separated by 15–30 m. The experiment followed a split-plot design, with the predator treatment applied as the split-plot factor. Within each plot, four 1 × 1 m predator treatment subplots were randomly arranged in a square configuration, spaced ~3 m apart (Figure 1A–C). These subplots included: (i) a shorebird exclosure—a 1 m^2 bamboo frame secured in the mud, with a 15 cm-high plastic mesh top—that allowed access to invertebrates, including mud snails; (ii) a full exclosure that excluded both shorebirds and adult mud snails, consisting of a plastic mesh cage enclosing both the top and sides (Figure 1B), which permitted access by smaller invertebrates; and (iii) an open plot—a bamboo frame with no mesh or physical barriers (Figure 1C)—serving as a reference for the combined effect of shorebird and adult mud snail presence. The mesh tops were secured with reusable cable ties, allowing access to the exclosures for sample collection (Figure 1B). With the full mesh cages, which excluded both shorebirds and adult mud snails, the side mesh was buried 15–20 cm below the sediment surface to prevent adult snails from burrowing in. However, the plastic mesh (6 mm^2 openings) did not restrict access to the exclosure by juvenile mud snails or similarly sized invertebrates. Ten plots (five per row) were randomly assigned to a fertilizer treatment (details below), resulting in 10 replicate subplots for each of the six predator–fertilizer treatment combinations. Each subplot contained nine replicate 33 cm × 33 cm sampling locations. Plots were randomly selected for sample collection, with no plot—nor any subplot or location—sampled

more than once. On the rare occasion that exclosures were dislodged, they were replaced within ~1 day. These mesh structures only minimally reduce light penetration and do not significantly affect water flow through subplots at high tide (Cheverie *et al.* 2014).

To ensure sustained nutrient enrichment over the 2-month experiment, pouches containing slow-release Miracle-Gro Shake 'N Feed fertilizer (formulated for flowering trees and shrubs) were buried approximately 4–5 cm below the sediment surface prior to the July 3–4, 2020 sampling and then replaced prior to the July 30–31 round. Each pouch, made of SurgiTube tubular cotton gauze, contained 56 g of fertilizer composed of 18% nitrogen (13% urea, 4.6% ammoniacal, and 0.4% water insoluble), 6% phosphorus pentoxide (P_2O_5), and 12% potassium oxide (K_2O). Each subplot designated for fertilization was treated with 16 evenly spaced pouches (Figure 1C), amounting to 896 g of fertilizer. As such, each subplot was enriched with up to 161 g of nitrogen, 23 g of phosphorus, and 89 g of potassium per month over the two-month experiment. Enrichment levels were based on the median experimental enrichment (i.e., 150 g/m^2 of nitrogen) reported across 47 soft-sediment studies (Douglas et al. 2016), delivering a balanced mix of essential nutrients to stimulate microalgal growth.

Three replicate plots were sampled during each of three periods—early summer (July 3–5), mid-summer (July 30–31), and late summer (August 27–28)—with a different set of plots used during each sampling event. These dates represent different durations of nutrient enrichment and coincided with variable densities of shorebirds, mud snails, and infaunal invertebrates. Early July served as a temporal control for the fertilizer and predator treatments, providing a baseline for microbial and infaunal community variability. By late July, nutrient enrichment had been ongoing for

approximately one month, with shorebirds and mud snails present at low densities or absent. By late August, nutrient enrichment had occurred for about two months, and shorebird and mud snail densities were substantially higher, though past their seasonal peak by about two weeks.

This experiment was designed to examine spatial variability in biofilm microalgal assemblages (top 2 mm of sediments) at two scales: individual samples (≥3 m apart) and plots (≥15 m apart). Temporal variability was assessed intra-seasonally, across monthly sampling intervals. Sediments were sampled for chlorophyll *a* (a proxy for microalgal biomass), environmental DNA (to assess the relative abundances of bacterial, microbial eukaryotic, and meiofaunal taxa), and macrofauna (identified, size-classed, and enumerated using a microscope). Microalgal variability was examined in relation to the presence and duration of nutrient enrichment, the experimentally manipulated abundances of epibenthic predators, the natural densities of macro-infauna, and the natural relative abundances of bacteria, heterotrophic microbial eukaryotes, and meiofauna (Table S2).

2.3.2 Sample collection and processing

A metal contact corer, cooled with liquid nitrogen, was used to collect biofilm sediment disks (4.7 cm diameter, 2 mm thick). Samples were stored in the dark at −20 °C during fieldwork and subsequently at −80 °C until chlorophyll *a* extraction and quantification. Sediment disks were crushed and freeze-dried under low-light laboratory conditions at −50 °C and 0.05 mBar for a minimum of 5 hours using a FreeZone 1 L Benchtop Freeze Dry System (Labconco Corporation, USA). For chlorophyll *a* extraction, 10–20 mg of each freeze-dried, homogenized sample was mixed with at least 5 mL of solvent—a 3:2

(v/v) mixture of 90% acetone and dimethyl sulfoxide—and incubated in the dark at room temperature for 30 minutes (Shoaf & Lium 1976). This process was repeated once. Chlorophyll *a* concentrations were measured using a Trilogy Laboratory Fluorometer (Turner Designs, USA) (Welschmeyer 1994). A linear calibration curve was constructed using a purified chlorophyll *a* standard (Sigma-Aldrich, USA) by plotting measured fluorescence (relative fluorescence units, RFU) against known pigment concentrations (µg/L) (Welschmeyer 1994). To determine the chlorophyll *a* concentration in each extract (µg/L), the y-intercept (RFU) of the calibration curve was subtracted from the measured fluorescence (RFU), and the result was divided by the slope of the calibration curve (expressed as RFU × L/µg). Finally, chlorophyll *a* content in sediment (µg/g) was calculated by accounting for both the total solvent volume and dry mass of sediment used in the extraction.

Sediment core samples for environmental DNA (eDNA) extraction, amplification, and sequencing were collected using tip-less plastic syringes (2–4 cm diameter, 10–13 cm long) and stored at −80 °C until processing. Total eDNA was extracted from the top 2 mm of each sediment core using a silica-membrane spin-column method and quantified as described in Section 2.2. Bacteria- and eukaryote-specific barcode genes were then amplified and sequenced (see Section 2.2). Macro-infauna from the unconsolidated layer of mudflat sediments (typically the top 5 cm) were sampled using a 5.5 cm diameter corer. Corers were pushed into the sediment until resistance was encountered. Within a day of collection, sediments were sieved through a 250 µm mesh to retain invertebrates, which were subsequently preserved in 95% ethanol until further processing. *Corophium* amphipods, *Macoma* (syn. *Limecola*) clams, and

several families of errant (i.e., mobile predators) and sessile (i.e., sedentary deposit feeders) polychaete worms were taxonomically identified and enumerated using a dissecting microscope. *Corophium volutator* individuals were further classified as juveniles (<4 mm in length) or adults (>4 mm).

Field observations of epibenthic mud snails and shorebirds were conducted to assess population densities and habitat usage. Juvenile and adult mud snail densities were determined by counting individuals within each subplot. Adult mud snails found inside full mesh cages were removed during monitoring visits, which occurred at least twice weekly. Shorebird habitat usage in open control subplots was estimated using a footprint coverage index, calculated as the percentage of grid cells in a 1 m^2 quadrat (subdivided into a 10 × 10 grid) that contained a shorebird footprint (Robar & Hamilton 2007). Shorebird footprints were rarely observed within exclosures. Finally, to prevent puddle formation near experimental plots and therefore minimize disturbance to sediment conditions, footprints created during sampling were smoothed out with a squeegee at the end of each day.

2.4 Analysis of rRNA gene amplicon sequence data and environmental metagenomes

I analyzed the amplicon sequence data using QIIME 2 (Quantitative Insights Into Microbial Ecology, v2022.2) (Bolyen *et al.* 2019) implemented on the Microbiome Helper platform (v2.3.0) (Comeau *et al.* 2017). This workflow included quality control and taxonomic annotation of rRNA gene amplicon sequence reads. Reads were quality-filtered, retaining only those with a Phred score ≥30 over ≥90% of sequenced bases. Chimeric (mixed PCR product) and erroneous (sequencing error-derived) reads were

discarded. Amplicon sequence variants (ASVs, unique sequences) were inferred using Deblur and associated denoising tools (Amir *et al.* 2017). Only ASVs with read counts ≥0.1% of the average read count per sample were retained and taxonomically annotated using a Naive Bayes classifier trained on publicly available reference databases. Bacterial V6–V8 rRNA gene reads were annotated using the SILVA (v138.1) database (Quast *et al.* 2013), while eukaryotic V4 rRNA gene reads were annotated using the PR2 (Protist Ribosomal Reference, v4.14.0) database (Guillou *et al.* 2013). Following quality filtration, an average of 14,088 reads per sample were retained across 114 eukaryotic 18S rRNA gene samples, and 23,448 reads per sample across 81 bacterial 16S rRNA gene samples.

I assigned microbial taxa to functional groups at the best achievable resolution using curated databases, annotation tools, and published literature. Bacterial genera and species were functionally annotated using the FAPROTAX database (Functional Annotation of Prokaryotic Taxa) (Louca *et al.* 2016). Eukaryotic assignments were based on Archibald et al. (2017) and Adl et al. (2019) for microbial eukaryotes broadly, Fiore-Donno et al. (2019) for Cercozoa and Endomyxa specifically, and the FUNGuild database for fungi (Nguyen *et al.* 2016), among other sources. I defined microalgae (photosynthetic microbial eukaryotes) to be inclusive of spores and other microscopic propagules of all photosynthetic eukaryotes with macroscopic or filamentous life stages, excluding land plants. Relative abundances of meiofaunal taxa were calculated as the percentage of rRNA gene reads assigned to each taxon out of the total meiofaunal reads per sample. Similarly, percent relative abundances of bacterial nitrogen and sulfur cyclers, and among microbial eukaryotes, fungi, protistan consumers, protistan parasites,

and microalgae, were calculated relative to the total reads per sample within each biological community.

Metagenomic sequence data from six sediment core samples collected in summer 2021 were analyzed using the SqueezeMeta pipeline (v1.6.2) (Tamames & Puente-Sánchez 2019) using default parameter settings. Typically, this process involves aligning short overlapping DNA sequences (reads) to assemble longer contiguous sequences (contigs), often comprising several genes, which are then annotated taxonomically and functionally. Reads are subsequently mapped back onto these annotated contigs to estimate taxon abundance, which can be expressed as relative read abundance (i.e., the percentage of mapped reads associated with each taxonomic group), coverage depth (i.e., the average number of mapped reads covering each nucleotide position within a contig), or normalized metrics such as RPKM (Reads Per Kilobase per Million mapped reads). However, for my analysis, taxonomic assignments were performed directly on the raw sequence reads rather than on assembled contigs. Similarity searches against the GenBank nr database (Clark *et al.* 2016) were conducted using the DIAMOND sequence aligner (v2.0.15) (Buchfink *et al.* 2021), and taxonomic identities were assigned using a Last Common Ancestor (LCA) algorithm as described in Tamames & Puente-Sánchez (2019). Relative read abundance for each taxon was calculated as the proportion of reads assigned to that taxon out of all taxonomically classified reads in the sample. This approach was necessary because the majority of reads in each of the six metagenomic samples were not represented in contig assemblies, indicating low assembly or mapping efficiency. The limited contig recovery likely resulted from the high microbial diversity of the mudflat sediments combined with the relatively shallow sequencing depth

employed. High biological diversity in environmental samples introduces numerous phylogenetically diverse sequences, complicating contig assembly, while shallow sequencing reduces genome coverage, further hindering robust reconstruction (Knight *et al.* 2018).

2.5 Statistical analyses

2.5.1 Spatial and temporal variability of the microphytobenthos

I characterized intra-seasonal variability in cyanobacterial relative abundance among bacteria using estimates from 36 DNA metabarcoding samples in summer 2020—nine per sediment depth across four depth intervals—and six metagenomic samples in summer 2021. I used the PRIMER v6 statistical package (Clarke & Gorley 2006) with the PERMANOVA v1.0.3 add-on (Anderson *et al.* 2008) to analyze the spatial and temporal dynamics of microalgal assemblages across multiple scales and their relation to biotic and environmental variations. Five response variables were used to assess microalgal dynamics: the univariate metrics chlorophyll *a* content, relative abundance, and ASV richness, and the multivariate metrics ASV presence-absence and species composition (see Introduction for details). Euclidean distances were used to create resemblance matrices for the univariate metrics, and Bray-Curtis similarities were used for the multivariate metrics (Anderson 2017). PERMANOVA and PERMANCOVA (Permutational Multivariate Analysis of Variance and Covariance, respectively) tests were conducted using these resemblance matrices (Clarke & Gorley 2006) and are suitable for both univariate and multivariate analyses (Anderson 2017). Spearman's rank correlation coefficient (ρ) was used to test for independence between covariates, which were considered independent and included in PERMANCOVA models if $\rho \leq 0.95$.

Spearman's rho ranged from −0.30 to 0.10 for pairwise correlations between environmental covariates, and −0.71 to 0.73 for biotic covariates. For PERMAN(C)OVA tests, *P* values were generated using permutations, except with temporal planned contrasts, where Monte Carlo randomization was used due to low numbers of unique permutations. An alpha level of 0.05 was applied to all analyses, with adjustments for temporal planned contrasts made using the Benjamini-Hochberg procedure (Benjamini & Hochberg 1995). Components of variation estimates were used to quantify the proportion of microalgal variability attributable to each source of variation (covariates, factors, interaction terms, and contrasts) in a PERMAN(C)OVA model (Clarke & Gorley 2006; Searle *et al.* 1992). I used Spearman's rho to characterize the strength and direction of significant relationships between univariate microalgal metrics and covariates. I used the RELATE test (Clarke & Gorley 2006) to investigate associations between microalgal metrics (see Table S6).

In 2019 and 2020, sediment cores were subsampled at two depths: the microalgal biofilm in the top 2 or 5 mm, and below the biofilm within the 0.75–1.25 cm interval. In 2021, a cross-sectional sample of the top 1 cm of sediments was obtained from each core. I performed a PERMANOVA to assess vertical variability in microalgal species composition within surficial mudflat sediments, both intra-seasonally and interannually. Depth (2 levels), Year (2 levels: 2019 and 2020), and Sub-season (3 levels: early, mid, and late summer) were used as fixed factors, with Plot (nested in Year and Sub-season) as a random factor. Each of the 12 combinations of Depth, Year, and Sub-season had three replicate plots, each with a single sample. Due to a significant three-way interaction of the fixed factors, a separate PERMANOVA was also performed for each

year. To ensure comparability of microalgal species composition across years, the relative abundance of each species (or the best taxonomic rank available) was averaged across the two surficial depth layers of sediment cores collected in 2019 and 2020. This approach standardized the data to better reflect a composite top 1 cm profile across all three years prior to analyzing temporal dynamics. The same procedure was applied to microalgal ASV richness and relative abundance. I performed a PERMANOVA with planned contrasts to assess intra-seasonal, inter-seasonal, and interannual variations in microalgal species composition. Sub-seasonYear (11 levels spanning late spring to early fall from 2019 to 2021; Table S1A) was used as a fixed factor, with Plot (nested in Sub-seasonYear) as a random factor. Each level of Sub-seasonYear had 2–3 replicate plots (3 in 2019 and 2020, but 2 in 2021), each with a single sample. Following a significant main effect or interaction involving Depth, or a significant temporal contrast, I performed a SIMPER (Similarity Percentages) analysis (Clarke 1993) to identify species or higher-level taxa contributing to significant variations in the composition of microalgal assemblages.

2.5.2 Microalgal intra-seasonal variability in relation to biotic and environmental covariates

I performed PERMANCOVA tests to assess whether microalgal relative abundance, ASV richness, and species composition varied significantly with sediment physico-chemical properties. Sub-season (4 levels: early summer, mid-summer, late summer, and early fall in 2019) and Depth (2 levels: top 0.5 cm and 0.75–1.25 cm) were used as fixed factors, with Plot (nested in Sub-season; 3 replicates) as a random factor. Sediment temperature (°C), redox potential (mV; proxy for oxygen content), pH, and penetrability

(mm; proxy for water content) were included as covariates in PERMANCOVA models. However, interactions between covariates, or between covariates and factors, were excluded from models. A parallel set of PERMANOVA tests was subsequently performed, restricted to the three summer sub-seasons.

I performed PERMANCOVA tests to assess whether chlorophyll *a* content, microalgal relative abundance, ASV richness, ASV presence-absence, and species composition in biofilms (top 2 mm of sediments) varied significantly with biotic and nutrient-related variables (Table S2). Sub-season (3 levels: early, mid, and late summer in 2020) and Fertilizer (2 levels: unfertilized and fertilized) were used as fixed factors, with Plot (nested in Sub-season and Fertilizer) as a random factor. The factor Plot had three levels corresponding to predator exclusion treatments—control subplot (open to both shorebirds and mud snails), shorebird exclosure, and full mesh cage—with three replicates per level. Refer to Table S2 for an exhaustive list of epifaunal, macro-infaunal, meiofaunal, microbial eukaryotic, and bacterial covariates included in PERMANCOVA models. Covariates were ordered along an organismal size gradient—from shorebirds to bacteria—to reflect ecological hierarchy and ensure that larger, higher-trophic organisms were generally given explanatory priority in PERMANCOVA models. Bacterial taxa, though often functionally versatile, were assigned to mutually exclusive groups for analysis. Specifically, taxa shared between nitrate-reducing and either sulfate-reducing or sulfur compound–oxidizing groups were assigned to the latter two, reflecting the broader functional versatility of sulfur-metabolizing bacteria at the Grande Anse mudflat. I excluded covariate interaction terms from PERMANCOVA models, except for two-way interactions involving the manipulated biotic covariates (i.e., shorebirds and

adult mud snails) or the factor Fertilizer. To account for differing units among biotic and environmental covariates—including percent habitat usage, density (individuals per m^2), and relative read abundance for biotic covariates—each covariate was standardized to a common scale using Z-score normalization (mean of 0 and standard deviation of 1), prior to analysis.

2.5.3 Classification of biotic and nutrient-related variables into ecological classes

Biotic covariates were classified as positive or negative ecological variables based primarily on significant positive or negative correlations with chlorophyll *a* content (Table S9). Where no significant correlation with chlorophyll *a* content existed, significant correlations with microalgal ASV richness were considered. In the absence of significant correlations with both chlorophyll *a* content and microalgal ASV richness, significant correlations with microalgal relative abundance were used. Chlorophyll *a* content (biomass proxy) and ASV richness (alpha-diversity index) were prioritized over microalgal relative abundance because they represent absolute rather than relative measures. The sessile polychaete family Spionidae, which was not significantly correlated with any of the above metrics, was classified as having putative net-negative influence on microalgal assemblages based on reported deposit-feeding behaviours (Jumars *et al.* 2015). Fertilizer application increased chlorophyll *a* content; as such, the factor Fertilizer and its interaction terms were classified as positive ecological variables. By estimating the relative influence of positive and negative ecological variables, this thesis provides a broad-scale perspective on the nature of the relationships shaping benthic microalgal dynamics, in addition to taxon-level insights.

3 Results

3.1 Rarity and taxonomic ambiguity of benthic cyanobacteria in the summer

Cyanobacteria were rare in the summer, characterized by taxonomically unresolved diversity, and were almost exclusively detected in the thin biofilm layer at the surface of Grande Anse mudflat sediments. In the top 2 mm of mudflat sediments, the relative abundance of cyanobacteria was consistently at or near zero during a metabarcoding survey in summer 2020. Cyanobacteria accounted for none of the bacterial reads in early summer (n = 12 samples across the four depths examined in the top 7 cm of sediments) and 0.05%, on average, in mid and late summer 2020 (n = 24). In the sedimentary subsurface, cyanobacteria were detected in just 11% of samples, compared to all biofilm samples, in mid and late summer. Metagenomic samples from 2021 estimated the relative read abundance of cyanobacteria in the top 1 cm of mudflat sediments at 0.1% in mid and late summer. Metagenomics also detected taxonomically diverse cyanobacterial assemblages, contrasting their low abundance. However, most cyanobacteria (57% of those reads) remained taxonomically unresolved beyond the phylum level, compared to Gammaproteobacteria (69% of reads were unresolved), Deltaproteobacteria (72%), Alphaproteobacteria (49%), Bacteroidetes (38%), and Planctomycetes (55%), five of the most relatively abundant bacterial lineages in sediments.

Among the resolved cyanobacterial lineages from metagenomic samples collected in 2021, 95% of reads, on average, were accounted for by five orders, namely Oscillatoriales and Nostocales (each 12% of cyanobacterial reads), as well as Pseudanabaenales, Synechococcales, and Chroococcales (each 5–6%). Furthermore,

within these orders, 42–62% of reads, on average, were taxonomically unresolved, and this was highest among Nostocales, within which the family Nostocaceae (22% of reads) and genera *Nostoc* and *Richelia* were otherwise the most abundant. Within Oscillatoriales, the family Oscillatoriaceae, including genera *Moorena* and *Okeania*, represented 22% of reads, while the family Microcoleaceae, including genera *Microcoleus*, *Planktothrix*, and *Symploca*, represented 15% of reads. Other abundant lineages included the genus *Leptolyngbya* (10% of Pseudanabaenales reads) within the more dominant family Leptolyngbyaceae, *Synechococcus* (21% of Synechococcales reads) within Synechococcaceae, and potentially toxigenic *Microcystis* species of Microcystaceae (20% of Chroococcales reads). Both the metabarcoding and metagenomic approaches consistently estimated the low relative abundance of cyanobacteria in Grande Anse mudflat sediments in the summer, with the metagenomic approach providing additional but limited taxonomic resolution of these rare cyanobacterial assemblages. In contrast to the rarity of photosynthetic cyanobacteria in sediments, even within the surface biofilm, photosynthetic microbial eukaryotes (i.e., microalgae) and chemosynthetic bacterial lineages had higher relative abundances.

3.2 Composition of diverse benthic microalgal assemblages

Benthic microalgal assemblages in surficial sediments (ca. top 1 cm) of the Grande Anse mudflat accounted for 33% of microbial eukaryotic rRNA gene reads on average and were mostly comprised of Bacillariophyta (diatoms), which represented 77% of microalgal reads, on average, across the summers of 2019–2021. Microalgal assemblages were also comprised of Chlorophyta (green algae) and Dinophyceae (dinoflagellates), each representing 10–11%, and other lineages including

prasinodermophytes, photosynthetic cercozoans, non-diatom ochrophytes, and cryptophytes, each representing less than 1% of microalgal reads (Figure 2A). Polar centric Mediophyceae (including genera *Thalassiosira* and *Minidiscus*) and raphid pennates (including genera *Gyrosigma*, *Navicula*, *Nitzschia*, *Surirella*, and *Cylindrotheca*) together accounted for most diatoms and 72% of microalgal reads (Figure 2B). Unresolved *Thalassiosira* species alone accounted for 29% of microalgal reads, and a further 1% of microalgal reads were known *Thalassiosira* species. *Delphineis* within the araphid pennates and *Paralia* of radial centric basal Coscinodiscophyceae were similarly among the most abundant diatom genera (Figure 2B). Chlorophyta were dominated by *Picochlorum* (class Trebouxiophyceae), which comprised 6% of microalgal reads, with unresolved Pycnococcaceae (class Pyramimonadophyceae) comprising 3% (Figure 2C). Chlorophyta were also comprised of *Pyramimonas* and *Pterosperma* (class Pyramimonadophyceae), *Ulva* microscopic matter (likely including propagules) and *Dilabifilum* (class Ulvophyceae), *Chlamydomonas* (class Chlorophyceae), and *Micromonas* (class Mamiellophyceae) (Figure 2C). Dinophyceae were dominated by *Alexandrium* (order Gonyaulacales) and, to a lesser extent, *Lepidodinium* (order Gymnodiniales), which comprised 6% and 2% of microalgal reads, respectively (Figure 2C). Dinophyceae were also comprised of *Gonyaulax* (order Gonyaulacales), *Gymnodinium* (order Gymnodiniales), *Protodinium*, *Yihiella*, *Ansanella*, and *Pelagodinium* (order Suessiales), and *Prorocentrum* (order Prorocentrales) (Figure 2C). Across ASVs, the median taxonomic classification confidence was 87% for *Thalassiosira*, with *Minidiscus* (85%), *Gyrosigma* (>99%), *Navicula* (96%), *Nitzschia* (84%), *Picochlorum* (96%), *Alexandrium fundyense* (>99%), and *Lepidodinium* (80%), as indicated. *Paulinella chromatophora* (Cercozoa; 91%

confidence) and *Prasinoderma coloniale* (Prasinodermophyta; >99% confidence) were also present in sediments (Figure 2C).

Within the pool of microalgal ASVs, diatoms accounted for 60% of ASVs compared to 77% of reads, green algae for 19% of ASVs compared to 11% of reads, and dinoflagellates for 11% of ASVs, similar to the 10% of reads they accounted for, on average, across three summers (Figure 3). While polar centric Mediophyceae, including the typically planktonic *Thalassiosira spp*., represented 52% of diatom reads and 40% of microalgal reads (i.e., relative abundance), raphid pennates, including the typically benthic *Gyrosigma limosum*, accounted for 67% of diatom ASVs and 40% of microalgal ASVs (i.e., richness and alpha-diversity). Among green algae, Trebouxiophyceae and Pyramimonadophyceae, which accounted for 50% and 36% of green algal reads, respectively, together represented just 46% of ASVs. This contrasts with Ulvophyceae, Chlorophyceae, and Mamiellophyceae, which were about 3–7 times more relatively diverse than abundant. Among dinoflagellates, Gonyaulacales, which accounted for 68% of Dinophyceae reads, accounted for just 23% of Dinophyceae ASVs, with the other four orders of Dinophyceae being 1.3–23 times more relatively diverse than abundant. Additionally, non-diatom ochrophytes, which comprised just 0.4% of microalgal reads, accounted for 7% of microalgal ASVs (Figure 3).

3.3 Depth-dependent variability of surficial benthic microalgal assemblages

Within the limited spatial scope of surficial mudflat sediments, microalgal taxa exhibited pronounced depth-specific distribution patterns. The effect of sediment depth (i.e., top 2–5 mm versus ca. 1 cm depth) on the species composition of surficial benthic microalgal assemblages varied between years, specifically 2019 versus 2020, and

between sub-seasons in a given year, specifically early and mid-summer versus late summer in 2020 (Table 1). Microalgal species composition in the top 2–5 mm of sediments differed from the 1 cm depth by 46–50% on average, except in late summer 2020 when these surficial sediment depths were 82% dissimilar (Table 2). Microalgal species composition was also more variable across the summer in the top 2–5 mm relative to the 1 cm depth in 2019 and 2020 (Figure 4A). Additionally, even within the relatively more buffered environment of the 1 cm depth, intra-seasonal heterogeneity in microalgal species composition within the summer increased from 2019 to 2020 (Figure 4A). Diatom species consistently contributed the most to differences in microalgal species composition between the surficial sediment depths (Table S3A–D). The average relative abundance of diatoms was not substantially different between depths across summer 2019 (67–70%) and in early and mid-summer 2020 (81–87%). However, diatom relative abundance decreased with depth in late summer 2020 from 98% of microalgal reads in the top biofilm to 68% at the 1 cm depth, closer to the overall average. The different diatom lineages were much more variable across depth layers (Figure 4A). The average relative abundance of raphid pennate diatoms was 62% lower at the 1 cm depth relative to the biofilm layer across the two summers (2019 and 2020), while that of polar centric Mediophyceae was 1.8 times higher (Figure 4B). This pattern of diatom depth distribution, while being consistent over time, was not equally apparent over time (Figure 4B). Peak variability between sediment depths occurred in late summer 2020 when raphid pennate diatoms decreased by 87% (93% to 12%), and polar centric Mediophyceae increased by 835% or just over ninefold (4% to 39%), with depth (Figure 4B).

The depth patterns of green algae displayed notable temporal variability (Figure 4B), with overall spatio-temporal patterns being more pronounced across annual than depth scales. In late summer 2019, green algae had higher relative abundance in the top 2–5 mm compared to the 1 cm depth, on average (33% vs. 19%) (Figure 4B). However, their average relative abundance was lower in the top 2–5 mm in mid and late summer 2020 (1% vs. 4%) and did not show significant variation between depths at other times. When significant depth differences were observed for dinoflagellates, this group exhibited higher relative abundance at the 1 cm depth compared to the top 2–5 mm sediment layer, like in mid-summer 2019 (28% vs. 8%) and late summer 2020 (28% vs. 2%) (Figure 4B). At the genus and species levels, the contribution of unresolved *Thalassiosira* (polar centric diatom) to depth differences was similar between years, accounting for 10–16% of the overall average dissimilarity in microalgal species composition (Table 2). However, annual disparities were evident in the contribution of *Gyrosigma limosum* (raphid pennate diatom), which contributed 13% of depth differences in summer 2019, compared to 20–21% across sub-seasons in summer 2020 (Table 2). Similarly, unresolved *Picochlorum* (green algae) contributed 11% in 2019 versus 0.7–6% in 2020 (Table 2). Notably, the contribution of *Picochlorum* to depth differences decreased over the summer of 2020, while that of *Alexandrium fundyense* (dinoflagellate) was highest in late summer 2020 (Table 2). The relative contributions of other microalgal species to depth variability were similarly dynamic over time (Table S3A–D).

3.4 Seasonal and annual variability of surficial benthic microalgal assemblages

The temporal patterns of microalgal relative abundance (among microbial eukaryotes) and microalgal ASV richness were uncoupled from one another within surficial sediments (i.e., averaging over the top 1 cm) and characterized by out-of-sync peaks, depressions, and plateaus (Figure 5). In 2019, the average relative abundance of microalgae notably peaked in early summer, increasing by 138% relative to the late spring baseline and again in early fall, rising by 105% relative to the preceding mid and late summer (Figure 5). In 2020, microalgal relative abundance instead peaked in late summer at a level 69% higher than the early and mid-summer plateau (Figure 5). There were no corresponding peaks in the ASV richness of microalgae in line with the intra-seasonal, inter-seasonal, or interannual variability of microalgal relative abundance (Figure 5). Instead, microalgal ASV richness was generally temporally stable relative to microalgal relative abundance in 2019 and 2020, but more variable in 2021 (Figure 5). Specifically, while microalgal relative abundance remained statistically unchanged across mid and late summer in 2021, ASV richness was highest early in mid-summer, before decreasing by 48% to below the 3-year summer average for the rest of the season (Figure 5).

Similar to microalgal relative abundance and ASV richness (to a lesser extent), the species composition of surficial benthic microalgal assemblages exhibited temporal dynamicity (Figure 6A). Significant interannual variation in microalgal species composition was observed during the summer, pooling over sub-seasons (Table 3). However, despite observable intra-seasonal summer patterns within each year, pooling across years revealed no significant intra-seasonal variability in microalgal species

composition during the summer (Table 3). This is attributable to the different intra-seasonal taxonomic patterns in each year. In 2019, raphid pennate diatoms peaked at 64% of microalgal reads, on average, in early summer (compared to 14% in mid and late summer), dinoflagellates at 18% in mid-summer (compared to 1% and 8% in early and late summer, respectively), polar centric Mediophyceae at 48% in late summer (compared to 29% in early and mid-summer), and green algae at 29% during mid and late summer (compared to 3% in early summer) (Figure 6B). In 2020, the relative abundance of polar centric Mediophyceae decreased from 53% in early summer to 21% in late summer, while raphid pennate diatoms peaked at 53% in late summer (Figure 6B). Araphid pennates represented just 2% of microalgal reads on average in early summer, versus 9% in mid and late summer, with green algae reaching a seasonal peak of 8% in early summer and dinoflagellates accounting for 7% of reads, on average, in early summer versus 15% in late summer 2020 (Figure 6B). Microalgal species composition also varied significantly during the transitions from spring to summer and summer to fall in 2019 (Table 3).

Microalgal species composition was 43–54% dissimilar on average between the summers of 2019 and 2021 (Table 4), with a more pronounced interannual variability occurring between 2019 and 2020 than between 2020 and 2021 (Figure 6A). Specifically, from 2019 to 2020, the average relative abundance of diatoms in mid and late summer increased from 55% to 84%, while green algae decreased from 29% to 3% (Figure 6A–B). A similar trend was evident between 2019 and 2021. Conversely, from 2020 to 2021, while diatoms remained comparatively stable in their average relative abundance, green algae increased from 3% to 8% in mid and late summer (Figure 6B).

Among diatoms, raphid and araphid pennates decreased from 2020 to 2021 (Figure 6B), whereas polar centric Mediophyceae exhibited higher relative abundance in 2021 (Figure 6B). This pattern contrasts with the 2019 to 2020 period, during which the different diatom lineages generally exhibited similar patterns of increased relative abundance over time (Figure 6B). From 2019 to 2021, dinoflagellates maintained a stable relative abundance in mid and late summer, comprising 12–13% of microalgal reads on average (Figure 6B). At the genus and species levels, unresolved *Thalassiosira* (polar centric diatom) contributed more to the variation between 2020 and 2021 than between 2019 and 2020 (23% vs. 11% of overall dissimilarity). The contributions of *Gyrosigma limosum* (raphid pennate diatom) and *Alexandrium fundyense* (dinoflagellate) were generally consistent over that time frame (12–15% and 6%, respectively). Unresolved *Picochlorum* (green algae) contributed more to the variation between 2019 and 2020 than between 2020 and 2021 (9% vs. 2% of overall dissimilarity) (Table 4; Figure 6). The relative contributions of other microalgal species to inter-seasonal and interannual variability were similarly dynamic (Table S4A–E).

Microalgal species composition was equally dissimilar (50–51%), on average, across the spring–summer and summer–fall transitions (Table 4). Nevertheless, distinct taxonomic patterns and contributors to dissimilarity underpinned significant variations between the seasonal transitions (Table 4; Figure 6A–B). During the transition from spring to summer, diatoms notably increased in average relative abundance by 37% (69% to 95%), while dinoflagellates decreased substantially by 92% (15% to 1%), and green algae declined by 74% (13% to 3%) (Figure 6B). In contrast, during the transition from summer to fall, diatom relative abundance remained stable at 64%, whereas

dinoflagellates increased by 190% (8% to 23%) and green algae decreased by 53% (26% to 11%) (Figure 6B). Across both transitions, raphid pennate diatoms increased in relative abundance, specifically 3 times (21% to 64%) from late spring to early summer and 2.7 times (14% to 36%) from late summer to early fall (Figure 6B). Conversely, polar centric Mediophyceae decreased by 39% (46% to 28%) and 48% (48% to 25%), respectively, during the seasonal transition periods (Figure 6B). Dinoflagellates, green algae, and polar centric Mediophyceae contributed more to microalgal variability during the summer-to-fall transition than the spring-to-summer transition (Table 4). The opposite pattern was apparent with raphid pennate diatoms, which were the top contributors to compositional differences in microalgal assemblages during the spring-to-summer transition (Table 4). Considering all aspects, the composition of surficial benthic microalgal assemblages revealed intricate patterns across multiple temporal scales (Figure 7). Notably, the prominence (i.e., high relative abundance) of the largely unresolved *Thalassiosira* genus served as one of the few stable fixtures within these assemblages over successive summers, even as lineages with marginal and minimal representation (i.e., low relative abundances) displayed remarkable persistence through time. Nonetheless, distinct shifts in the taxonomic makeup of microalgal assemblages were evident in response to drivers operating at intra-seasonal, inter-seasonal, and interannual scales (Figure 7). I subsequently investigated potential environmental and biotic drivers of benthic microalgal dynamics.

3.5 Environmental contributions to benthic microalgal dynamics

Among the sediment environmental variables measured, pH emerged as the variable most consistently associated with significant variability in benthic microalgal

assemblages, albeit with only a marginal contribution (5–9%) to the overall microalgal variability (Table 5A). Specifically, in the period from early summer to early fall 2019, pH, which ranged from 7.1 to 7.7 within surficial sediments, was positively correlated with microalgal relative abundance and ASV richness (Table 5A; Figure S1). Additionally, microalgal species composition and the presence or absence of microalgal ASVs varied significantly in association with pH variability (Table 5A). Sediment temperature, which ranged from a peak of 23.4 °C in early summer to a low of 11.6 °C in early fall, was also positively correlated with microalgal ASV richness and significantly associated with variability in microalgal species composition during the same period (Table 5A). Notably, when solely considering summer dynamics, sediment temperature alone accounted for 44% of the variability in microalgal ASV richness (Table 5B). In contrast to pH and temperature which had weak but significant associations with microalgal metrics, the redox potential within surficial sediments (−123 mV to 268 mV range) and mudflat sediment penetrability (92 mm to 192 mm range), which are proxies for oxygen and water content, respectively, were not significantly associated with any microalgal metric (Table 5A–B; Figure S1). Sediment conditions were relatively homogeneous across the mudflat surface in each sub-season, except early summer for temperature and late summer for penetrability. Conversely, sediment pH conditions were patchier in mid and late summer than the preceding and superseding sub-seasons, while there was no consistent pattern for redox potential.

While pH appears to have lineage-specific associations with microalgal abundance, the relationship with sediment temperature appeared more species-specific. Sediment pH was significantly positively correlated (ρ = 0.41 to 0.53) with polar centric

(specifically *Minidiscus*) and raphid pennate (specifically *Navicula* and *Surirella*) diatom genera but negatively correlated (ρ = −0.44 to −0.46) with non-diatom microalgae, specifically *Chlamydomonas* and *Micromonas* (green algae), *Ansanella granifera* and *Lepidodinium* (dinoflagellates), and *Prasinoderma coloniale* (prasinodermophyte). Among diatoms, sediment temperature was positively correlated with *Paralia* (ρ = 0.67) but negatively associated with *Fistulifera saprophila* (ρ = −0.48). Among cercozoans, it was positively correlated with *Paulinella chromatophora* (ρ = 0.46) but negatively associated with *Gymnochlora stellata* (ρ = −0.44). Sediment temperature was also positively correlated with *Chlamydomonas* (green algae) (ρ = 0.43) but negatively associated (ρ = −0.47 to −0.48) with *Pyramimonas* (green algae) and *Alexandrium* (dinoflagellate).

3.6 Biotic and nutrient-related drivers of benthic microalgal dynamics

In the biofilms (top 2 mm sediments) of the Grande Anse mudflat, biotic and nutrient-related variables accounted for 43–88% of microalgal variability across several key metrics during the summer of 2020. Biotic variables, ranging in size and function from shorebirds to bacteria, were 1.7–3.4 times less relatively important to the variability in chlorophyll *a* content than other microalgal metrics (Figure 8A). Specifically, these variables accounted for less of the variation in chlorophyll *a* content (26%) than microalgal relative abundance (88%) and microalgal ASV richness (53%) (Table 6; Figure 8A). Additionally, biotic variables accounted for 43% of the variability in the presence or absence of microalgal ASVs, hereafter ASV presence-absence, which contrasts with 52% for microalgal species composition, a metric based on the relative abundances of microalgal species. From a biotic perspective, much of the disparity

between the variability of microalgal relative abundance and other microalgal metrics was attributable to shorebirds (15% vs. 1–5% of the variability), adult mud snails (30% vs. 5–12%), ostracods (9% vs. 0.1–4%), and protistan consumers (22% vs. 0–12%), which each accounted for more of the variability in microalgal relative abundance than any other metric (Table 6). Conversely, nutrient-related variables, and to a lesser extent, temporal factors, accounted for the lower biotic contribution to the variability in chlorophyll *a* content, relative to other microalgal metrics (Figure 8A). There was no significant association between variations in chlorophyll *a* content and microalgal relative abundance (Table S7). However, chlorophyll *a* content varied significantly, albeit weakly, with microalgal ASV richness and species composition.

The biotic and nutrient-related variables associated with increased microalgal abundance or richness, in absolute or relative terms (hereafter referred to as 'positive ecological variables'), were comparable to, or more important than negative ecological variables in accounting for microalgal variability in all but one metric, ASV presence-absence (Figure 8A; see Section 2.5.3 and Figure 8B for variable classifications). Positive ecological variables included shorebirds, adult mud snails, and protistan parasites, among other variables (Table 6; Figure 8B), and accounted for more of the variation in microalgal species composition than ASV presence-absence (37% vs. 18%) (Figure 8A). In contrast, juvenile mud snails, adult and juvenile *Corophium* amphipods, protistan consumers, and bacteria, among others, appeared to have net negative ecological relationships with microalgae. These variables accounted for more of the variation in microalgal relative abundance (40%) and ASV richness (36%) than chlorophyll *a* content (14%) (Figure 8A). Temporal factors, operating on a monthly scale

within the summer, accounted for as much as 36% of microalgal variability across metrics, whereas spatial factors, operating on a scale of metres to tens of metres, accounted for up to 46% (Figure 8A). Chlorophyll *a* content was more temporally dynamic than microalgal relative abundance and ASV richness (Figure 9). Conversely, microalgal ASV richness was more spatially variable than chlorophyll *a* content and microalgal relative abundance. Microalgal species composition was more temporally dynamic but less spatially variable than microalgal ASV presence-absence (Figure 10). Microalgal species composition was also more spatially variable than chlorophyll *a* content.

Epifauna, particularly adult mud snails, were the top biotic contributors to microalgal variability (Table 6; Figure 8A). Significant effects of shorebird activity and mud snail density on microalgal assemblages were evident across metrics, distinguishing these variables from the over two dozen other sources of microalgal variability examined (Table 6). Juvenile mud snails, however, were not significantly associated with variability in microalgal ASV richness, the only exception to this notable pattern, and were much less influential than adult mud snails across all microalgal metrics (0–4% vs. 5–30% of the variability). Epifauna explained 45% of the variability in microalgal relative abundance, compared to 20% for species composition and 10–12% each for chlorophyll *a* content, ASV richness, and ASV presence-absence (Figure 8A). Despite accounting for similar proportions of the variability in chlorophyll *a* content, juvenile mud snail density was associated with a decrease in chlorophyll *a* content, in contrast to adult mud snail density, which, like shorebirds, was associated with increased

chlorophyll *a* content, microalgal relative abundance, and ASV richness (Table 6; Figures 9 and S2A).

Macro-infauna (including *Corophium* amphipods and polychaete worms) and meiofauna (including nematodes and ostracods) accounted for similar proportions of microalgal variability, 2–8% and 3–11%, respectively, across metrics (Figure 8A). However, while the contribution of macro-infauna to microalgal variability was lowest with microalgal relative abundance, that of meiofauna was highest for this metric. Adult *Corophium* amphipods were significantly associated with variability in most microalgal metrics, including chlorophyll *a* content ($\rho = -0.41$; Table S8A) and microalgal ASV richness ($\rho = -0.27$; Table S8C), but not microalgal relative abundance (Figure S2B), whereas juveniles were significantly associated with microalgal relative abundance ($\rho = -0.23$; Table S8B; Figure S2B) and species composition (Table 6). *Macoma* clams were negatively correlated with chlorophyll *a* content, albeit weakly, and significantly associated with variability in microalgal species composition (Table 6; Figure S2B). Among polychaetes, Nereididae were associated with decreased chlorophyll *a* content ($\rho = -0.48$; Table S8A; Figure S2C); Phyllodocidae were positively correlated with microalgal relative abundance, albeit weakly, and significantly associated with species composition; and Capitellidae were significantly associated with both microalgal ASV presence-absence and species composition (Table 6), as well as with increased chlorophyll *a* content ($\rho = 0.42$; Table S8A). The sessile polychaete family Spionidae was the only biotic variable not significantly associated with variability in any microalgal metric (Table 6; Figure S2C). Additionally, none of the polychaete families were significantly associated with variability in microalgal ASV richness.

The associations between meiofaunal taxa and microalgal metrics were generally weak when significant (Table S8A–C; Figure S2D). Nematodes and turbellarians were the only meiofaunal taxa positively correlated with chlorophyll *a* content, whereas ostracods were uniquely negatively associated with microalgal ASV richness among the meiofauna, while also positively correlated with microalgal relative abundance (Table 6). The contribution of ostracods to variability in microalgal relative abundance was the largest by a meiofaunal taxon to microalgal dynamics. Ostracods were significantly associated with variations in both microalgal ASV presence-absence and species composition, unlike copepods, which were only associated with species composition (Table 6).

Microbial eukaryotes accounted for just 4% of the variability in chlorophyll *a* content, compared to 28% for microalgal relative abundance and 12–14% each for microalgal ASV richness, ASV presence-absence, and species composition (Figure 8A). Among microbial eukaryotes, protistan consumers were associated with decreased microalgal relative abundance (22% of the variation, $\rho = -0.87$; Table S8B) and ASV richness (12% of the variation, $\rho = -0.54$; Table S8C) but were not significantly associated with variability in chlorophyll *a* content or species composition (Table 6; Figure S2E). Fungi were significantly associated with decreased chlorophyll *a* content, accounting for 3% of the variation ($\rho = -0.60$; Table S8A), and decreased microalgal relative abundance, accounting for 5% of the variation ($\rho = -0.45$; Table S8B). Unlike both protistan consumers and fungi, protistan parasites were significantly associated with microalgal species composition, accounting for 13% of the variability, their largest contribution to microalgal dynamics (Table 6). Protistan parasites were also positively

correlated with chlorophyll *a* content, albeit weakly (Table S8A; Figure S2E), but negatively associated with microalgal relative abundance ($\rho = -0.55$; Table S8B).

Bacterial nitrogen and sulfur cyclers accounted for 2–15% of microalgal variability across metrics (Table 6). All bacterial functional groups examined were significantly associated with variability in microalgal species composition; however, this pattern did not extend to other microalgal metrics (Table 6; Figure S2F). Similar to adult mud snails and protistan consumers, sulfur compound-oxidizing bacteria were top biotic contributors to variations in microalgal ASV richness (accounting for 11%) and ASV presence-absence (6%), respectively (Table 6). Specifically, the increased proportion of variation accounted for by biotic variables in microalgal ASV richness (alpha-diversity index), relative to chlorophyll *a* content (biomass proxy), is attributable to negative interactions with protistan consumers and sulfur-compound oxidizing bacteria. Sulfur compound-oxidizing bacteria, although negatively associated with microalgal ASV richness—a unique pattern among bacteria—were positively correlated with microalgal relative abundance and not significantly associated with chlorophyll *a* content (Table 6). They contrasted sulfate-reducing bacteria, which were negatively associated with chlorophyll *a* content, albeit weakly, but not microalgal relative abundance. Conversely, nitrate-reducing bacteria were negatively associated with microalgal relative abundance ($\rho = -0.45$; Table S8B), but not with chlorophyll *a* content.

Nutrient enrichment through fertilizer application significantly increased chlorophyll *a* content but notably did not have an observable effect on microalgal relative abundance, ASV presence-absence, or species composition (Table 6; Figures 9–10). Nutrient-related variables explained 32% of the variability in chlorophyll *a* content,

only slightly behind the proportion accounted for by temporal factors (36%; Figure 8A). The effect of fertilizer application on chlorophyll *a* content was conditional on adult mud snail density, shorebird activity, and the duration of fertilizer application (Table 6). Though marginally non-significant, the interactive effects of fertilizer application and epifaunal density and activity, notably accounted for 17% of the variability in microalgal ASV richness (Table S8B). Specifically, in late summer 2020, after 8 weeks of fertilizer application, microalgal ASV richness was, on average, 41% higher in fertilized sediments compared to unfertilized sediments where only shorebirds were excluded, 30% lower in fertilized sediments where both shorebirds and adult mud snails were excluded, and comparable between fertilizer treatments where neither epifaunal predator was excluded (Figure 9). That contrasts with chlorophyll *a* content, which, after 8 weeks of fertilizer application, but not 4 weeks, was higher in fertilized compared to unfertilized sediments, regardless of the presence of epifaunal predators (Figure 9). Specifically, fertilizer application caused chlorophyll *a* content to increase by 42% overall: 46% in shorebird exclosures, 44% in exclosures for both shorebirds and adult mud snails, and 36% in open plots (Figure 9). Essentially, nutrient enrichment more substantively influenced chlorophyll *a* content than microalgal ASV richness, and microalgal ASV richness than microalgal relative abundance, ASV presence-absence, and species composition, considering the interactive and non-interactive contributions of biotic variables (Figure 8A).

Half of the variability in microalgal species composition was attributable to a robust base of biotic variables, encompassing all epifauna and bacteria examined, most macro-infauna, half of the meiofaunal taxa, and solely protistan parasites among

microbial eukaryotes (Figures 10–12; Table S8E). Spatial and temporal factors accounted for almost the entirety of the other half of the variability in microalgal species composition (Figure 8A). In essence, biotic variables and spatio-temporal factors interacted with benthic microalgal assemblages in species-specific manners, contrasting with the seemingly broad-spectrum effects of nutrient enrichment (Table 6). Differences in the co-occurrence patterns of raphid pennate and polar centric diatoms played a defining role in the species-specific dynamics of microalgal assemblages (Figures 11–12). Epifauna, adult *Corophium* amphipods, ostracods, and sulfate-reducing bacteria were positively correlated with the most abundant raphid pennate diatom genera but negatively associated with polar centric diatoms and other microalgae (Figure 11). In contrast, protistan parasites, Capitellidae (sessile polychaete worm), and to some extent, nitrate-reducing bacteria, displayed the inverse pattern. Juvenile and adult *Corophium* amphipods were significantly correlated with different sets of microalgal genera, whereas *Macoma* clams were negatively associated with raphid pennates but not significantly correlated with other microalgae (Figure 11). Copepods were positively correlated with a few select microalgal genera, while sulfur-oxidizing bacteria were negatively associated with representatives from several microalgal lineages (Figure 11).

Shorebirds and adult mud snails were notably significantly correlated with 79% and 89% of the most abundant microalgal genera, respectively, compared to 54% for juvenile mud snails, 29% for adult *Corophium* amphipods, and 11% for juvenile *Corophium* (Figure 11). Similarly, significant microalgal-bacterial relationships existed with 71% of genera for sulfate reducers, 57% for nitrate reducers, and 29% for sulfur compound oxidizers. The top protistan parasitic/parasitoid lineages, namely Syndiniales

Group I Clade 1, *Cryothecomonas aestivalis* (Cercozoa) (Drebes *et al.* 1996), and *Pirsonia* clade (Pseudofungi), specifically the taxonomically unresolved members, were significantly correlated with 36%, 46%, and 61% of top-occurring microalgal genera, respectively, across diverse microalgal lineages (Figure 12). Conversely, Syndiniales Group II Clade 1 was significantly correlated with just a single algal genus, while several dinoflagellate genera, most notably *Alexandrium*, were not significantly correlated with any specific protistan parasitic lineage (Figure 12). None of the most abundant microalgal genera had a consistent positive or negative correlation with all biotic variables (Figure 11); however, this pattern was a common feature of microalgal-parasitic relationships (Figure 12).

4 Discussion

4.1 Implications of cyanobacterial rarity on microbial primary production

The rarity of cyanobacteria at the Grande Anse mudflat (also known as Buck's Flat) on Canada's Atlantic coast is consistent with reports from a previous microscopy-based survey at the mudflat by Trites et al. (2005) that sporadically detected cyanobacteria. My finding is also consistent with reports from a macrotidal flat on the French Atlantic coast where Cyanobacteria accounted for <0.01% of the over 31,000 microphytobenthic cells enumerated (Méléder *et al.* 2007). This raises intriguing questions about the underlying ecological mechanisms shaping cyanobacterial abundance in north temperate mudflats. Cyanobacterial rarity at Grande Anse may, in part, be attributable to the high silt content in the sediments, which is among the highest for Bay of Fundy mudflats (Gerwing *et al.* 2015a). Watermann et al. (1999) observed that select species of cyanobacteria and diatoms had variable growth rates when incubated in monospecific cultures under

different temperature and sediment grain size conditions. Specifically, cyanobacterial biomass was highest in sandy sediments (63–200 μm particle size) at 15 °C, whereas diatom biomass peaked in finer sediments—consisting of 50% mud (<63 μm) by weight—at both 10 °C and 15 °C. Therefore, if the sediment grain size is a primary determinant of cyanobacterial abundance in Bay of Fundy mudflats, the relative abundance estimates from the Grande Anse site may represent the lower threshold for mudflats in the region and may not accurately reflect cyanobacterial abundance in coastal phytoplankton. Further research over a larger spatial scale is needed, which should be inclusive of benthic habitats with varying sediment types and planktonic habitats across a turbidity gradient in the Bay of Fundy.

Cyanobacteria often form dense mats in sandy intertidal sediments, while photoautotrophic assemblages of intertidal mud usually occur as biofilms (Stal 2003). These biofilms are thin, often diatom-dominated cell layers within extracellular polymeric substance matrices that offer protection against desiccation and other intertidal stressors (Stal 2003). Benthic diatoms typically migrate downward in mudflat sediments with incoming tides, reducing their risk of resuspension (Consalvey *et al.* 2004; Pinckney *et al.* 1994). However, mat-forming cyanobacteria would face severe challenges to their sustained growth and long-term viability due to the turbulent and disruptive hydrodynamic conditions of the hypertidal Bay of Fundy (Archer 2013; Desplanque & Mossman 2004). Pan et al. (2017) observed that mature cyanobacterial mats in the lower supratidal zone of tidal flats in the Bahía Blanca estuary (Argentina) could not substantially reestablish themselves following physical disruption. Instead, pennate diatoms drove the post-disturbance recolonization.

Cyanobacteria's distinct temperature preferences and competitive interactions with diatoms likely influence their temporal dynamics and latitudinal distribution. Evidence from mixed and monospecific cultures (Watermann *et al.* 1999) suggests that cyanobacteria are less tolerant of colder temperatures and have stronger preferences for relatively warmer temperatures when compared to diatoms. Specifically, filamentous cyanobacteria outcompeted diatoms for space and other resources at an incubation temperature of 25 °C over 30 days in a laboratory experiment. Conversely, diatoms dominated at 10 °C and 15 °C in the sand- and mud-based cultures. Additionally, Scholz & Liebezeit (2012d) found that among tidal flat isolates, cyanobacteria outcompeted diatoms in low-diversity mixed-species cultures, while a diatom outgrew the cyanobacteria in more complex, nitrogen-limited cultures. These findings suggest that diatoms and potentially other microalgae in temperate ecosystems, such as the Bay of Fundy mudflats, may competitively exclude much of the cyanobacterial assemblages, unlike in tropical regions where environmental conditions better favour cyanobacterial proliferation. Furthermore, a study screening 25 diatom and 10 cyanobacterial isolates from the Solthörn tidal flat (Lower Saxon Wadden Sea, Germany) for phytochemical compounds and biological activity, in single-species and mixed cultures, concluded that competitive interactions between cyanobacteria and diatoms are mediated by the production of complex allelochemicals (Scholz & Liebezeit 2012d). The relative climatic stability of tropical regions contrasts starkly with the pronounced seasonality of temperature and other environmental variables in temperate habitats (Addo-Bediako *et al.* 2000), likely further influencing cyanobacterial dynamics. The cyanobacterial relative abundance estimate in my thesis likely reflects summer conditions, including temperature, thus it also likely represents the annual peak at the Grande Anse mudflat,

similar to reports from the Tagus estuary (Portugal) (Jesus *et al.* 2009) and Wadden Sea (Germany) (Vogt *et al.* 2018) intertidal sediments. However, further sampling across different seasons, including during typical microalgal bloom periods, would provide valuable insights into the temporal dynamics of cyanobacteria and their interactions with microalgae in the Bay of Fundy.

Microalgal biomass in the Grande Anse mudflat sediments positively responded to increased nitrogen, phosphorus, and potassium availability, as will be discussed in greater depth later, raising questions about potential nutrient limitations in the mudflat. This is particularly relevant for cyanobacteria, which are also sensitive to nutrient levels in tidal flats for growth (Armitage & Fong 2004). Cyanobacteria are associated with nitrogen fixation in marine ecosystems (Hoffman 1999; Hutchins & Fu 2017), serving as an important source of bioavailable nitrogen essential for microbial metabolism, especially in nutrient-limited habitats. Therefore, their rarity in the Grande Anse mudflat sediments raises intriguing questions about which bacterial or archaeal lineages may fulfill that crucial ecological role. However, recent characterizations of nitrogen-fixing organelles of cyanobacterial origin (i.e., nitroplasts) in haptophytes suggest that microalgae may contribute, even minimally, to nitrogen fixation in mudflat sediments (Massana 2024). Although cyanobacteria are not the only bacterial phylum capable of primary production, they are predominantly composed of oxygenic photoautotrophs, which is unique among bacterial lineages (Thiel *et al.* 2018). The rarity of cyanobacteria at the Grande Anse mudflat suggests that other groups drive microbial primary production. Although extensive knowledge exists about the broad identities of eukaryotic microorganisms involved in photosynthesis in these mudflats, further

research is needed to explore the contributions of chemosynthetic bacteria to mudflat trophic dynamics. In this regard, while all autotrophs fix carbon dioxide, photoautotrophs like cyanobacteria derive energy from light, while chemoautotrophs—such as many bacterial lineages detected in Grande Anse sediments during my sampling program, but not examined in this thesis—obtain energy for biosynthesis through the oxidation of inorganic compounds, including hydrogen gas, sulfides, ammonia, nitrite, and ferrous iron (Kelly & Wood 2006). The higher relative abundance of diatom-dominated benthic microalgal assemblages in Grande Anse mudflat biofilms compared to cyanobacteria (33% of microbial eukaryotic reads vs. <0.1% of bacterial reads) likely indicates the differential contributions of these oxygenic photosynthesizers to primary production in Bay of Fundy mudflats.

Cyanobacterial diversity, which was mostly taxonomically ambiguous, was almost exclusively detected in the top 2 mm of sediments during sampling for my thesis, unlike microalgae in the same mudflat (Kalu *et al.* 2023). Given the steep oxygen gradients in these sediments (Gerwing *et al.* 2015b; Glud 2008), the vertical distribution of cyanobacteria in Grande Anse sediments may indicate their increased sensitivity to oxygen conditions or the decreased formation and persistence of resting stages compared to microalgae (Ellegaard & Ribeiro 2018). The observed taxonomic ambiguity of cyanobacteria in the present study underscores the need for deeper exploration of uncharacterized cyanobacterial diversity in these understudied ecosystems to expand the coverage of reference taxonomic databases. Despite this ambiguity, the consistency between the metabarcoding and metagenomic approaches in quantifying the low relative abundance of cyanobacteria in Grande Anse mudflat sediments validates the observed

patterns. Ultimately, cyanobacterial dynamics in the studied Bay of Fundy mudflat likely result from a deterministic interplay of sediment grain size, climate, hydrodynamic forces, nutrient availability, and competitive interactions, ranked by their probable contribution. However, further research is necessary for a more holistic understanding of these processes and their implications on mudflat ecosystem functioning. Cyanobacteria exist within the rare microbial biosphere (Pascoal *et al.* 2021) of the Grande Anse mudflat and likely other Bay of Fundy mudflats, and therefore may not by themselves constitute a core functional component of the sediment microbiome.

4.2 Divergent ecological strategies of dominant diatoms as drivers of spatio-temporal heterogeneity in mudflat microalgal assemblages

4.2.1 Diatom diversity, motility, and ecological dominance

Based on three consecutive summers of amplicon sequence data, microalgal presence in Grande Anse surface sediments was neither imposing nor negligible; instead, it represented one-third of reads in a eukaryotic microbiome also populated by notable predatory and parasitic contingents (Kalu *et al.* 2023). Diatoms dominated the microalgal assemblages at the Grande Anse mudflat, accounting for 77% of reads and 60% of ASVs. This pattern mirrors findings from intertidal mudflats in the Severn estuary (United Kingdom), where diatoms accounted for 95% of living epipelic microalgal cells (Underwood 1994), and from a macrotidal flat in the Bay of Bourgneuf (French Atlantic coast), where they accounted for 97% of taxa and cells in the photic zone microphytobenthos (Méléder *et al.* 2007). Forster et al. (2016) similarly reported that diatoms comprised the vast majority of phototrophic diversity in European coastal sediments, while Broman et al. (2017) found that diatoms dominated eukaryotic

metatranscriptomes (gene expression) in coastal Baltic Sea sediments during spring and fall. At the Grande Anse mudflat (Buck's Flat) specifically, Trites et al. (2005) observed live diatom densities exceeding 200,000 cells per gram of wet sediment in early July. At Grande Anse, diatom dominance in the sequence data aligns with expectations that they serve as important food sources for benthic fauna and contribute substantially to nutrient cycling and carbon sequestration. Given their dominance, even modest shifts in the abundance or diversity of these diatoms raise the possibility of cascading effects that extend upward through the food web and outward across the broader mudflat ecosystem.

Polar centric and raphid pennate diatoms jointly accounted for at least 90% of diatom reads, on average, and ASVs at Grande Anse over three consecutive summers, highlighting their co-dominance. Notably, although raphid pennate diatoms were 20% less abundant than polar centric diatoms based on sequence reads, they were nearly three times more diverse, accounting for two-thirds of diatom ASVs. Piredda et al. (2018) observed a similar pattern in European coastal habitats. This pattern may indicate relatively broader niche exploitation and specialization by raphid pennate diatoms in mudflats. Yet, the co-dominance of polar centric and raphid pennate diatoms remains notable given their divergent spatiotemporal dynamics across multiple scales, as well as their contrasting ecological relationships. Trade-offs associated with planktonic versus benthic origin, passive versus active mobility, and the relative reliance on physiological versus behavioral stress responses may help explain the observed patterns.

Raphid pennate diatoms in the Grande Anse mudflat were mostly represented by the genera *Gyrosigma* (specifically the 55–85 µm long and 8.5–12.5 µm wide *G. limosum*) (Sterrenburg & Underwood 1997), *Navicula* (*N. cryptocephala*), *Surirella*,

Nitzschia, and *Cylindrotheca*. These diatoms are typically epipelic, living on or near the sediment surface. Together with other raphid pennate genera, they are well-established components of the microphytobenthos in European intertidal systems, including the Colne estuary (United Kingdom) (Thornton *et al.* 2002), Severn estuary (UK) (Underwood 1994), Menai Straits (UK) (Happey-Wood & Jones 1988), Tagus estuary (Portugal) (Ribeiro *et al.* 2003, 2013), Westerschelde estuary (the Netherlands) (Sahan *et al.* 2007), Marennes-Oléron Bay (French Atlantic coast) (Haubois *et al.* 2005), and Solthörn tidal flat (southern North Sea, Germany) (Scholz & Liebezeit 2012b). In contrast, centric and/or tychoplanktonic diatoms are prominent in other tidal flats. For example, in two temperate intertidal mudflats in the hypertidal upper Bay of Fundy (Canada), *Paralia sulcata* and *Brockmanniella brockmanni* together accounted for half of the live diatom cells from late June to late August (Trites *et al.* 2005). Similarly, viable *Thalassiosira* cells were well represented in a monsoon-influenced tropical sandflat (Dias Beach, India) (Mitbavkar & Anil 2006).

The specialized raphe system in raphid pennate diatoms enables gliding motility and vertical migration in benthic environments. The literature supports a dual mechanism of diatom gliding motility in which the actin–myosin system associated with the raphe—a slit in the silica frustule—generates propulsive force, while mucilage secreted from vesicles associated with the raphe provides the adhesion to the substrate necessary for traction and effective motility (Chen *et al.* 2019; Edgar & Pickett-Heaps 1983; Poulsen *et al.* 1999). This motility may facilitate biotic and abiotic stress avoidance (Consalvey *et al.* 2004), resource acquisition, and post-disturbance recovery. For instance, positive chemotactic responses to silicate and phosphate, although not to

nitrate or ammonium, have been reported in the raphid pennate diatom *Seminavis robusta*, under nutrient limitation (Bondoc *et al.* 2016, 2019). Pan et al. (2017) found that, following experimental physical disturbance of tidal flat sediments in the Bahía Blanca estuary (Argentina), pennate diatoms drove rapid recolonization of sediment biofilms, thereby stabilizing sediments and facilitating subsequent microbial colonization—a process attributed in part to their motility. Underwood & Paterson (1993) similarly found that, in the absence of grazing macrofauna, intertidal epipelic diatom assemblages in the Severn estuary (UK) showed full recovery within six days of a biocide (formaldehyde) treatment, likely due to their rapid recolonization capacity. These attributes likely confer on raphid pennate diatoms a competitive advantage over non-motile microalgae in the dynamic and environmentally heterogeneous conditions prevailing on mudflats. The motility of raphid pennate diatoms—with maximum speeds occurring during tidal exposure (Happey-Wood & Jones 1988)—contrasts with the passive dispersal mechanisms typical of many polar centric diatoms, including tidal transport, sedimentation, and both biotic and abiotic sediment reworking.

The non-motile genera *Thalassiosira* and *Minidiscus* (Kaczmarska *et al.* 2009) (specifically M. *spinulatus* and M. *comicus*) comprised most polar centric diatoms in Grande Anse mudflat sediments. These taxa likely occur as a mix of vegetative cells and resting stages, presumably of planktonic origin—that is, taxa that spend most of their life cycle, including reproduction, in the water column. Cells deposited cyclically can accumulate in intertidal sediments over time. These genera feature prominently in coastal and offshore marine systems globally, specifically the nanoplankton for *Minidiscus* and both the nano- and microplankton for *Thalassiosira*. This is the case for

Minidiscus in the northwestern Mediterranean Sea (Leblanc *et al.* 2018), where they play important roles in nutrient and biomass export to the seafloor, particularly during spring blooms; *Thalassiosira* in the Gulf of Naples (Mediterranean Sea) (Montresor *et al.* 2013), as well as in Jiaozhou Bay (China) where their alpha diversity showed distinct temporal patterns between the 0.2–10 µm and 10–200 µm size fractions; and both genera in the Western English Channel (Arsenieff *et al.* 2020). In their global survey of diatom diversity, Malviya et al. (2016) identified *Thalassiosira* as one of the most abundant and diverse planktonic genera, alongside *Chaetoceros* and others, and noted its uniform global distribution. In their meta-analysis of 86 coastal sites across North America and Europe, Carstensen et al. (2015) found that *Thalassiosira* species were among the frequent contributors to phytoplankton blooms, with diatoms, including many centric diatoms, dominating 58% of events. These were primarily monospecific spring blooms that occurred across a broad range of salinity and temperature conditions, reflecting the ecological versatility of these diatoms. Additionally, the order Thalassiosirales, which includes the genera *Thalassiosira* and *Skeletonema*, dominated diatom metatranscriptomes (gene expression) in coastal Baltic Sea sediments in the spring and fall, accounting for over 95% of RNA reads (Broman *et al.* 2017). However, a previous microscopy-based survey of diatoms at my study site, Grande Anse (Buck's Flat), found that while centric tychoplanktonic species were abundant within live diatom assemblages, *Thalassiosira* cells were present but not numerous (Trites *et al.* 2005).

The high relative abundance of unresolved *Thalassiosira* species at the Grande Anse mudflat—averaging 29% of microalgal reads—raises intriguing questions about their specific identity, physiological state, adaptive strategies underlying their

dominance, and ecological roles, both locally and potentially across the Bay of Fundy. These metabolically ambiguous *Thalassiosira* mudflat diatoms can be studied using established workflows. Specifically, flow cytometric sorting—to isolate target microalgal cells from sediments based on size, photopigment complement, viability, and metabolic activity (the latter two assessed via fluorescent dyes)—paired with downstream culturing, microscopy, metagenomics, and/or phylogenetic analysis can be highly effective (Martinez-Garcia *et al.* 2012; Sieracki *et al.* 2019). Beyond the intrigue, the high abundance and consistent (100%) detection of *Thalassiosira*, spanning multiple temporal scales, demonstrate remarkable persistence and suggest a robust allochthonous pelagic influence at the Grande Anse mudflat. Such pelagic contributions can be ecologically important. For example, van Oevelen et al. (2006) estimated that microbial primary production accounted for 80% of carbon inputs to the Molenplaat, a temperate estuarine tidal flat in the Netherlands, with deposited phytoplankton exceeding local microphytobenthic contribution by 50%.

4.2.2 Life-history strategies, resting stage formation, and implications for mudflat ecosystem function

Evolutionary and ecological trade-offs between polar centric and raphid pennate diatoms—specifically in their reliance on physiological versus behavioral adaptive strategies—are evident across multiple functional traits. For instance, regarding photoprotection, Barnett et al. (2015) reported that diatom growth forms in intertidal sediments differ in their reliance on physiological mechanisms, with motile epipelic diatoms exhibiting the lowest capacity for thermal dissipation of excess light energy (i.e., non-photochemical quenching) and non-motile epipsammic diatoms exhibiting the

highest. Additionally, Jesus et al. (2023) noted the use of vertical migration and cell orientation as behavioural strategies to modulate light exposure among motile epipelic diatoms. The formation of resting stages is another instance of this trade-off, which, as a physiological and morphological stress response, is a trait more commonly associated with centric than pennate diatoms (McQuoid & Hobson 1996). This is supported by observations from Pacific Arctic sediments, where viable resting stage assemblages included 12 centric genera but only 7 pennate genera, with centric taxa numerically dominant at most stations sampled (Fukai *et al.* 2021).

Coastal sediments have long been described as reservoirs or 'seed banks' of microbial diversity, partly due to the widespread accumulation of microalgal resting stages, including those of diatoms and dinoflagellates, built up over successive bloom cycles (Itakura *et al.* 1997; Lewis *et al.* 1999). Triggered by environmental cues, including light, temperature, nutrients, and oxygen, the formation of resting stages involves physiological and morphological transformations that enhance the resilience of diatoms to perturbations and resource limitations (McQuoid & Hobson 1996; Sugie & Kuma 2008; Wang *et al.* 2024). For instance, tidal burial of microalgal cells deep within intertidal mudflat sediments may induce this response; however, Broman et al. (2017) concluded that low light rather than low oxygen limits metabolic activity and vegetative growth in resting diatoms. Additionally, diatom resting stages may seed algal blooms upon germination, similar to processes proposed for dinoflagellates (Anderson *et al.* 2012).

The ability to form long-lived resting stages may contribute to the ecological success of diatoms at the Grande Anse mudflat. Diatom resting stages can remain viable

for millennia, although their viability decreases with age (Sanyal *et al.* 2022). A germination experiment on *Chaetoceros muelleri* resting stages, a colony-forming centric diatom retrieved from subsurface Baltic Sea sediments, found that 41% of resting stages aged ≤80 years germinated, compared to just 12% aged ~6600 years (Sanyal *et al.* 2022). Diatom resting stages may also remain viable longer than those of dinoflagellates. McQuoid et al. (2002) found that planktonic diatom resting stages remained viable down to 40 cm in Koljö Fjord (Sweden) sediments—corresponding to decades—while dinoflagellate resting stages, which were over 99% less abundant, remained viable up to less than half that depth. Among polar centric diatoms, resting stage viability appears to be species-specific and may be prolonged by low-temperature conditions (Itakura *et al.* 1997).

Diatoms demonstrate remarkable versatility in nitrogen metabolism. They are adapted to utilize a wide range of nitrogen sources and are unique among photosynthetic eukaryotes in possessing a complete urea cycle (Armbrust *et al.* 2004; Parker *et al.* 2008). This versatility likely confers an advantage over other microalgae in variable, nutrient-limited environments. Moreover, some diatoms utilize dissimilatory nitrate reduction to ammonium (DNRA), an anaerobic energy-generating pathway not observed in dinoflagellates or green algae (Kamp *et al.* 2015). DNRA conserves bioavailable nitrogen within sediments and has been hypothesized to aid the energy-intensive transition of vegetative cells to resting stages, in nitrate-storing diatom species like *Thalassiosira weissflogii* (polar centric) and *Amphora coffeaeformis* (raphid pennate), during the onset of dark and anoxic conditions (Kamp *et al.* 2013, 2015). Recent findings by Stenow et al. (2024) demonstrate that diatom resting stages can perform

DNRA using extracellular nitrate, sustaining energy generation beyond the initial transition. These resting stages, previously considered largely metabolically inactive, can also assimilate urea and acetate from sediments to sustain cellular maintenance (Stenow *et al.* 2024). Thus, even as resting stages, diatoms may continue to influence sediment biogeochemistry.

Changes associated with microalgal resting stage formation include cytoplasmic acidification, which has been implicated in the reduced respiration and photosynthesis observed in this life stage (Wang *et al.* 2024). Transcriptomic analysis revealed the downregulation of photosynthesis-related genes in the centric marine diatom *Chaetoceros socialis* during resting stage formation induced by nitrogen limitation (Pelusi *et al.* 2023). This has clear implications for benthic primary productivity. However, the relatively quick transition from vegetative to resting stages contrasts with the much longer timeline (weeks, months, or even longer) over which photopigment degradation may occur, allowing for the resumption of photosynthetic activity after extended periods of dark and anoxic conditions (Kamp *et al.* 2013). Furthermore, Broman et al. (2017) found that resting diatoms in coastal Baltic Sea sediments remained transcriptionally active in dark and oxygen-deficient conditions, with a notable portion of their activity related to photosystem maintenance. Consequently, if most polar centric diatoms at Grande Anse exist as resting stages, it follows that raphid pennate diatoms may contribute more to net primary production, while resting polar centric diatoms may resume high production under more favourable environmental conditions. This interpretation is supported by the depth distribution patterns observed at Grande Anse, which were pronounced—up to 82% dissimilar—despite the expected

homogenizing influence of semi-diurnal tidal currents in this hyper-tidal coastal system. Specifically, raphid pennate diatoms were consistently more abundant in the top 2–5 mm of sediment—where most primary production is expected due to shallow light and oxygen penetration into the mud—than in the 1 cm layer, where polar centric diatoms were most abundant based on relative read proportions. Kalu et al. (2023) reported that this contrasting distribution among diatoms extends, at minimum, to a depth of 7 cm at Grande Anse. Accordingly, while raphid pennate diatoms may drive primary production and energy flow near the sediment surface, cyclically deposited polar centric diatoms may disproportionately contribute to buried organic carbon stores—an important consideration for carbon burial modelling in the Bay of Fundy and similar coastal systems.

4.2.3 Broader relevance for Bay of Fundy mudflats

The Grande Anse mudflat is among the siltiest in the Bay of Fundy (Gerwing *et al.* 2015a), a notable point given that sediment grain size is a key predictor of diatom genus occurrence (Mitbavkar & Anil 2002). This relationship is further supported by observations from the Tagus estuary (Portugal), where, over a two-year period and across stations with variable grain size profiles, mud content was strongly correlated ($\rho = 0.86$) with the composition of intertidal benthic diatom assemblages (Ribeiro *et al.* 2013). Building on this, Méléder et al. (2007) reported that hydrodynamic processes strongly influenced microphytobenthic variability across a temperate macrotidal mudflat, both spatially and on a monthly scale across a 15-month period. Kalu et al. (2023), providing further nuance, found that 73% of observed microbial eukaryotic ASVs were shared between two mudflats separated by 21 km of coastline in the upper

Bay of Fundy. This overlap in biodiversity could be interpreted as preliminary evidence of a high level of connectivity within hypertidal sections of the Bay of Fundy, likely mediated by hydrodynamic processes. On this basis, the diatom and other microalgal diversity detected at Grande Anse may represent a substantial subset of a regional species pool, with site-specific factors like grain size shaping local composition. To clarify the influence of hydrodynamic processes on microalgal depth patterns, future work should assess the vertical extent of semi-diurnal sediment resuspension. It will also be important to quantify the relative densities of vegetative cells and resting stages of polar centric diatoms in Bay of Fundy mudflats, to better understand life stage dynamics and their implications for ecosystem function. Finally, microalgal dynamics should be examined concurrently in the mudflat benthos and coastal plankton to quantify benthic–pelagic linkages through time and across tidal regimes in the Bay of Fundy.

4.3 Insights into the ecological roles of green algae, dinoflagellates, and rare microalgae in diatom-dominated microphytobenthos

Phototrophic dinoflagellates are dominant in the photic zone of temperate and tropical oceans (Singer *et al.* 2021; de Vargas *et al.* 2015) but only accounted for 10% of microalgal reads—mostly *Alexandrium* (specifically *A. fundyense*) and *Lepidodinium*—at Grande Anse. *Lepidodinium* species are green dinoflagellates, harboring secondary plastids of pedinophycean green algal origin (Watanabe *et al.* 1990). They exhibit temperature-sensitive mixotrophic growth (Chen *et al.* 2024; Liu *et al.* 2021, 2022). They preferentially consume nitrogen-rich prey under suboptimal nutrient regimes (Liu *et al.* 2022; Ng *et al.* 2017). Additionally, although resting cyst formation has not been

confirmed, *Lepidodinium* species have been found to persist in the water column at low abundances outside bloom events (Roux *et al.* 2023).

Certain dinoflagellates, including many *Alexandrium* species, form resting stages (Anderson *et al.* 2012). If metabolically active in Grande Anse sediments, *Alexandrium* may face chemical interference from diatoms, as hypothesized by Townsend et al. (2005) for coastal and offshore waters of the northern Gulf of Maine. Parasitoid activity may also suppress *Alexandrium*, as observed in a Nauset Marsh pond (Massachusetts, USA), where *Amoebophrya* spp.—parasites of dinoflagellates belonging to the Syndiniales Dino-Group-II-Clade-4 lineage (Adl *et al.* 2019)—induced daily mortality equivalent to ~30% of the standing A. *fundyense* population during bloom decline (Velo-Suárez *et al.* 2013). At the same time, *Alexandrium* at the Grande Anse mudflat may be stimulated by facilitative interactions with bacteria. For instance, Ferrier et al. (2002), working with isolates from near-surface waters in the outer Bay of Fundy during an *Alexandrium* bloom, found that among bacterial assemblages introduced into axenic *Alexandrium fundyense* cultures, only the gammaproteobacterium *Alteromonas* sp. (family Alteromonadaceae)—also detected at the Grande Anse mudflat in my study—substantially enhanced A. *fundyense* growth. Specifically, the bacterium increased daily growth rates of A. *fundyense* by 63% based on autofluorescence signals, and 5.5-fold based on cell counts, highlighting the magnitude of the biological response in addition to the specificity of this microalgal–bacterial interaction (Ferrier *et al.* 2002).

In the water column, the pronounced occurrence of *Alexandrium* parasites in sequence data has often marked host activity, specifically bloom events. In Northport Bay (New York, USA) waters, *Amoebophyra* spp. accounted for up to 17% of total algal

and heterotrophic dinoflagellate reads during the peak of the spring *Alexandrium* bloom (~25,300 cells L^{-1}) in early to mid-May, when water temperature averaged 15.8 °C (Hattenrath-Lehmann & Gobler 2017). Additionally, the relative read abundance of *Amoebophyra* spp. was approximately 2.5 times higher in Northport Bay plankton cultures amended with cell-free *Alexandrium* culture medium, compared to unamended control cultures, following a 48-hour incubation (Hattenrath-Lehmann & Gobler 2017). However, at Grande Anse, *Alexandrium* was not significantly correlated with any protistan parasite lineage, despite a global correlation with parasite relative abundance. This contrasts with *Lepidodinium*, which was positively correlated with several protistan parasite lineages at Grande Anse. However, across three summers of sequence data, *Alexandrium* showed no significant correlation with any parasitic lineage in Grande Anse surficial sediments. Bacterial groups may also respond to *Alexandrium*-associated compounds. For instance, Flavobacteriia relative read abundance increased 2.9-fold in amended cultures, with *Owenweeksia* spp. alone rising 11-fold (Hattenrath-Lehmann & Gobler 2017). At Grande Anse, *Alexandrium* was positively correlated with nitrate-reducing bacteria, albeit weakly, but not significantly correlated with sulfur-metabolizing bacteria.

Resting stage formation may enable *Alexandrium* species to occupy relatively deeper sediments. At Grande Anse, dinoflagellate depth patterns were not consistent; however, when dinoflagellate relative abundance significantly differed between the top 2–5 mm and 1 cm depth layers, it trended toward increased magnitude with depth. The accumulation of resting stages in sediments may contribute to the ecological success of *Alexandrium* by extending the geographic range of future bloom events (Anderson *et al.*

2014). McGillicuddy et al. (2005) demonstrated that the germination of resting cysts from coastal sediments—which releases vegetative cells into the water column—is pivotal in shaping *Alexandrium fundyense* seasonal population dynamics, a finding reinforced by Martin et al. (2014), who concluded that cyst-derived cells play an important role in A. *fundyense* bloom initiation in the Bay of Fundy. Moreover, Butman et al. (2014) revealed that tidal and wave-induced resuspension erodes the top millimeter of these cyst-rich coastal sediments—where cyst germination rates are predicted to be high (Anderson *et al.* 2005) and may follow an endogenous circannual rhythm (Matrai *et al.* 2005)—mobilizing cysts into the water column and thereby increasing the pool available for germination. As such, the presence of *Alexandrium* species, which were previously found to be toxigenic at Grande Anse (Kalu *et al.* 2023), is notable, due to the potential for harmful algal blooms seeded by mudflat dinoflagellates to disrupt marine food webs and adversely impact human health (Anderson *et al.* 2012). Specifically, *Alexandrium fundyense* isolates from the Bay of Fundy have been shown to produce paralytic shellfish poisoning (PSP) toxins (Orr *et al.* 2011).

Green algae, which are diverse and abundant in freshwater ecosystems (Debroas *et al.* 2017) and comprise most of the piconanoplankton (0.8 to 5 mm cell size fraction) in the photic zone of global oceans (de Vargas *et al.* 2015), were primarily represented by the genus *Picochlorum* and unresolved members of the Pycnococcaceae family at Grande Anse. The trebouxiophycean green alga *Picochlorum* is recognized for its stress tolerance and high productivity, potentially granting it broader resilience to certain environmental perturbations than most other microalgae. *Picochlorum* is described as being coccoid, with a diameter of 1.5–3 μm, mostly lacking flagella, and reproducing

through autospore formation (Henley *et al.* 2004), as well as composed of 20–23% lipids (Gates *et al.* 2024). *Picochlorum* can achieve growth rates >40 g m^{-2} day^{-1} (Krishnan *et al.* 2021), in part due to an efficient photosystem II (Gates *et al.* 2024), and can tolerate salinities up to 107.5 g L^{-1} as well as temperatures as high as 40 °C (Dahlin *et al.* 2019). *Picochlorum* possesses a highly reduced, gene-dense genome complemented by horizontally acquired, stress-adaptive genes of recent evolutionary origins that are upregulated in response to high temperature and variable salinity conditions (Foflonker *et al.* 2015, 2018; Krasovec *et al.* 2018). Specifically, 4 °C above its 38 °C optimal growth temperature, *Picochlorum* downregulates genes involved in core metabolic and biosynthetic processes—such as photosynthesis and carbon fixation—while upregulating pathways that recycle or remove damaged proteins and cellular waste, suggesting a shift away from energy-intensive processes and a strategic reallocation of resources to enhance survival (Barten *et al.* 2022). *Picochlorum* also displays a rapid increase in cell volume, accumulates fatty acids that serve as energy reserves, and alters its pigment and fatty acid composition under high temperature conditions (Barten *et al.* 2022). The genome of *Picochlorum* is also organized into functional clusters for rapid stress response and enriched in pathways supporting versatile carbon and nitrogen metabolism (Foflonker *et al.* 2015, 2016). Together, these studies suggest that *Picochlorum* may be able to apply the same genetic toolkit to tackle various environmental challenges to their persistence in intertidal mudflat sediment.

The pico- to nano-sized (1.5–3 μm diameter) marine prasinophyte green algae *Pycnococcus provasolii* (Fawley *et al.* 1999), a thecate scaleless coccoid, and *Pseudoscourfieldia marina* (Moestrup & Throndsen 1988), a scaly biflagellate, were

thought to represent the only extant genera and known species of the family Pycnococcaceae. However, recent plastomic and mitogenomic evidence suggests that they are morphologically distinct growth forms or alternate life-cycle stages of the same species (Crépeault *et al.* 2024), with other evidence suggesting adaptations to low-light conditions in planktonic environments (Guillard *et al.* 1991). A study in the subtropical region of the northwestern North Pacific Ocean detected these algae in the epipelagic zone throughout the year and found that they sank to the abyssopelagic zone during blooms (Shiozaki *et al.* 2019). This suggests that blooms of this green alga in the outer Bay of Fundy may be detected as sedimentary deposits in mudflats in the region.

Rare microalgal taxa, those with low relative abundance, included non-diatom ochrophytes, prasinodermophytes, photosynthetic cercozoans, and cryptophytes. Despite their low abundance, rare microbial taxa can exert an outsized influence on ecosystem processes such as nutrient turnover (Jousset *et al.* 2017; Pascoal *et al.* 2021). Microalgal lineages that I found to be rare or moderately abundant in Grande Anse mudflat sediments have been reported as being dominant in other ecosystems. For instance, Chrysophyceae (golden algae) and Chlorophyta (green algae) are among the most abundant phototrophic protists in soils, particularly in arid environments (Bates *et al.* 2013; Oliverio *et al.* 2020). They, along with Cryptophyta and other groups, are also key members of freshwater phytoplankton (Debroas *et al.* 2017; Garner *et al.* 2022). Genomic evidence shows that Prasinodermophyta, a newly classified phylum within Viridiplantae, exhibits adaptations to nutrient-poor, low-light oceanic conditions (Hasegawa *et al.* 1996; Li *et al.* 2020; Yang *et al.* 2023). Additionally, while not typically dominant in any ecosystem, the photosynthetic amoeboid cercozoans detected

in Grande Anse sediments—paulinellids (Gabr *et al.* 2020; Macorano & Nowack 2021; Nowack & Grossman 2012) and chlorarachniophytes (Ishida *et al.* 1996; Keeling 2017)—hold importance for understanding the evolutionary history of photosynthetic organelles (Reyes-Prieto *et al.* 2007). Rare microalgal lineages at Grande Anse showed greater ASV richness than would be expected from their low relative abundances, contributing disproportionately to microalgal diversity. For instance, non-diatom ochrophytes were approximately 18 times more relatively diverse than abundant. Additionally, the presence of high-occurring but low-abundance microalgal genera at Grande Anse is peculiar. The ecological roles of rare microalgae in Grande Anse remain speculative. They may enhance complexity and functional redundancy in the benthic environment, contributing to ecosystem resilience against environmental disturbances. Alternatively, their presence may reflect successive inadvertent depositions of dormant phytoplankton onto sediments. Regardless, the general scientific consensus appears to be that rare microbial species collectively serve as a rich reservoir of genetic and metabolic potential (Jousset *et al.* 2017).

4.4 Potential mechanisms governing spatio-temporal associations between benthic environmental conditions and microalgal assemblages

4.4.1 Temporal perspective

Spatial and temporal variability in microalgal assemblages at Grande Anse was evident across multiple scales. Although species composition generally varied over the summer, these intra-seasonal patterns were not consistent over several years, potentially reflecting interannual variations in underlying biotic and environmental drivers. Additionally, interannual variability in microalgal species composition occurred irrespective of the

summer sub-season, suggesting that drivers operating at the interannual scale may have a greater influence over microalgal dynamics than those at the intra-seasonal scale. A six-year study in intertidal sites in the Netherlands and the United Kingdom found that interannual variation in microphytobenthos biomass—estimated using the Normalized Difference Vegetation Index (NDVI)—was likely driven by wind-wave effects on photoperiod and sediment resuspension, as well as by air temperature (van der Wal *et al.* 2010). Additionally, de Jonge et al. (2012) found that mean annual air temperature in the Ems estuary (the Netherlands)—which ranged from 7.5 °C to 10.2 °C across eight non-consecutive years between 1976 and 1999—explained nearly 60% of the interannual variation in estuary-wide mean annual microphytobenthos chlorophyll *a* content, which varied by about 2.5-fold during that time. As such, while not directly examined in my thesis, interannual air temperature variability in the Bay of Fundy region may have influenced corresponding variability in benthic microalgal assemblages.

Whereas diatom lineages exhibited both spatial and temporal variability in their relative abundances, green algal relative abundance showed greater variability across years than across depths at Grande Anse. I found that green algal relative abundance in surficial sediments decreased by about 90% from 2019 to 2020, while diatom relative abundance increased by 53%, specifically in mid and late summer. During roughly the same period, Mogle (2021) observed that the lipid content of microalgal biofilms—important for shorebird nutrition—in the top 2 mm of Grande Anse sediments decreased by at least 50%, coinciding with reported hotter and drier conditions at the mudflat in 2020 compared to 2019. The co-occurrence of declining green algae—dominated by the thermotolerant genus *Picochlorum* (Barten *et al.* 2022)—and reduced lipid content is

notable. However, its interpretation is complicated by the physiological characteristics of different microalgal groups. Diatoms, dinoflagellates, and green algae—but generally not cyanobacteria—include species capable of substantial lipid accumulation (Morales *et al.* 2021) and of forming resilient, long-lived resting stages (Ellegaard & Ribeiro 2018). Additionally, lipid content is typically higher in diatom resting stages than vegetative cells (Doucette & Fryxell 1985), and often increases in response to environmental stress, with lipids serving as an energy reserve (Zulu *et al.* 2018). Furthermore, resting stage formation is more widespread among centric than pennate diatoms (McQuoid & Hobson 1996). As such, while diatom relative abundance increased overall at Grande Anse from 2019 to 2020, only a 7% increase was observed in the relative abundance of polar centric diatoms, which are more likely to form lipid-rich resting stages. Conversely, the relative abundance of raphid pennate diatoms increased by 22%, and araphid pennate diatoms increased more than ninefold. These patterns highlight the complex interplay between intrinsic and extrinsic factors likely shaping microalgal interannual dynamics.

In 2019, during both the spring–summer and summer–fall transitions—each marked by similarly sized shifts in microalgal species composition (50–51% dissimilarity)—I observed significant transitions from polar centric to raphid pennate diatoms. This pattern suggests that raphid pennate diatoms are better adapted to the heightened environmental heterogeneity characteristic of these seasonal transitions. However, given that intra-seasonal summer patterns in microalgal species composition varied between years, it remains unclear whether these inter-seasonal patterns are unique to 2019 or part of a broader, recurring trend. In the context of the spring–summer transition, insights from the macro-tidal Brouage mudflat (French Atlantic coast) suggest

that thermo-inhibition plays an important role in shaping microalgal dynamics, alongside modest contributions from local biotic factors (Savelli *et al.* 2018). I observed a steep decline in dinoflagellate relative abundance from spring to summer at Grande Anse, followed by a sharp increase from summer to fall. This pattern may reflect offshore blooms during spring and fall, as well as a preference for cooler temperatures that promote dinoflagellate cyst germination—a common precursor to bloom events (Anderson *et al.* 2012; Carstensen *et al.* 2015). In contrast to diatoms and dinoflagellates, green algal relative abundance declined during both seasonal transitions but rebounded in the summer between transitions, suggesting that they are most competitive under summer conditions.

4.4.2 Physico-chemical perspective

Sediment temperature increases from 16.6 °C to 23.4 °C during the summer at Grande Anse were positively correlated with microalgal ASV richness and accounted for 44% of the variation, highlighting the potentially important role of intra-seasonal temperature variability in shaping microalgal dynamics, alongside the well-studied inter-seasonal trends. Scholz & Liebezeit (2012a) examined the species-specific growth rates of 25 benthic diatoms from the Solthörn tidal flat (southern North Sea, Germany). They concluded that variability in temperature within typically observed ranges (i.e., 10 °C to 30 °C) had minimal impacts on their growth, likely due to their high adaptability; however, extreme temperatures (i.e., 4 °C and 40 °C) had a significant effect. Additionally, Blanchard et al. (1997) found that 25 °C was the optimum temperature, while 38 °C was the maximum temperature for photosynthesis, for epipelic diatoms from an intertidal mudflat in the Marennes-Oleron Bay (French Atlantic coast).

Furthermore, Cartaxana et al. (2015) found that elevated temperatures reduced the biomass and photosynthetic performance of intertidal microphytobenthos, as well as shifted their species composition from microalgae to cyanobacteria, but also from motile epipelic *Navicula* and *Gyrosigma* diatoms, and partially towards reportedly tychoplanktonic representatives of *Minidiscus* and *Thalassiosira* diatoms. Jacobs et al. (2021) concluded that uncharacteristically high air temperatures in northwestern Europe during 2018 and 2019—with a focus on the Ems-Dollard estuary—which often exceeded optimum levels for photosynthesis, explained summer declines in microphytobenthos biomass better than grazing, highlighting the dominant role thermo-inhibition can play in similar systems. Also, Thaler & Kaczmarska (2009) found that *Gyrosigma orbitum*—which they discovered in salt marsh and tidal flat sediments at Pecks Cove (upper Bay of Fundy, Canada's Atlantic coast)—was associated with high sediment temperature and water content, as well as with the shallow anoxic sediment layer, thus providing insights into G. *limosum* at the Grande Anse mudflat. The findings of Vieira et al. (2013) specify that the photosynthetic capacity of intertidal microphytobenthos in a north temperate estuary increased with temperature up to 35 °C, with efficiency decreasing at 42 °C. Collectively, these studies help contextualize the patterns observed at Grande Anse. Specifically, sediment temperatures fell within typical moderate ranges, and microalgal ASV richness increased as temperatures approached optimal conditions.

Environmental associations with microalgal metrics in Grande Anse surficial sediments from early summer to early fall suggest that sediment pH and temperature play a greater role than oxygen and water content in shaping microalgal dynamics at

these spatio-temporal scales. Specifically, redox potential and sediment penetrability—proxies for oxygen and water content—were not significantly associated with variability in microalgal assemblages at Grande Anse. This is likely due to the spatial and temporal scales examined in my thesis, within which these environmental variables do not exhibit strong and/or consistent patterns. Specific redox regimes are essential for archaeal- and bacterial-mediated elemental cycling (Falkowski *et al.* 2008), which in turn supports microalgal growth. However, although steep redox gradients typically occur across a sediment depth profile, variability between the top 2–5 mm and 1 cm depth layers was not consistent across the mudflat and over time. Additionally, while substantial variation in sediment water content occurs between different mudflats in the Bay of Fundy (Gerwing *et al.* 2015a), this is not expected within a single mudflat.

Within a narrow 0.6-point range near neutral pH, sediment pH was positively correlated with microalgal ASV richness at Grande Anse mudflat, and significantly correlated with other microalgal metrics, suggesting that even relatively small pH fluctuations may influence benthic microalgal assemblages, albeit modestly. This aligns with the findings of Hinga (2002), specifically that small deviations within and around the natural pH baseline can significantly impact phytoplankton growth in coastal waters. However, Taraldsvik & Myklestad (2000) found that the centric diatom *Skeletonema costatum* maintained nearly stable growth when cultivated within the pH range of 6.5 to 8.5, with a notable decline only observed at pH 9. Additionally, Lundholm et al. (2004) observed that the growth of several species of the raphid pennate diatom genera *Pseudo-nitzschia* and *Nitzschia* ceased at pH 8.7–9.1. Sediment pH at Grande Anse appeared more spatially variable across the mudflat surface in mid and late summer—coinciding

with the observed infaunal and epifaunal—than in early summer and early fall. One interpretation of this pattern is that patchily distributed mudflat fauna influence microalgal assemblages not only through direct grazing, but also indirectly, by modifying the benthic environment at a hyper-local scale. Accordingly, observed associations between pH and microalgal metrics at Grande Anse may be confounded by concurrent biological activities.

At Grande Anse, I found that pH was positively correlated with certain diatoms but negatively correlated with non-diatom microalgae; however, for temperature, correlations with microalgal species and genera often differed within lineages. These findings suggest that, within the observed ranges of these environmental variables, pH may act as a broad environmental filter, discriminating between major microalgal lineages. Conversely, temperature may drive finer-scale species sorting within microalgal lineages across a range of species-specific thermal optima. This aligns with Liu et al. (2024), who found that within the *Thalassiosira* diatom population of Jiaozhou Bay (Yellow Sea, China), different species exhibited distinct seasonal patterns associated with the temperature seasonality of the coastal waters. My findings also align with Lundholm et al. (2004), who concluded that pH tolerance in marine diatoms may be size-dependent, potentially indicating that pH effects may vary based on broad microalgal traits.

4.5 Nuances in the top-down and bottom-up regulation of mudflat microalgal assemblages

4.5.1 Microalgal biomass

DNA metabarcoding provided taxonomic resolution to variability in microalgal biomass, as estimated by chlorophyll *a* content, at the Grande Anse mudflat. Chlorophyll *a* content was weakly correlated with microalgal ASV richness and species composition, but not significantly with microalgal relative abundance. This pattern is consistent with findings from a survey of 366 freshwater lakes across Canada, spanning various trophic states, where chlorophyll *a* concentration in surface waters was significantly correlated with phytoplankton composition (Garner *et al.* 2022). At Grande Anse, the observed correlations suggest that shifts in the robustness of the microalgal species pool may contribute to, or be indicative of, corresponding variation in the standing crop available to benthic consumers. A similar argument could be made regarding taxonomic composition. Both chlorophyll *a* content and microalgal ASV richness trended toward increased magnitude in response to nutrient enrichment, albeit to differing degrees and conditional on other factors. Similarly, chlorophyll *a* content and microalgal species composition both exhibited pronounced monthly variability during the summer, further pointing to shared, though not identical, structuring influences.

Benthic microalgal assemblages at the Grande Anse mudflat exhibited larger variability between subplots than between plots, after controlling for biotic and nutrient-related influences. Spatial averaging at broader scales likely dampens finer-scale stochastic noise. This pattern suggests that microalgal spatial variability is more strongly shaped by variables operating at the scale of metres than by broader-scale influences spanning tens of metres. Presence–absence patterns of microalgal ASVs—which offer finer resolutions than the species composition metric—exhibited larger spatial variability than other microalgal metrics, likely reflecting the peak influence of stochastic forces.

Findings from the Grande Anse mudflat suggest that variability in chlorophyll *a* content is more closely associated with nutrient-related and temporal variables than biotic ones. Given the breadth of biotic variables considered—20 in total—and the fact that environmental variables, beyond nutrient enrichment, were not analyzed alongside them, this temporal signal likely reflects the influence of intra-seasonal shifts in sediment environmental conditions during the summer. These results point to environmental heterogeneity as a primary driver of microalgal biomass, while biotic interactions appear more influential in shaping species composition. Notably, epifauna, infauna, heterotrophic microbial eukaryotes, and bacteria each accounted for 11–15% of the variability in microalgal ASV richness, suggesting an intricate web of biotic influences collectively shaping over half of the observed variability in benthic microalgal diversity.

4.5.2 Nutrient enrichment

Nutrient enrichment is often expected to favour fast-growing microalgal species capable of rapidly exploiting available resources (Conley *et al.* 2009). Nutrients, specifically nitrate and phosphate, were also reported to be more influential in shaping the abundance of centric diatoms than that of pennate diatoms in a tropical intertidal sand flat (Mitbavkar & Anil 2002). However, specific responses to nutrient enrichment can be dependent on local environmental conditions (Rabalais *et al.* 2009). Broader comparisons underscore the variability of responses. Scholz & Liebezeit (2012b), for instance, describe a nutrient regime at the Solthörn tidal flat (southern North Sea, Germany) shaped by variability in silicate and ammonium content, nitrogen limitation in the summer, and phosphorus limitation in other seasons. At this tidal flat, the nutrient

regime exerted influence over the composition of benthic diatoms in surface sediments. Garner et al. (2022) surveyed Canadian lakes spanning latitudinal climatic gradients and trophic states and found that microalgal assemblages in high-nutrient lakes had lower diversity, lower turnover, and higher compositional similarity compared to low-nutrient systems.

At the Grande Anse mudflat, nutrient enrichment significantly increased chlorophyll *a* content as expected; however, although microalgal ASV richness trended toward increased magnitude, the change was not statistically significant. Notably, this effect was modulated by adult mud snail density, shorebird activity, and the duration of enrichment, highlighting how the local ecological context and perturbation regime shape microalgal responses. Furthermore, the delayed 42% increase in chlorophyll *a*, which appeared between four and eight weeks after fertilizer application, indicates a degree of resilience in benthic microalgal assemblages. Microalgal assemblages may be capable of buffering short-term perturbations, albeit within limits. The presence of a microalgal response to nutrient enrichment raises questions about whether the Grande Anse mudflat is nutrient-limited. For context, in nutrient-replete sediments—where microbial activity is not constrained by nutrient availability—additional nutrient inputs would not typically be expected to stimulate further microbial growth. Future research should employ experimental manipulations that vary nutrient type, quantity, and enrichment frequency, as well as organismal density, to elucidate mechanisms underpinning the causal relationships observed.

While nutrient enrichment at Grande Anse increased microalgal biomass within 8 weeks, it did not significantly alter species composition, suggesting that species

composition is more resilient to nutrient fluctuations than biomass. This may be due to a uniform stimulation of microalgal species or to compensatory mechanisms regulating microalgal assemblages. My findings differ from past reports of species-specific responses. For instance, Underwood et al. (1998) noted species-specific responses of estuarine epipelic diatoms to ammonium enrichment, even without changes in the overall biomass. Underwood & Provot (2000) reported on the differential tolerances, sensitivities, and preferences of certain epipelic diatoms to a range of ammonium (10–4000 μM) and nitrate (10–2000 μM) concentrations, across a salinity gradient (10–35 ‰), in axenic cultures. Du et al. (2017) observed species-specific associations between microphytobenthic species and a range of environmental factors in intertidal flats in the La Rochelle (France) coastal area, including a positive correlation between cell densities of the raphid pennate diatom *Gyrosigma limosum* and sediment concentrations of heavy metals. In their study, Hagerthey et al. (2002) attributed the lack of an observable nutrient response by most intertidal benthic diatoms, in the absence of grazers, to the broad adaptations of the microalgal assemblages to environmental heterogeneity. Although this thesis employed DNA metabarcoding, future studies could incorporate metatranscriptomics to investigate the metabolic processes underpinning the compositional resilience of microalgal assemblages to nutrient enrichment.

4.5.3 Epifaunal interactions

Shorebird activity and adult mud snail density had significant effects on all examined microalgal metrics and were also significantly correlated with most of the dominant microalgal genera. In addition, juvenile mud snails, adult *Corophium* amphipods, ostracods, protistan parasites, and sulfur-oxidizing bacteria were significantly associated

with most metrics. These patterns underscore the breadth of biotic interactions shaping benthic microalgal assemblages, spanning organismal size classes and functional groups. Among these, epifauna—particularly adult mud snails—emerged as key structuring agents within the mudflat. Significant associations with microalgal species composition suggest that epifauna, along with *Corophium* amphipods, *Macoma* clams, certain polychaetes, copepods, and ostracods, may engage in selective grazing on benthic microalgae or other taxon-specific interactions. This contrasts with the seemingly broad-scale effect of nutrient enrichment.

Benthic microalgae constitute an important component of the diet of shorebirds during their migratory stopover at mudflats. Gerwing et al. (2016) used DNA metabarcoding of fecal samples to reconstruct the diets of Semipalmated Sandpipers in the upper Bay of Fundy and found that diatoms were present in all samples, far more than any other prey item, including *Corophium* amphipods, the next most common. Microalgae, detritus, and other components of sediment biofilms can comprise about half of Semipalmated Sandpiper diets in the Chignecto Bay (upper Bay of Fundy, Canada) (Quinn & Hamilton 2012), with larger estimates for Western Sandpipers in the Fraser River estuary (British Columbia, Canada) (Kuwae *et al.* 2008; Mathot *et al.* 2010). Additionally, Booty et al. (2020) found that shorebirds can impact sediment-water nutrient flux, including by decreasing sediment nitrate uptake. They concluded that this process may contribute to decreases in microphytobenthos biomass that can be attributed to the presence of shorebirds.

At the Grande Anse mudflat, shorebird activity was significantly associated with increased chlorophyll *a* content and microalgal ASV richness. Although shorebirds are

expected to decrease microalgal biomass through direct predation, they may also indirectly cause it to increase by preying on invertebrate grazers of benthic microalgae (Daborn *et al.* 1993). Additionally, physical disturbance of sediments from shorebird activity, including pecking, may stimulate nutrient cycling and increase nutrient availability for microalgal growth (Middelburg 2018). Furthermore, shorebird faeces extracts have been found to increase the pigment content and growth rate of diatom cells in culture (Jauffrais *et al.* 2015). Hicklin & Smith (1979) found that *Corophium amphipods* were among the top invertebrate prey by volume in the digestive tracts of Semipalmated Sandpipers and several other migratory shorebirds in the Minas Basin (upper Bay of Fundy, Canada). Additionally, Hamilton et al. (2006) reported that shorebird predation decreased *Corophium* density in a Bay of Fundy mudflat by about 80%. However, past manipulative experiments in upper Bay of Fundy mudflats, including Grande Anse, have yielded no evidence that the cumulative negative effects of shorebird and mud snail presence on *Corophium* abundance cascade down the mudflat food web into increased microalgal biomass at the sediment surface (Cheverie *et al.* 2014; Hamilton *et al.* 2006). Instead, compensatory interactions have been observed.

Shorebird activity was positively correlated with the relative abundances of several raphid pennate diatom genera but negatively correlated with other microalgae at Grande Anse. This suggests that shorebirds consume raphid pennate diatoms less than other microalgae, stimulate their growth more, or both. Scholz & Liebezeit (2013) found that among 25 pennate diatom species in the Solthörn tidal flat (southern North Sea, Germany), most had temperature-dependent lipid contents at 19.4% to 25.6% of dry weight, except some species of the genera *Amphora*, *Gyrosigma*, and *Pleurosigma*,

which had lipid compositions of over 30%. This contrasts with the centric diatoms *Thalassiosira weissflogii* and *Chaetoceros muelleri*, which had lipid contents of 26.9% and 22.2%, respectively (Lin *et al.* 2018). Thus, if shorebird predation at Grande Anse were highly sensitive to species-specific lipid content, the observed correlation patterns might have been reversed for raphid pennate and polar centric diatoms. It is also worth noting that microalgal resting stages typically have higher lipid content than vegetative cells (Doucette & Fryxell 1985), and that resting stages at Grande Anse are more likely to consist of polar centric than raphid pennate diatoms (McQuoid & Hobson 1996). Similarly, resting stages are more likely to comprise planktonic-origin *Alexandrium* dinoflagellates and *Picochlorum* green algae than benthic-origin raphid pennate diatoms. The motility of raphid pennate diatoms may also have contributed to their relative success, as previously discussed.

The effect of mud snails on microalgal assemblages at Grande Anse was likely driven by a combination of density-dependent grazing on microalgal prey, interference competition and predation on other microalgal grazers, and bioturbation-enhanced nutrient cycling (Hamilton *et al.* 2006). The findings of Connor et al. (1982) suggest that at low snail densities, snail activity tends to increase microalgal abundance because bioturbation stimulates growth more than grazing reduces biomass. Consistent with this, Cheverie et al. (2014) found that medium (~117 individuals/m^2) and high (~213 individuals/m^2), but not low (~45 individuals/m^2), snail densities resulted in decreased chlorophyll *a* content at the Grande Anse mudflat. The findings of Blanchard et al. (2000) add further nuance to this density-dependent interaction. They observed that average ingestion rates of the mud snail *Hydrobia ulvae* on epipelic diatoms decline as

the mud snail density increases in microcosm experiments. However, this reduction may reflect population saturation effects, and the observation may not apply to Grande Anse, where mud snail densities remain well below the peaks documented in other upper Bay of Fundy mudflats (Gerwing *et al.* 2015a).

I found that juvenile mud snails, with a peak density of about 50 individuals per m^2 at Grande Anse, were associated with less variability across fewer microalgal genera compared to the adults, which peaked at around 15 individuals per m^2. Notably, juveniles were negatively correlated with chlorophyll *a* content, whereas adults exhibited a positive association. These findings likely reflect the influence of organismal size on bioturbation-enhanced nutrient cycling and grazer density on prey abundance. Specifically, these findings suggest that juveniles may exert stronger top-down pressures on microalgal biomass than adults due to their higher density, while contributing less to bottom-up effects due to their smaller size.

Cheverie et al. (2014) demonstrated that even at low densities, mud snails can negatively impact the abundance of *Corophium* amphipods, top infaunal grazers of microalgae. This suggests that the positive correlation I found between adult mud snails and chlorophyll *a* content at Grande Anse may partly result from a trophic cascade, where adult snails reduce *Corophium* populations through predation or competition. While mud snails may selectively feed on larger, migratory microalgal cells (Connor *et al.* 1982), I found that both juvenile and adult mud snails at Grande Anse were positively correlated with the motile raphid pennate diatoms but negatively correlated with other microalgae. This suggests that selective grazing by mud snails may vary with ecological context.

4.5.4 Macro-infaunal interactions

Stable isotope evidence from intertidal flats in Korea's southern coastal bays indicates that benthic microalgae constitute a dominant source of organic matter for macrozoobenthic consumers (Kang *et al.* 2003). At Grande Anse, macro-infaunal and meiofaunal taxa accounted for similar proportions of microalgal variability, suggesting that they play comparably important roles in shaping benthic microalgal dynamics. This contrasts with the Molenplaat intertidal flat (Scheldt estuary, the Netherlands), where macro-infauna exerted stronger grazing pressure on primary producers than meiofauna (van Oevelen *et al.* 2006). There, macrobenthos (deposit and suspension feeders) accounted for 15% of total secondary production (excluding birds and fish) and derived 75% of their carbon from grazing on microphytobenthos and phytoplankton, consuming 18% of gross microbial primary production (van Oevelen *et al.* 2006). In contrast, meiobenthos (nematodes, copepods, ostracods, and foraminifera) obtained 68% of their carbon from microphytobenthos and none from phytoplankton but contributed only 2% to secondary production and consumed just 5.9% of available microphytobenthic carbon (van Oevelen *et al.* 2006). At Grande Anse, adult *Corophium* amphipods, *Macoma* clams, and Nereididae (Polychaeta) were negatively correlated with chlorophyll *a* content but not significantly associated with microalgal relative abundance, implying that the ratio of autotrophic to heterotrophic microbial eukaryotic abundance, based on amplicon sequence reads, remained unchanged. Together, these patterns suggest that while infauna likely reduce microalgal biomass through grazing, they may not do so preferentially, instead consuming a broad range of microbial eukaryotic prey in sediments.

At Grande Anse, *Macoma* clams exhibited a weak negative correlation with chlorophyll *a* content and accounted for just 1.2% of the observed variation. Based on $\delta^{13}C$ mixing models, Kanaya et al. (2008) found that benthic diatoms comprise 59–100% of the diet of the surface deposit feeder *Macoma contabulata* across stations in a Japanese estuarine mudflat (Sendai Bay). Similarly, Daggers et al. (2020b) used $\delta^{13}C$ and $\delta^{15}N$ isotopic signatures to estimate that microphytobenthos contributes up to 60% of the diet of *Macoma balthica* in muddy sediments, and up to 90% in sandy sediments of tidal flats in the Westerschelde estuary and Oosterschelde tidal basin (the Netherlands). In a complementary study, stable isotope data from the Dutch Wadden Sea showed that benthic primary production contributes anywhere from <5% to >95% of the dietary carbon of the most commonly occurring clam species, with a similarly broad range reported for other bivalve mollusks (Christianen *et al.* 2017). Together, these studies highlight the potential for substantial dietary plasticity within and among bivalve species, as well as across sediment environments—patterns that may also characterize populations at the Grande Anse mudflat and broadly across the Bay of Fundy.

Adult *Corophium* amphipods, which were negatively correlated with both chlorophyll *a* content and microalgal ASV richness at Grande Anse, accounted for only 1.1% and 3.1% of the respective variations—an unexpectedly small amount given their known grazing capacity. Amphipod abundance is often inversely correlated with microalgal abundance. Trites et al. (2005) found that the cell density of live diatoms was significantly higher in sediments at Grande Anse (Buck's Flat) than at Daniel's Flat, another Bay of Fundy mudflat with higher *Corophium* abundances. Similarly, Aikins & Kikuchi (2002) found an inverse seasonal pattern in Gamo Lagoon (Japan), with high

amphipod and low microalgal densities characterizing warmer months (June to September), and the opposite pattern evident in colder months (December to March). While experimental evidence generally supports these patterns, the dependence of *Corophium* on microalgae as a food source can differ markedly across habitats. Laboratory feeding experiments by Gerdol & Hughes (1994b) found that *Corophium volutator* individuals can consume the equivalent of 4000 nano-sized centric diatoms per hour, with Smith et al. (1996) estimating daily consumption rates at between 2150 and 3767 diatom cells in estuarine sediments in Essex, United Kingdom. Reflecting this, diatom densities were 76% to 142% higher in insecticide-treated areas of two *Corophium*-dominated intertidal mudflats (southeastern England) due to reduced grazing (Gerdol & Hughes 1994a). Notably, stable isotope data from Dutch intertidal flats suggest that benthic primary production may contribute as little as <5% to the diet of *Corophium* in the Wadden Sea (Christianen *et al.* 2017), but as much as 52% in muddy sediments of the Westerschelde estuary and 90% in sandy sediments of Oosterschelde tidal basin (Daggers *et al.* 2020b).

Aikins & Kikuchi (2002) found that amphipods can have a feeding preference for both centric and pennate diatom species. They further reported that among pennate diatoms, tube-dwelling amphipods like *Corophium* species showed a preference for *Nitzschia* (4.1–34 µm long) (Trobajo *et al.* 2013) over *Gyrosigma* (45–175 µm long) (Sterrenburg 1994) species, suggesting size-based selection. Laboratory experiments by Hagerthey et al. (2002) similarly showed that *Corophium volutator*, collected from an intertidal mudflat in the Westerschelde estuary (the Netherlands), preferentially fed on the larger, more dominant epipelic *Navicula* species (4.2–21 µm long) (Wetzel *et al.*

2015) than smaller epipsammic diatoms. At the Grande Anse mudflat, correlation patterns with microalgal genera are consistent with adult *Corophium* amphipods consuming raphid pennate diatoms less than polar centric diatoms and other microalgae—a pattern also observed for epifauna. *Thalassiosira* species, the dominant polar centric diatoms at Grande Anse, span a wide range of cell diameters, from 2.5 μm for T. *spinulata* to 120 μm for T. *thailandica* (Hoppenrath *et al.* 2007; Park *et al.* 2016). Due to the prevalence of taxonomically unresolved *Thalassiosira* reads in the Grande Anse sequence data, the likely size distribution of polar centric diatoms in the mudflat remains unclear. This ambiguity makes it difficult to determine whether my findings support size-selective grazing by *Corophium* and whether cell size contributed to the observed correlation patterns. Clarifying this relationship could be a focus of future research. Building on this, juvenile *Corophium* were significantly correlated with a different, albeit smaller, pool of microalgal genera than adult *Corophium* at Grande Anse, hinting at ontogenetic shifts in dietary preferences and biological interactions. This also suggests that, like juvenile mud snails, juvenile *Corophium* may exert less influence over microalgal dynamics than their adult counterparts.

At Grande Anse, Nereididae was the only polychaete family negatively correlated with chlorophyll *a* content, consistent with previous research supporting a strong trophic link between the microphytobenthos and this errant polychaete family. For instance, stable isotope analysis by Herman et al. (2000) identified *Nereis* spp. as among the macrofauna most dependent on microphytobenthos for nutrition in a Westerschelde estuary tidal flat (the Netherlands). Similarly, Smith et al. (1996) estimated that individual *Nereis diversicolor* consume between 5476 and 12184 diatom

cells daily in an Essex (UK) estuary, up to 3.2 times more than *Corophium volutator*. Further supporting this dependence, natural abundance $\delta^{15}N$ and $\delta^{13}C$ data combined with a ^{15}N-enrichment experiment in a Plum Island estuary (Massachusetts, USA) marsh mudflat showed that the microphytobenthos comprised over 84% of the diet of *Nereis diversicolor* (Galván *et al.* 2008). Comparable dietary contribution estimates, ranging from 67% to 92%, have also been reported for *Nereis diversicolor* in Dutch tidal flats with varying mud content (Daggers *et al.* 2020b).

Nereididae were, on average, more dense than adult *Corophium* amphipods at the Grande Anse mudflat during summer 2019. Given this and the findings of Smith et al. (1996), it is reasonable to expect that Nereididae exert stronger predatory pressure and regulatory influence on microalgal biomass than *Corophium*. However, my findings provide only marginal support for this expectation. Specifically, at Grande Anse, Nereididae accounted for 2.4% of the variation in chlorophyll *a* content, compared to 1.1% for adult *Corophium*. Additionally, the negative correlation with chlorophyll *a* content was slightly stronger for Nereididae than adult *Corophium* ($\rho = -0.48$ vs. -0.41).

In contrast to the errant Nereididae, the density of the sessile polychaete family Capitellidae at Grande Anse was positively correlated with chlorophyll *a* content. This pattern is somewhat unexpected given findings from an estuarine mudflat in Idoura Lagoon (Sendai Bay, Japan), where benthic diatoms accounted for an estimated 28–50% of the diets of the deep-deposit-feeding capitellids *Heteromastus* and *Notomastus*, as inferred from $\delta^{13}C$ isotopic data (Kanaya *et al.* 2008). Nevertheless, variation in polychaete–microalgal relationships across polychaete families and lifestyles (sessile versus errant) is not unusual and have been observed in other systems. For instance,

stable carbon isotope data from Dutch Wadden Sea intertidal flats showed that benthic primary production contributed 63% to over 95% of the diets of the most common errant polychaetes (including nereidids), compared to less than 5% to 84% for sessile taxa, with capitellids at the lower end on the range (Christianen *et al.* 2017). None of the polychaete families at Grande Anse were significantly associated with variability in microalgal ASV richness, suggesting that their effects, when present, may be limited to the most abundant and accessible microalgal taxa. Notably, Spionidae—a sessile, deposit-feeding lineage and the least abundant of the polychaete families—showed no significant association with any microalgal metric. This suggests that not all benthic organisms exert a measurable influence on microalgal dynamics and further highlights the potential importance of density-dependent interactions.

4.5.5 Meiofaunal interactions

Moens et al. (2002) identified microalgae as a core dietary component for nematodes across tidal flats and a salt marsh in the Schelde estuary (the Netherlands), based on $\delta^{13}C$ signatures. Similarly, Riera et al. (1996) found that in an intertidal mudflat at Marennes-Oléron Bay (France), the carbon and nitrogen isotopic ratios of microphytobenthos most closely matched the theorized dietary signature of nematodes. Supporting this, estimates from the Molenplaat intertidal flat (Scheldt estuary, the Netherlands) indicate that nematodes obtained 49% of their carbon from microphytobenthos (van Oevelen *et al.* 2006). However, Pinckney et al. (2003) found that benthic microalgal production in a subtropical intertidal mudflat (Terrebonne Bay estuary, Louisiana, USA) was not limited by meiofaunal grazing, even during seasonal peaks in grazer activity. At Grande Anse, nematode relative abundance, inferred from sequence reads, was positively correlated

with chlorophyll *a* content, albeit weakly. This pattern is unexpected, as consumers typically exhibit negative co-occurrences with their prey (Boyse *et al.* 2025). However, microcosm experiments have shown that nematodes can stimulate microalgal growth (D'Hondt *et al.* 2018; Hubas *et al.* 2010).

The dietary plasticity of nematodes is key to their success across ecosystems (Schratzberger *et al.* 2019), enabling them to shift their diets in response to increased organic inputs. The findings of Moens et al. (2002) reinforce this conclusion. Nematode diets broadened across sediment types, ranging from primarily microalgae in sandy sediments to the inclusion of particulate organic matter in silty sediments, and additionally, cordgrass detritus in marsh environments (Moens *et al.* 2002). Moens et al. (2002) also documented high assimilation of pre-labelled algal detrital carbon by nematodes, indicating reliance on both live microalgae and detrital sources. Accordingly, nematodes at Grande Anse may instead rely on alternative nutritional sources such as bacterial prey, sedimentary detritus, and suspended particulate organic matter.

Pascal et al. (2008) found that in laboratory feeding experiments using nematodes from the Brouage mudflat (French Atlantic coast) and pre-labelled microbial prey, nematodes preferentially consumed the diatom *Navicula phyllepta* over bacteria. Additionally, at a bacterial abundance of 10.5×10^8 cells mL^{-1} and N. *phyllepta* abundance of 113.7 μg Chl a · g dry sed^{-1}, nematode assimilation of algal carbon was 11.6 times higher than that of bacterial carbon (Pascal *et al.* 2008). This disparity was not due to bacterial limitation but rather to the continued uptake of algal carbon with increasing availability, whereas bacterial carbon uptake plateaued (Pascal *et al.* 2008).

Accordingly, complete bacterial replacement of microalgal resources for nematodes at Grande Anse appears unlikely.

The positive correlation between nematode relative abundance and chlorophyll *a* content may indicate shared structuring forces—such as favourable microhabitat conditions—or facilitation through unidentified mechanisms. It may also reflect resource tracking, whereby nematode abundance increases with microalgal availability, but grazing remains insufficient to produce a measurable decline in chlorophyll *a* content. In line with this, Montagna et al. (1995) found that although nematode grazing rates can increase with microphytobenthos photosynthetic activity, this does not necessarily translate into population growth for nematode species. The dietary plasticity of nematodes likely further weakens their top-down control of microalgae. Nematode dynamics at Grande Anse may reflect broader patterns among mudflat consumers that, while deriving nutrition from microalgae, do not strongly regulate microalgal dynamics.

At Grande Anse, neither ostracods nor copepods—both motile grazers—were significantly associated with chlorophyll *a* content; however, both taxa showed significant associations with microalgal species composition. Additionally, ostracods were negatively correlated with microalgal ASV richness, albeit weakly. These findings align with past research. Montagna et al. (1995) found that ostracod grazing rates did not change with increased microphytobenthos photosynthetic rate. Azovsky et al. (2005) documented selective feeding by copepods on two pennate diatoms from the genera *Nitzschia* and *Navicula*, which comprised up to 97% of their gut content but only 10% of the ambient diatom biomass in an intertidal sandflat in the White Sea (Russia). De Troch et al. (2006), using a microcosm experiment, found that copepod grazing efficiency on

diatoms was significantly reduced in silty sediments compared to coarser substrates. Additionally, benthic copepods likely also feed on bacteria and flagellated protists (Azovsky *et al.* 2005). While neither ostracods nor copepods likely exert substantial control over microalgal assemblages, the outsized contribution of ostracods to variations in microalgal relative abundance and species composition is notable.

4.5.6 Microbial interactions

The relative abundance of protistan consumers at Grande Anse was not significantly associated with variation in chlorophyll *a* content or microalgal species composition. This was unexpected, given the amplicon sequence data from the top 7 cm of sediments at two Bay of Fundy mudflats, including Grande Anse, indicated that 43% of protistan consumers were eukaryvores and an additional 14% fed on both eukaryotes and bacteria (Kalu *et al.* 2023). Furthermore, in the Molenplaat intertidal flat (Scheldt estuary, the Netherlands), van Oevelen et al. (2006) found that microbenthos (flagellates and ciliates) had a carbon demand 78% higher than macrobenthos and over 13-fold higher than meiobenthos. Roughly 47% of that demand was met by extracellular polymeric substances and detritus from photosynthetic microorganisms, with negligible direct grazing on live algal cells. Additionally, Epstein et al. (1992) examined ciliates—a major group of protistan consumers—from a temperate sandy tidal flat (White Sea, Russia) and found, through light microscopy of digestive vacuoles, that they grazed 93% of dinoflagellate production but only 6% of diatom production, selectively targeting just 4 of 42 diatom species. However, the absence of a significant association with chlorophyll *a* content may be explained by the prevalence of indirect trophic linkages between protistan consumers and microalgae, mediated through detrital and dissolved organic

carbon pathways, as described by van Oevelen et al. (2006). Such indirect pathways may weaken direct consumer-resource coupling, obscuring taxon-specific associations between protistan consumers and microalgae like the one described in Epstein et al. (1992). Interestingly, protistan consumers were also negatively correlated with microalgal ASV richness ($\rho = -0.54$), as expected, and notably accounted for 12% of the observed variation. This pattern may reflect weak, diffuse, and non-selective grazing that disproportionately reduces rare microalgae across taxonomic groups without significantly impacting overall biomass.

Many marine fungi are known to break down complex biopolymers and form symbiotic relationships (Richards *et al.* 2012). Given these general ecological roles, I expected fungal relative abundance to be positively correlated with chlorophyll *a* content at Grande Anse. Such a pattern could reflect increased detrital processing triggered by the accumulation of planktonic phytodetritus, which may indirectly support microalgal growth through nutrient recycling. It could also reflect the processing of non-algal detritus or direct symbiotic interactions. However, fungi were not significantly associated with variability in microalgal species composition at Grande Anse, suggesting non-selective interactions. Additionally, fungal relative abundance was negatively correlated with chlorophyll *a* content, possibly reflecting resource competition. These findings provide limited insights into the nuanced interactions between benthic fungi and microalgae. Instead, they highlight persistent questions about the ecology of marine fungi (Amend *et al.* 2019), including whether benthic fungal DNA metabarcoding detections primarily reflect active organisms or dormant spores (Hyde *et al.* 1998), a distinction critical to understanding fungal contributions to benthic microalgal dynamics.

Parasite and host populations are often tightly coupled, exerting reciprocal effects on each other's dynamics, as demonstrated by Syndiniales–Dinophyceae associations in a eutrophic Chinese bay (Yan *et al.* 2023). Protistan parasites at Grande Anse were positively correlated with chlorophyll *a* content, suggesting coupled parasite–host dynamics, and were also significantly associated with microalgal species composition, underscoring the potential selectivity of some of these interactions. Specifically, based on relative read abundances, protistan parasites were negatively correlated with raphid pennate diatoms but positively correlated with polar centric diatoms and other microalgae. This pattern may indicate a preference for polar centric diatoms and other planktonic hosts, with infections likely occurring in the water column prior to their sediment deposition.

In surface waters of the Skidaway River estuary (Georgia, USA), diatoms accounted for nearly a quarter of significant associations with Syndiniales parasites—most of which were negative—whereas Syndiniales associations with Dinophyceae were predominantly positive (Anderson & Harvey 2020). Co-occurrence network analyses further revealed that Syndiniales broadly associate with multiple potential host taxa, suggesting ecological flexibility rather than strict host specificity (Anderson & Harvey 2020). At Grande Anse, the pseudofungal *Pirsonia* clade was significantly correlated with more top-occurring microalgal genera than any other parasitic lineage. While the *Pirsonia* clade is documented to target diatoms (Adl *et al.* 2019; Prokina *et al.* 2024), correlation patterns at Grande Anse suggest a broader host range and potentially opportunistic infections across diverse microalgal taxa. Other notable parasitic lineages at Grande Anse include the cercozoan *Cryothecomonas aestivalis*, which has been

reported to parasitize the centric diatom *Guinardia delicatula* in a temperature-dependent manner, ultimately killing the host in a parasitoid-like fashion (Catlett *et al.* 2023; Drebes *et al.* 1996; Peacock *et al.* 2014), and Syndiniales Group I Clade 1.

Reports from the Molenplaat intertidal flat (Scheldt estuary, the Netherlands) suggest that bacteria are key mediators of carbon flow, constituting 74% of heterotrophic biomass generation, albeit functioning more as organic sinks than trophic intermediaries due to low bacterivory (van Oevelen *et al.* 2006). At Grande Anse, bacterial nitrogen- and sulfur-cycling groups were significantly associated with variability in microalgal species composition. Sulfate-reducing bacteria (SRB) were significantly correlated with 71% of the top-occurring microalgal genera, highlighting their potential widespread influence. SRB were negatively correlated with chlorophyll *a* content. However, their positive correlations with raphid pennate diatom genera suggest that these diatoms are more resilient to potential negative interactions with SRB than other microalgae. Sulfur-oxidizing bacteria were negatively correlated with microalgal ASV richness, potentially reflecting resource competition.

Through food web reconstruction, van Oevelen et al. (2006) estimated that bacteria in the Molenplaat intertidal flat derived 57% of their carbon from deposited planktonic phytodetritus, with an additional 6% supplied by carbohydrate-rich extracellular polymeric substances (EPS) secreted by microphytobenthos. In turn, bacteria contributed to half of the respiration-driven inputs to the dissolved inorganic carbon (DIC) pool that microphytobenthos rely on for photosynthesis (van Oevelen *et al.* 2006), reinforcing the cycle of biogeochemical interdependence. Additionally, bacterial remineralization of organic matter releases inorganic nutrients—such as ammonium,

nitrate, phosphate, and sulfate—required by microalgae to synthesize essential biomolecules (Yao *et al.* 2019). In coastal sediments, this process is likely driven directly by organotrophic SRB and indirectly by hydrogenotrophic SRB engaged in syntrophic relationships with fermenters (Adam-Beyer *et al.* 2025; Crump & Bowen 2024; Dyksma *et al.* 2018; Plugge *et al.* 2011). As such, the strong linkages between microalgal genera and bacterial functional groups at the Grande Anse mudflat—particularly with abundant sulfate- and nitrate-reducing forms—were anticipated, though both the quantity and quality of available resources can be determinative in shaping interactions. This is highlighted by findings from the Molenplaat intertidal flat, where bacterial growth efficiencies on labile EPS were approximately twice as high as on semi-labile detritus (van Oevelen *et al.* 2006).

From a microcosm experiment, Risgaard-Petersen et al. (2004) found that microphytobenthos outcompete ammonia-oxidizing bacteria for nitrogen resources, thus exerting top-down pressure on bacterial abundance and nutrient cycling. Koedooder et al. (2019) found that diatom co-cultures are more productive than monocultures and more resistant to bacterial negative pressures on primary productivity. Additionally, microalgal-bacterial interactions led to compositional shifts in both assemblages, highlighting potential specificity of these interactions (Koedooder *et al.* 2019). My findings at Grande Anse, contextualized by past research, hint at the specificity and competitive nature of microalgal–bacterial interactions, underscoring the need for further research into microbial interactive dynamics in coastal habitats.

4.6 Conclusion

This thesis explored the diversity and ecological context of microbial primary producers in a hypertidal mudflat, as framed by their spatial, seasonal, and annual heterogeneity. Cyanobacteria were rare at the Grande Anse mudflat, indicating a negligible contribution to primary production relative to microalgae and chemosynthetic bacteria. This rarity may reflect the combined influences of temperature preferences and tolerances, high silt content, diatom competition, nutrient limitation, hydrodynamic disturbance, and the formation of short-lived and vulnerable resting stages. Against this backdrop, the strong representation of two key lineages in the sequence data underpins diatom dominance in microalgal assemblages: non-motile polar centric diatoms of planktonic origin and motile, behaviorally opportunistic raphid pennate diatoms of benthic origin. Although 20% less abundant, raphid pennate diatoms were three times more diverse than polar centric diatoms. The ability of some nitrate-storing diatoms—including those within the *Thalassiosira* genus—to sustain energy production and viability under dark, anoxic conditions via dissimilatory nitrate reduction to ammonium (DNRA) may confer a competitive advantage over other microalgae in sediments subject to frequent tidal burial. Beyond this, diatoms exhibit metabolic versatility by utilizing a wide range of inorganic and organic nitrogen sources, likely increasing their competitiveness in nutrient-limited and nutrient-variable environments. Consistent with this ecological flexibility, representatives of the polar centric genera *Thalassiosira* and *Minidiscus* occurred in every sample, as did the raphid pennate genus *Gyrosigma*, underscoring their local-scale ubiquity through time and suggesting a persistent allochthonous pelagic influence. These local insights, when viewed in the context of possible regional interconnectivity in the hypertidal upper Bay of Fundy—in terms of microbial species

presence and absence, as previously discussed by Kalu et al. (2023)—suggest that the observed diatom diversity may be mirrored in other regional mudflats, with their abundance likely modulated by site-specific biotic and environmental regimes. Yet, despite their prevalence at Grande Anse, most *Thalassiosira* remain taxonomically unresolved, leaving their species-level identities, adaptive strategies, and ecological roles open to further investigation.

Contrasting life-history strategies—reflecting trade-offs associated with planktonic versus benthic origin, passive tidal transport versus active raphe-enabled motility, and the relative reliance on physiological (e.g., formation of persistent resting stages) versus behavioural (e.g., motility) stress responses—may underlie the divergent spatiotemporal dynamics and ecological relationships of polar centric and raphid pennate diatoms. This was reflected in the pronounced depth-dependent distribution of microalgal species composition within the top centimetre of sediments at Grande Anse during summer, especially late in the season, despite the expected homogenizing influence of hypertidal mixing. Specifically, polar centric diatoms—presumably present as episodically sedimented resting stages—were generally more abundant in subsurface sediments and may contribute disproportionately to buried organic carbon pools, while raphid pennate diatoms—likely vegetative cells—occupied the topmost illuminated sediment layer to a greater extent and may contribute more to mudflat primary production than their polar centric counterparts. The active motility of raphid pennate diatoms confers a competitive advantage, potentially by enabling them to rapidly recolonize sediments post-disturbance, minimize desiccation and thermal stress, evade predators and parasites, and exploit nutrient-rich microhabitats. While dominant, the

richness of the microphytobenthos extended beyond diatoms. The toxigenic, bloom-forming dinoflagellate *Alexandrium* and the stress-tolerant, highly productive green alga *Picochlorum* were the best-represented non-diatom genera, suggesting potential ecological relevance, including functional redundancy and bloom initiation. Additionally, rare microalgal lineages—such as non-diatom ochrophytes, prasinodermophytes, photosynthetic cercozoans, and cryptophytes—contributed disproportionately to overall diversity, with some genera occurring frequently despite their low relative abundance, an intriguing pattern warranting further study.

Temporal variability in microalgal assemblages was evident across multiple scales: intra-seasonal, inter-seasonal, and interannual. Intra-seasonal variability in microalgal species composition was inconsistent across the three summers examined, likely reflecting interannual variability in environmental drivers like air temperature and wind-wave action. Sediment temperature (16–24 °C) was positively correlated with microalgal richness, accounting for 44% of the summer variability in an analysis that excluded biotic and nutrient-related variables, highlighting its potential intra-seasonal role alongside its established inter-seasonal influence. Shifts from polar centric to raphid pennate diatoms during the spring–summer and summer–fall transitions further suggest that raphid pennate taxa are better adapted to variable benthic regimes than their polar centric counterparts; however, it remains unclear whether this inter-seasonal pattern recurs annually. Beyond these temporal patterns, exploratory correlation analysis suggests that pH—even when variable within a narrow range—may act as a broad environmental filter, differentiating major taxonomic lineages, particularly diatoms, relative to other microalgae, in a manner analogous to how benthic organisms

differentially interact with raphid pennate and polar centric diatoms at Grande Anse. In contrast, temperature may drive finer-scale species sorting within these groupings.

DNA metabarcoding provided taxonomic resolution to variability in microalgal biomass estimated from chlorophyll *a* content. Shifts in microalgal richness and species composition only partly reflect variability in biomass—the standing crop available to consumers—and their partial decoupling suggests functional redundancy in maintaining this biomass. Biotic interactions with microalgal assemblages were taxon-dependent, accounted for 26–88% of intra-seasonal summer variability across different metrics, and reflected influences across a gradient of organismal sizes and functions. Specifically, most raphid pennate diatoms were positively correlated with shorebirds, mud snails, adult *Corophium* amphipods, ostracods, and sulfate-reducing bacteria, but negatively with protistan parasites and capitellids (sessile polychaete worms), while polar centric diatoms and other microalgae exhibited the opposite pattern. Mud snails appear to exert greater control over microalgal dynamics than any other organism in the mudflat. The net-positive effect of adult mud snails on microalgal biomass can likely be attributed to their low density, interference competition with microalgal grazers, and bioturbation-enhanced nutrient cycling that stimulates microalgal growth. As with adult mud snails, low shorebird activity likely shifted the grazing–bioturbation equilibrium in favour of net microalgal gain, albeit to a lesser extent. On a separate but related track, nutrient enrichment increased microalgal biomass by 42% over two months, driving assemblage-wide responses, with the magnitude of this effect modulated by adult mud snail density, shorebird activity, and the duration of enrichment.

The findings of this thesis raise intriguing questions relevant to microbial community assembly, primary production, and carbon sequestration in mudflats that warrant further investigation. How tightly coupled are benthic and planktonic microbial diversity and dynamics in the Bay of Fundy, and does this differ between microtidal and hypertidal systems? While local biotic interactions and environmental variability may govern short-term microalgal dynamics, broader regional or climatic influences likely dictate longer-term patterns. This raises the question: which variables best explain interannual variability in microalgal biomass and species composition, and how do biotic and environmental drivers compare across temporal scales? How relatively abundant are resting stages versus vegetative cells among raphid pennate diatoms, polar centric diatoms, phototrophic dinoflagellates, and green algae in mudflat sediments, and what might this reveal about the dominant contributors to primary production? The polar centric diatom genus *Thalassiosira* is well represented in the sequence data. This raises the question: what is the genomic basis of its ecological success in the mudflat? Which chemosynthetic bacteria supplement microalgal production? How do microalgal assemblages in mudflat sediments vary across geological time, and what might this reveal about the taxa most involved in long-term carbon sequestration and storage? Finally, what ecological relationships exist between benthic fauna—including epifauna, macro-infauna, and meiofauna—and heterotrophic microbial eukaryotes and bacteria?

Tables

Table 1. PERMANOVA results for variability in the species composition of microalgal assemblages (percentages of 18S rRNA gene reads), with depth (top 2–5 mm and 0.75–1.25 cm), within surficial sediments of a Bay of Fundy mudflat, both intra-seasonally (early, mid and late summer) and interannually in 2019 and 2020. Planned contrasts examined intra-seasonal variability in depth patterns. Significant and interpretable *P* values of fixed effects are indicated in bold. Estimates and percentages of variation accounted for by each source of variation are provided. Number of unique permutations = 975–999, except 267–273 for the factor Sub-season in year-specific analyses. Note that *P* values generated from permutations and Monte Carlo randomization resulted in similar inferences. df, degrees of freedom; MS, mean square.

Year	Source of variation	df	MS	Pseudo-*F*	*P*	Components of variation	
						Estimate	%
Both	Depth	1	8302	17.92	0.001	435.50	12.23
	Year	1	9797	13.47	0.001	503.88	14.15
	Sub-season	2	2617	3.60	0.002	157.50	4.42
	Depth × Year	1	3231	6.98	0.002	307.55	8.64
	Depth × Sub-season	2	820	1.77	0.116	59.39	1.67
	Year × Sub-season	2	6435	8.85	0.001	951.30	26.72
	Depth × Year × Sub-season	2	2113	4.56	**0.002**	549.89	15.45
	Plot(Year × Sub-season)	12	727	1.57	0.020	132.08	3.71
	Depth × Plot(Year × Sub-season)	12	463			463.20	13.01
2019	Depth	1	2327	4.90	**0.011**	205.77	11.44
	Sub-season	2	6556	11.84	**0.007**	1000.30	55.61
	Depth × Sub-season	2	710	1.49	0.211	78.13	4.34
	Plot(Sub-season)	6	554	1.16	0.355	39.14	2.18
	Depth × Plot(Sub-season)	6	475			475.34	26.43
2020	Depth	1	9206	20.41	0.003	972.78	38.83
	Sub-season	2	2497	2.77	0.028	265.98	10.62
	Depth × Sub-season	2	2223	4.93	**0.005**	590.54	23.57
	Early vs. mid-summer	1	1421	2.78	0.090		
	Mid vs. late summer	1	1998	8.64	**0.022**		
	Early vs. late summer	1	3249	5.31	**0.028**		
	Plot(Sub-season)	6	901	2.00	0.011	225.02	8.98
	Depth × Plot(Sub-season)	6	451			451.06	18.00

Table 2. Summary of SIMPER results showing the contributions of microalgal species to the significant variation between depths (top 2–5 mm and 0.75–1.25 cm), within surficial sediments of a Bay of Fundy mudflat, both intra-seasonally (early, mid and late summer) and interannually in 2019 and 2020 (see Table 1 for significant depth-related effects). The top 10 contributors, together contributing 66–81% of the overall dissimilarity between depths, are shown here. See Table S3A–D for all species contributing ≥1% of the overall dissimilarity between depths, and for their differential average relative abundances between depths. The overall average dissimilarity between the microalgal species composition at different depths, in a given year or sub-season and year, is indicated in bold.

Species	Overall average dissimilarity (%) between depths and contributions to that dissimilarity (%)			
	Summer 2019	Early summer 2020	Mid-summer 2020	Late summer 2020
Average dissimilarity	**50.11**	**45.91**	**49.89**	**81.81**
Bacillariophyta *Gyrosigma limosum*	13.41	20.56	21.14	20.23
Bacillariophyta *Thalassiosira* spp.	15.52	12.36	10.35	13.37
Dinophyceae *Alexandrium fundyense*	5.80	6.51	4.91	14.44
Bacillariophyta *Navicula cryptocephala*	0.35	10.34	3.62	11.38
Bacillariophyta *Delphineis* sp.	0.74	2.37	11.76	9.08
Bacillariophyta Raphid-pennate_X	13.16	2.72	2.17	1.87
Chlorophyta *Picochlorum* spp.	11.20	6.40	1.36	0.69
Bacillariophyta *Minidiscus spinulatus*	3.79	4.40	4.45	1.96
Bacillariophyta *Surirella* spp.	1.01	2.57	3.37	6.82
Bacillariophyta Polar-centric-Mediophyceae_X	0.75	7.87	2.38	1.64

Table 3. PERMANOVA results for variability in the species composition of microalgal assemblages (percentages of 18S rRNA gene reads) intra-seasonally in the summer, inter-seasonally during the spring–summer and summer–fall transitions, and interannually in 2019–2021, in surficial sediments (across the top 1 cm) of a Bay of Fundy mudflat, over 11 dates. Planned contrasts examined variability among sub-seasons or years. Critical *P* values for contrast were corrected using the Benjamini-Hochberg (B-H) procedure. Significant and interpretable *P* values of fixed effects are indicated in bold. Estimates and percentages of variation accounted for by each source of variation are provided, showing their relative importance to the overall temporal variability of microalgal species composition. Negative estimates were replaced with zeros (Searle *et al.* 1992). Number of unique permutations = 997 for the factor Sub-seasonYear, and 10–829 for the planned contrasts, as such *P* values were generated using permutations for the factor Sub-seasonYear, and Monte Carlo randomization for the planned contrasts.

Source of variation	df	MS	Pseudo-*F*	*P*	B-H $P_{critical}$	Components of variation	
						Estimate	%
Sub-seasonYear	10	2499	5.99	**0.001**			
Early, mid and late Summer: 2019 vs. 2020	1	5824	7.38	**0.001**	0.006	598.92	14.78
Mid and late summer: 2019 vs. 2021	1	2749	6.07	**0.004**	0.019	534.19	13.18
Mid and late summer: 2020 vs. 2021	1	3063	7.38	**0.005**	0.025	592.54	14.63
2019 and 2020: Early vs. mid-summer	1	2384	2.35	0.072	0.038	237.73	5.87
2019 and 2020: Early vs. late summer	1	1917	1.65	0.212	0.044	130.73	3.23
2019 and 2020: Mid vs. late summer	1	760	0.59	0.554	0.050	0.00	0.00
2019: Late spring vs. early summer	1	3410	13.80	**0.002**	0.013	1054.30	26.02
2019: Late summer vs. early fall	1	3089	8.14	**0.005**	0.031	903.09	22.29
Plot(Sub-seasonYear)	19	417					

Table 4. Summary of SIMPER results showing the contributions of microalgal species to significant inter-seasonal variability during the spring–summer and summer–fall transitions, and interannual variability in 2019–2021, in surficial sediments (across the top 1 cm) of a Bay of Fundy mudflat (see Table 3 for significant temporal contrasts). The top 10 contributors, together contributing 61–75% of the overall dissimilarity between sub-seasons or years, are shown here. See Table S4A–E for all species contributing ≥1% of the overall dissimilarity between sub-seasons or years, and for their differential average relative abundances between sub-seasons or years. The overall average dissimilarity between the microalgal species composition in different sub-seasons or years, is indicated in bold.

Species	Overall average dissimilarity (%) between sub-seasons or years and contributions to that dissimilarity (%)				
	2019 vs. 2020	2019 vs. 2021	2020 vs. 2021	Late spring vs. Early summer	Late summer vs. Early fall
Average dissimilarity	**53.86**	**42.59**	**43.36**	**50.69**	**50.46**
Bacillariophyta *Thalassiosira* spp.	11.00	11.38	22.77	13.10	26.41
Bacillariophyta *Gyrosigma limosum*	11.66	2.53	14.88	22.74	6.10
Dinophyceae *Alexandrium fundyense*	6.02	5.00	6.26	11.77	15.14
Chlorophyta *Picochlorum* spp.	9.22	15.50	2.17	4.70	6.85
Bacillariophyta Raphid-pennate_X	7.81	5.22	1.70	14.33	6.99
Chlorophyta Pycnococcaceae_X	4.75	7.66	1.23	3.15	5.89
Bacillariophyta *Navicula cryptocephala*	7.26	6.30	5.55	0.17	0.31
Bacillariophyta *Delphineis* sp.	5.12	3.37	5.86	1.31	1.10
Bacillariophyta *Minidiscus spinulatus*	2.36	1.92	2.60	2.92	2.77
Bacillariophyta *Surirella* spp.	3.53	2.07	4.34	0.51	2.03

Table 5. Summary of PERMANCOVA results for spatio-temporal variability in microalgal assemblages, in relation to covariability in environmental variables, in surficial sediments (top 5 mm and 0.75–1.25 cm) of a Bay of Fundy mudflat, from early summer to early fall 2019. Among the environmental covariates, temperature varies seasonally, redox potential (a proxy for oxygen content) and pH decrease with sediment depth, and penetrability is a proxy for sediment water content. Four response variables of microalgae were examined in univariate and multivariate analyses using two different timeframes (A–B). The percentages of microalgal variation accounted for by each source of variation are provided. Negative estimates were replaced with zeros (Searle *et al.* 1992). (+) and (–) indicate significant positive and negative Spearman's rank correlation coefficients, respectively, between a univariate response variable and a covariate. *** $P \leq 0.001$, ** $P \leq 0.01$, * $P \leq 0.05$, or non-significant effect or association. See Tables S5A–D and S6A–D for the complete PERMANCOVA results. Across response variables, environmental covariates typically accounted for 8–22% of microalgal variability, except in the summer when they accounted for 58% of the variation in microalgal ASV richness. Relative abundance, percent of 18S rRNA gene reads of microalgae among microbial eukaryotes. ASV richness (an alpha diversity index), number of microalgal amplicon sequence variants. ASV presence absence, presence or absence of microalgal amplicon sequence variants. Species composition, percentages of 18S rRNA gene reads for species of microalgae.

(A) Early summer to early fall (4 sub-seasons)

Source of variation	Estimate of components of variation (%) of microalgae			
	Relative abundance	ASV richness	ASV presence or absence	Species composition
Temperature	0.00	13.8 (+) *	2.67	3.88 *
Redox potential	1.45	0.00	0.00	0.68
pH	4.97 (+) *	8.13 (+) *	9.28 **	7.46 **
Penetrability	1.70	0.00	0.46	2.18
Sub-season	75.4 *	45.5	25.7	34.1
Depth	1.73	0.05	14.9 *	12.9 **
Sub-season × Depth	1.42	0.00	1.22	4.91
Plot(Sub-season)	9.30	20.4	2.51	10.9
Depth × Plot(Sub-season)	4.04	12.1	43.3	23.0

(B) Early, mid and late summer (3 sub-seasons)

Source of variation	Estimate of components of variation (%) of microalgae			
	Relative abundance	ASV richness	ASV presence or absence	Species composition
Temperature	14.4 (+) *	43.6 (+) *	3.20	7.17
Redox potential	0.00	0.00	0.00	0.00
pH	4.65	6.77	9.13 *	8.72 *
Penetrability	0.00	7.49	2.45	0.59
Sub-season	57.2	0.00	36.9	30.0
Depth	0.68	0.00	8.24	10.4 *
Sub-season × Depth	2.27	0.00	0.00	5.21
Plot(Sub-season)	16.2	20.4	0.66	18.5
Depth × Plot(Sub-season)	4.64	21.7	39.4	19.3

Table 6. Summary of PERMANCOVA results for spatio-temporal variability in microalgal assemblages, in relation to covariability in biotic and nutrient-related variables, and across epifaunal predator exclusion and nutrient enrichment treatments, in surficial biofilms (top 2 mm) of a Bay of Fundy mudflat, in summer 2020. Biotic covariates, spanning from shorebirds to nitrate reducing bacteria, represent various levels of the mudflat food web, and are reflective of predator-prey, parasite-host and resource availability interactions with microalgae (see Table S9 for classification of covariates). Five response variables of microalgae were examined in univariate and multivariate analyses. The percentages of microalgal variation accounted for by each source of variation are provided. Negative estimates were replaced with zeros (Searle *et al.* 1992). Shading indicates top (i.e. above-average) determinants of microalgal variability (i.e., components of variation estimate ≥4%). (+) and (–) indicate significant positive and negative Spearman's rank correlation coefficients, respectively, between a univariate response variable and a covariate. *** $P \leq 0.001$, ** $P \leq 0.01$, * $P \leq 0.05$, or non-significant effect or association. See Table S8A–E for the complete PERMANCOVA results. Across response variables, biotic and nutrient-related variables accounted for 43–88% of microalgal variability in the summer. Chlorophyll *a* content (a proxy for microalgal biomass), µg of chlorophyll *a* per g of sediment. Relative abundance, percent of 18S rRNA gene reads of microalgae among microbial eukaryotes. ASV richness (an alpha diversity index), number of microalgal amplicon sequence variants. ASV presence absence, presence or absence of microalgal amplicon sequence variants. Species composition, percentages of 18S rRNA gene reads for species of microalgae.

Source of variation	Estimate of components of variation (%) of microalgae				
	Chlorophyll *a* content	Relative abundance	ASV richness	ASV presence absence	Species comp.
Shorebirds	1.24 (+) **	14.7 (+) ***	3.03 (+) *	3.44 ***	5.23 ***
Adult mud snails	4.57 (+) ***	29.5 (+) ***	7.68 (+) **	6.27 ***	11.9 ***
Juvenile mud snails	4.28 (−) ***	0.67 (+) **	0	2.40 *	2.64 **
Adult *Corophium* amphipods	1.14 (−) **	0.08	3.13 (−) *	2.96 ***	2.06 ***
Juvenile *Corophium* amphipods	0	1.03 (−) ***	0	0.27	0.66 *
Macoma clams	1.20 (−) **	0	0	0.33	0.88 **
Nereididae (Polychaeta)	2.38 (−) ***	0.04	0.08	0.83	0.53
Phyllodocidae (Polychaeta)	0.10	0.49 (+) **	0	0.02	0.94 *
Capitellidae (Polychaeta)	1.28 (+) *	0.05	2.24	2.12 *	2.67 ***

Spionidae (Polychaeta)	0	0.11	0	0	0.04
Turbellaria (Platyhelminthes)	0.71 (+) *	0.44 (−) *	1.63	0.59	0.11
Nematoda	2.62 (+) **	0	0	0.78	0.26
Copepoda	0	0.78 (+) **	2.15	0.28	1.12 **
Ostracoda	0.09	9.46 (+) ***	4.00 (−) *	1.64 *	4.04 ***
Fungi	2.83 (−) **	4.75 (−) ***	1.68	3.06 *	0.69
Protistan consumers	0	22.1 (−) ***	12.2 (−) *	4.96 **	1.03
Protistan parasites	1.23 (+) *	1.87 (−) ***	0	4.06 ***	12.6 ***
Sulfate reducing bacteria	1.86 (−) **	0	3.24	0.98	0.76 *
Sulfur compound oxidizing bacteria	0	1.11 (+) **	11.3 (−) ***	6.05 ***	1.19 *
Nitrate reducing bacteria	0.65	0.42 (−) *	0.38	1.97 *	2.56 **
Sub-season	35.5 **	9.12 **	0	10.8	28.5 **
Fertilizer	3.18 *	0	0	0	0.08
Shorebirds × Fertilizer	3.70 **	0	3.42	0	0
Adult mud snails × Fertilizer	9.94 **	0	13.8	0	1.54
Sub-season × Fertilizer	15.5 **	0	0	0	0
Plot(Sub-season × Fertilizer)	1.88	0	2.18	9.96 **	7.70 ***
Residual	4.22	3.24	27.9	36.3	10.3

Figures

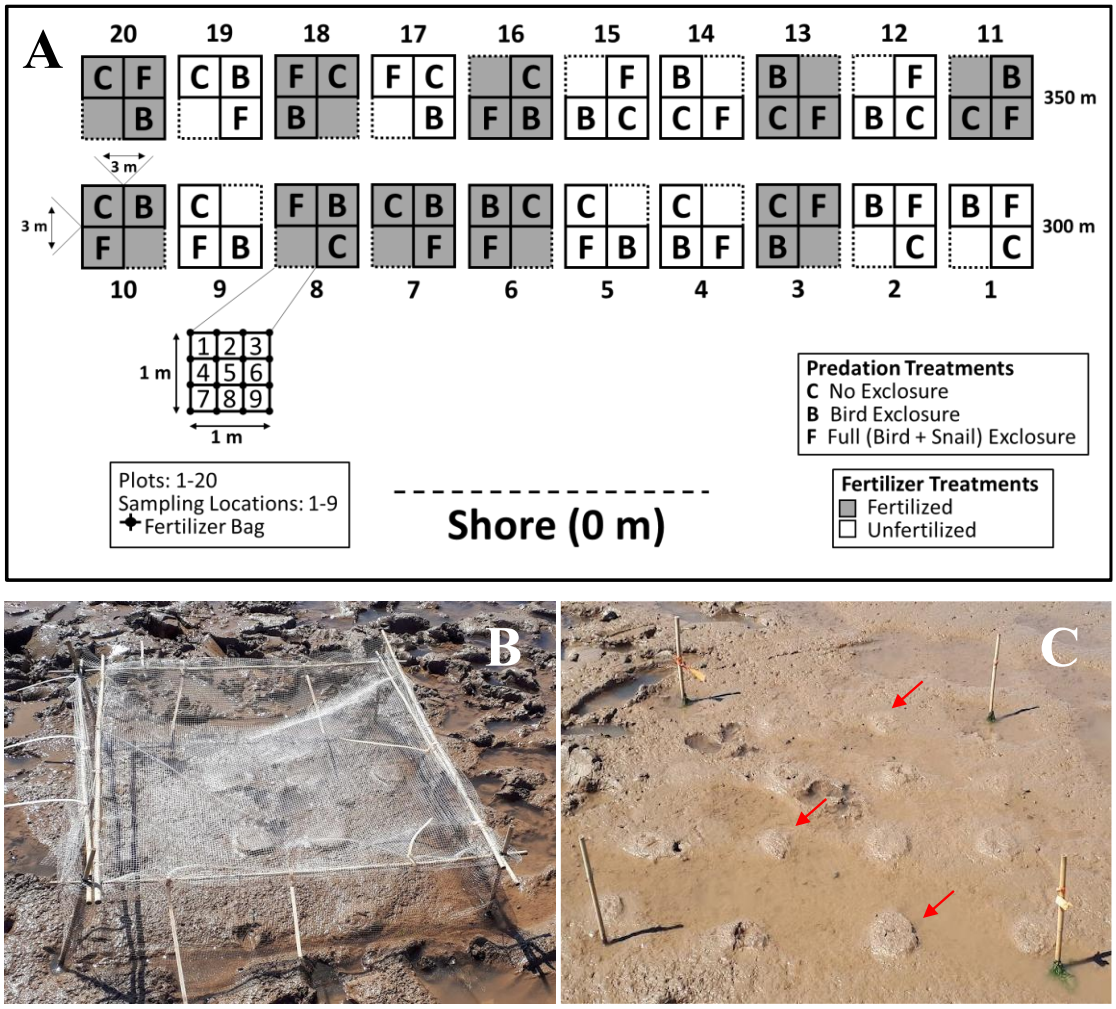


Figure 1. Experimental set-up (not to scale) at the Grande Anse mudflat in summer 2020 to investigate the trophic dynamics of benthic microalgal assemblages, in surficial biofilms, using epifaunal predator exclosures and fertilizer application. (A) Split-plot experimental design with the fertilizer and epifaunal predator exclosure treatments randomly assigned to plots and subplot, respectively. (B) Post-sampling photograph of a predator treatment subplot designed to exclude shorebirds and adult mud snails. (C) Post-sampling photograph of a predator treatment control (i.e., an open plot) showing the placement of buried fertilizer bags, indicated by the red arrows and sediment bumps.

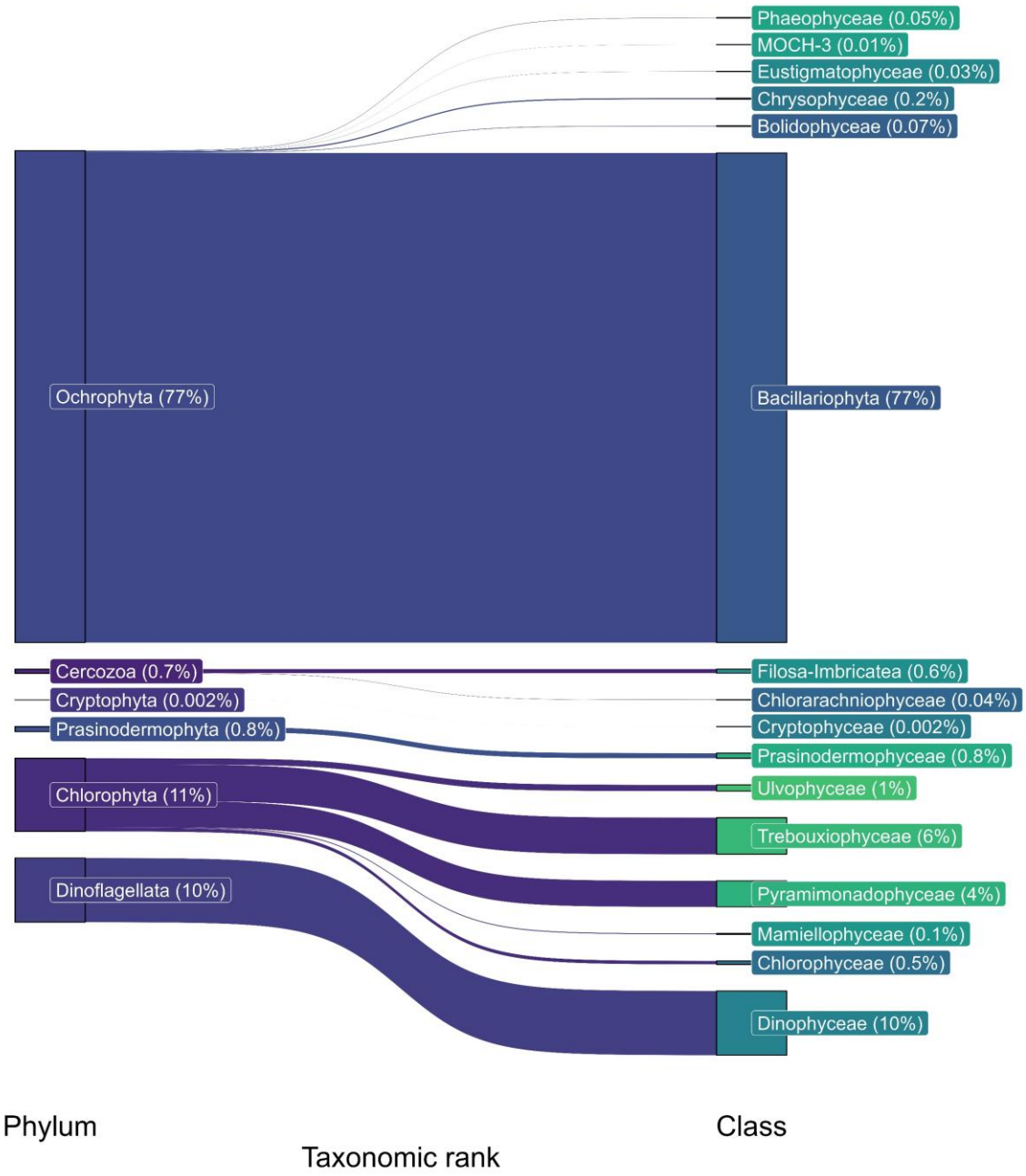


Figure 2A. Microalgal phyla and classes. See figure caption below.

Figure 2. Sankey flow diagrams showing the diversity and approximate average relative abundances (percentages of microalgal 18S rRNA gene reads) of taxa within microalgal assemblages, across taxonomic ranks, in surficial sediments (across the top 1 cm) of a Bay of Fundy mudflat during the summers of 2019 to 2021. To estimate relative abundances, surficial depth samples were averaged within each sediment core (see Methods section for details), and the 24 sediment cores collected across 9 dates during the summers (refer to Table S1A for sampling dates) were averaged. The diagrams provide (A) an overview of microalgal taxa, and inventories of (B) diatom, and (C) non-diatom lineages.

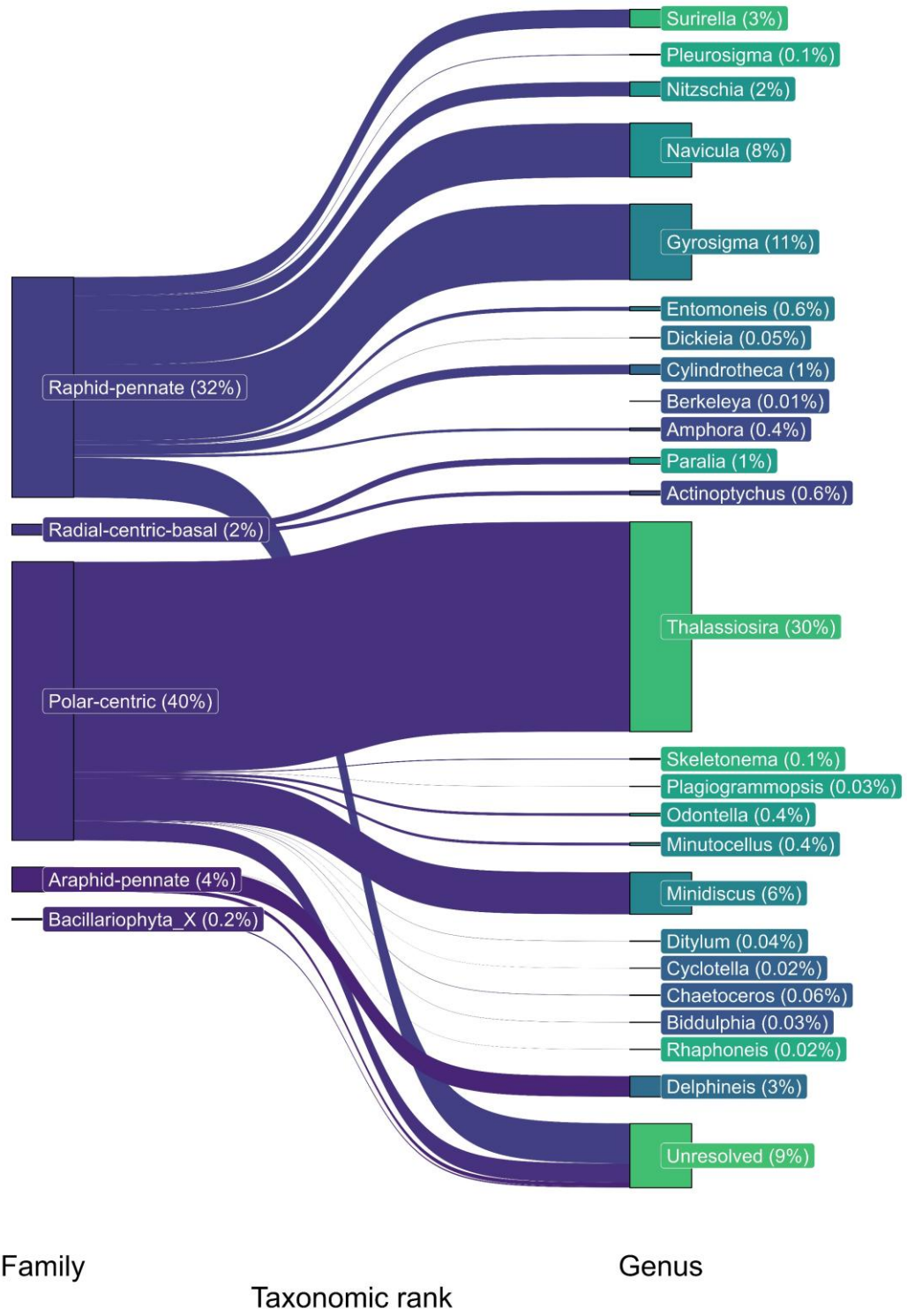


Figure 2B. Inventory of Bacillariophyta (diatom) genera. See figure caption above.

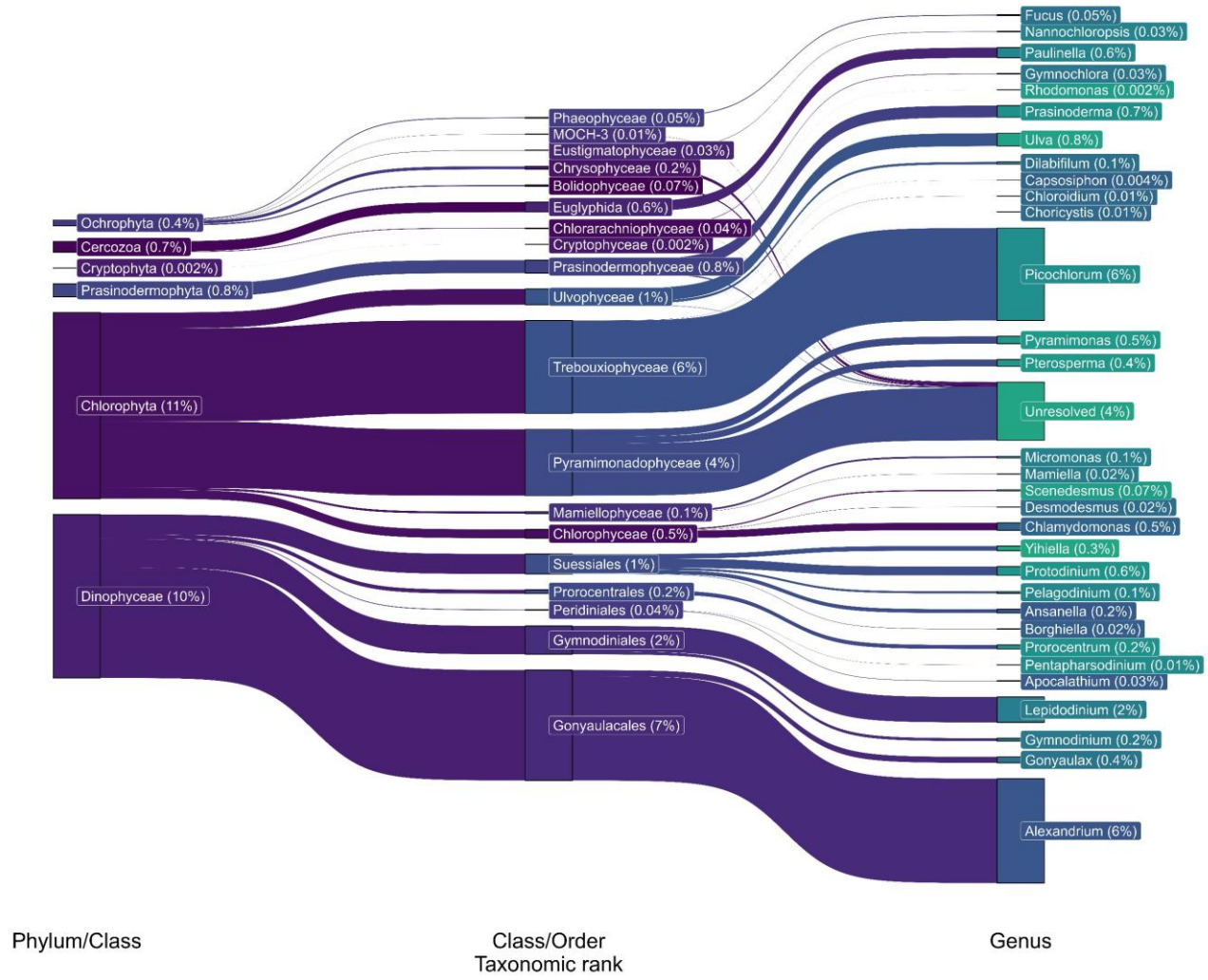


Figure 2C. Inventory of non-Bacillariophyta (non-diatom) microalgal genera. See figure caption above.

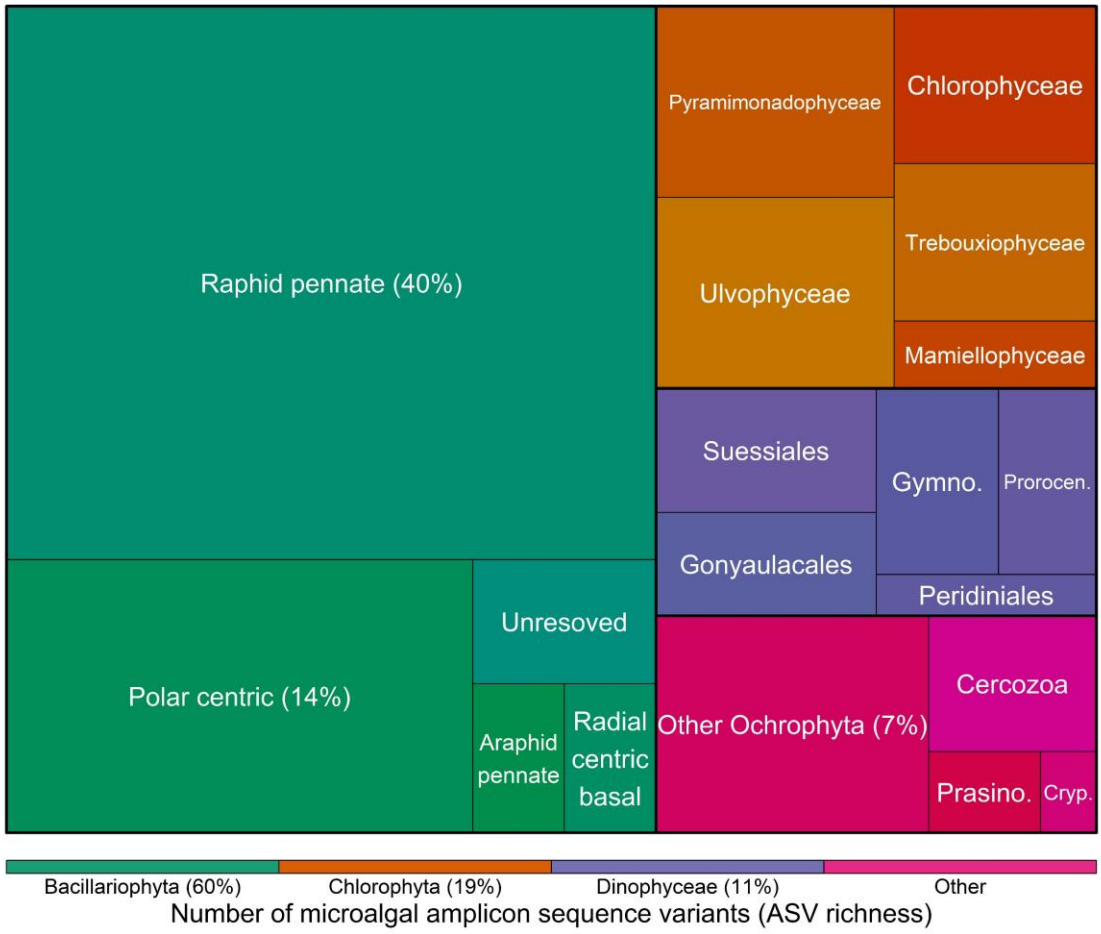
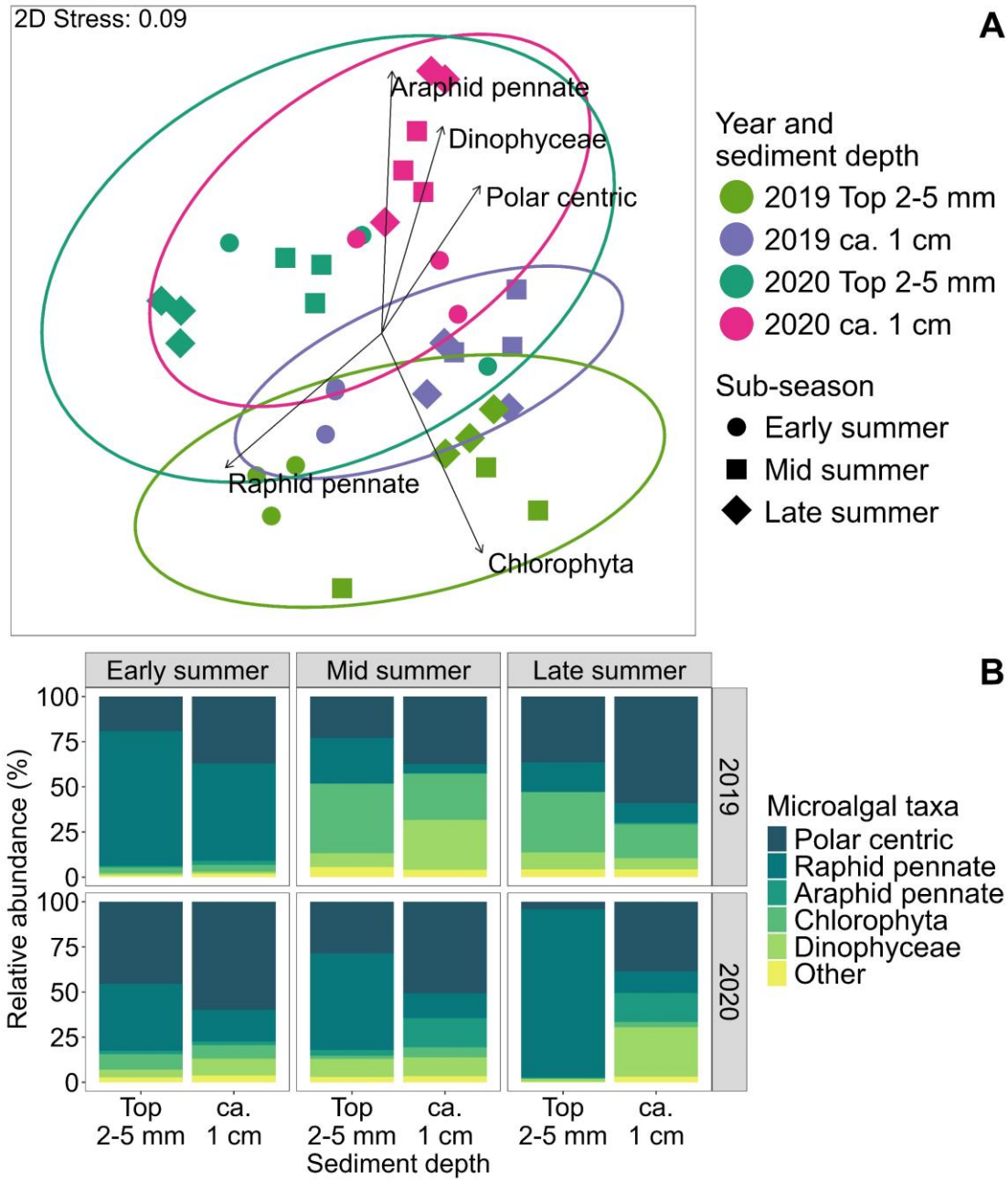


Figure 3. Treemap chart showing the relative contributions of various taxa to the pool of microalgal amplicon sequence variants within surficial sediments (across the top 1 cm) of a Bay of Fundy mudflat during the summers of 2019 to 2021. “Unresolved” refers to Bacillariophyta ASVs unassigned to any family. “Other Ochrophyta” refers to all lineages within that phylum with the exception of Bacillariophyta. Bacillariophyta, diatoms. Chlorophyta, green algae. Dinophyceae, dinoflagellates.

Figure 4. Depth variability in the composition of microalgal assemblages within surficial sediments of a Bay of Fundy mudflat, both intra-seasonally in the summer and interannually in 2019 and 2020. (A) Non-metric multidimensional scaling graph showing the relative placement of microalgal assemblages in ordination space. Each symbol represents the microalgal species composition of a sediment sample. The dispersion (multivariate variability) around the centroid (multivariate average) for each combination of year and sediment depth, pooling over sub-seasons, is captured by a 95% confidence ellipse. Each arrow indicates the direction of increasing relative abundance for a given microalgal lineage. (B) Average relative abundances of microalgal lineage (percentages of microalgal 18S rRNA gene reads). Polar centric Mediophyceae, raphid pennates and araphid pennates are lineages of Bacillariophyta (diatoms). Chlorophyta, green algae. Dinophyceae, dinoflagellates. Refer to Figure 2A–C for "Other" microalgal lineages.


A
2D Stress: 0.09
Araphid pennate
Dinophyceae
Polar centric
Raphid pennate
Chlorophyta
Year and sediment depth
2019 Top 2-5 mm
2019 ca. 1 cm
2020 Top 2-5 mm
2020 ca. 1 cm
Sub-season
Early summer
Mid summer
Late summer
B
Early summer
Mid summer
Late summer
2019
2020
Relative abundance (%)
100
75
50
25
0
Top 2-5 mm
ca. 1 cm
Sediment depth
Microalgal taxa
Polar centric
Raphid pennate
Araphid pennate
Chlorophyta
Dinophyceae
Other

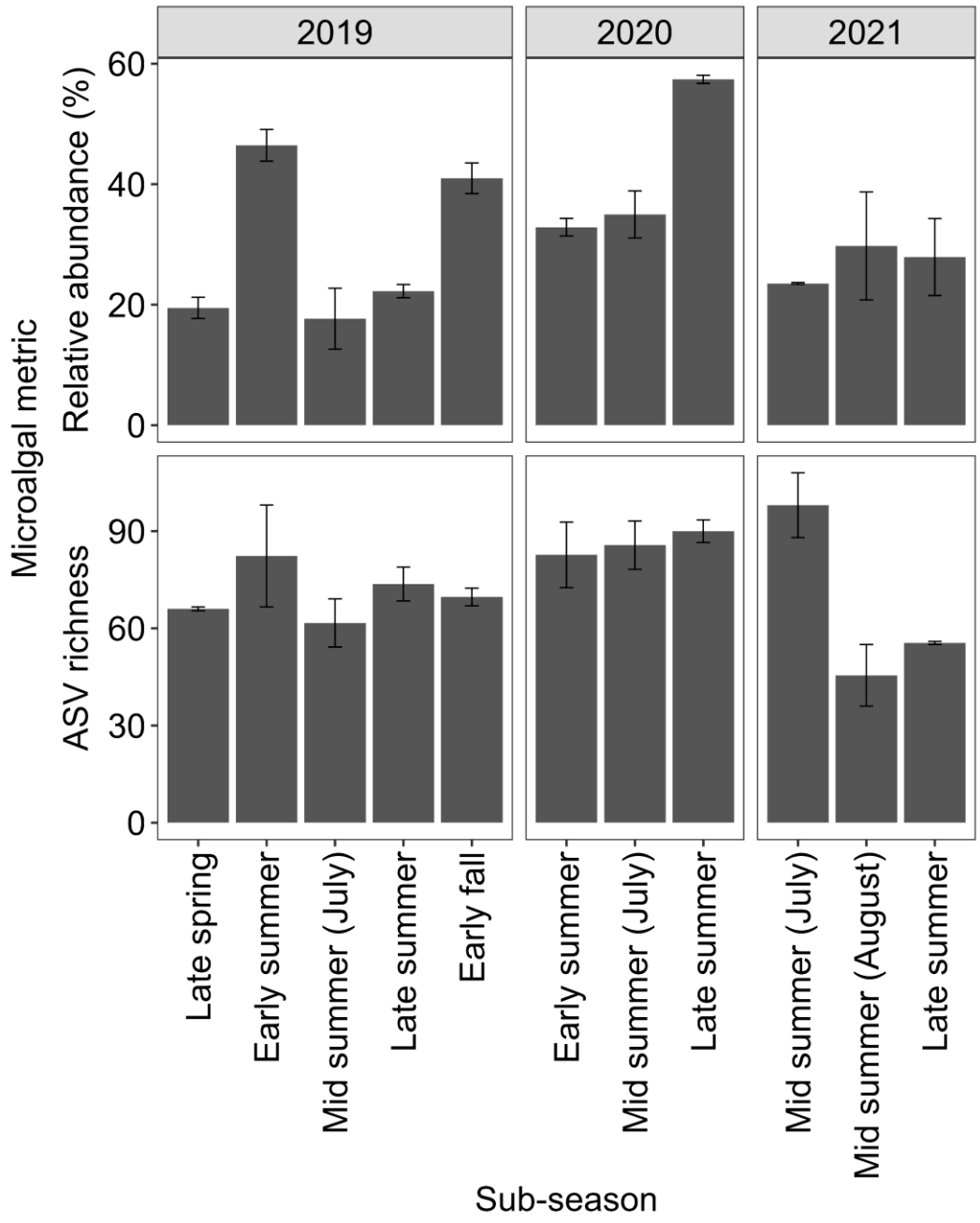


Figure 5. Temporal variability (mean ± SE) in relative abundance and ASV richness of benthic microalgae, intra-seasonally in the summer, inter-seasonally during the spring–summer and summer–fall transitions, and interannually in 2019–2021, in surficial sediments (across the top 1 cm) of a Bay of Fundy mudflat. n = 3 in 2019 and 2020, and n = 2 in 2021. The temporal dynamics of microalgal relative abundance and ASV richness are not coupled. Relative abundance, percent of 18S rRNA gene reads of microalgae among microbial eukaryotes. ASV richness (an alpha diversity index), number of microalgal amplicon sequence variants.

Figure 6. Temporal variability in the composition of microalgal assemblages, intra-seasonally in the summer, inter-seasonally during the spring–summer and summer–fall transitions, and interannually in 2019–2021, in surficial sediments (across the top 1 cm) of a Bay of Fundy mudflat. (A) Non-metric multidimensional scaling graph showing the relative placement of microalgal assemblages in ordination space. Each symbol represents the microalgal species composition of a sediment core. The dispersion (multivariate variability) around the centroid (multivariate average) for each year, averaging across sediment depths and pooling over sub-seasons, is captured by a 95% confidence ellipse. Refer to Table S1A for an overview of samples collected. Each arrow indicates the direction of increasing relative abundance for a given microalgal lineage. (B) Average relative abundances of microalgal lineage (percentages of microalgal 18S rRNA gene reads). Polar centric Mediophyceae, raphid pennates and araphid pennates are lineages of Bacillariophyta (diatoms). Chlorophyta, green algae. Dinophyceae, dinoflagellates. Refer to Figure 2A–C for “Other” microalgal lineages.

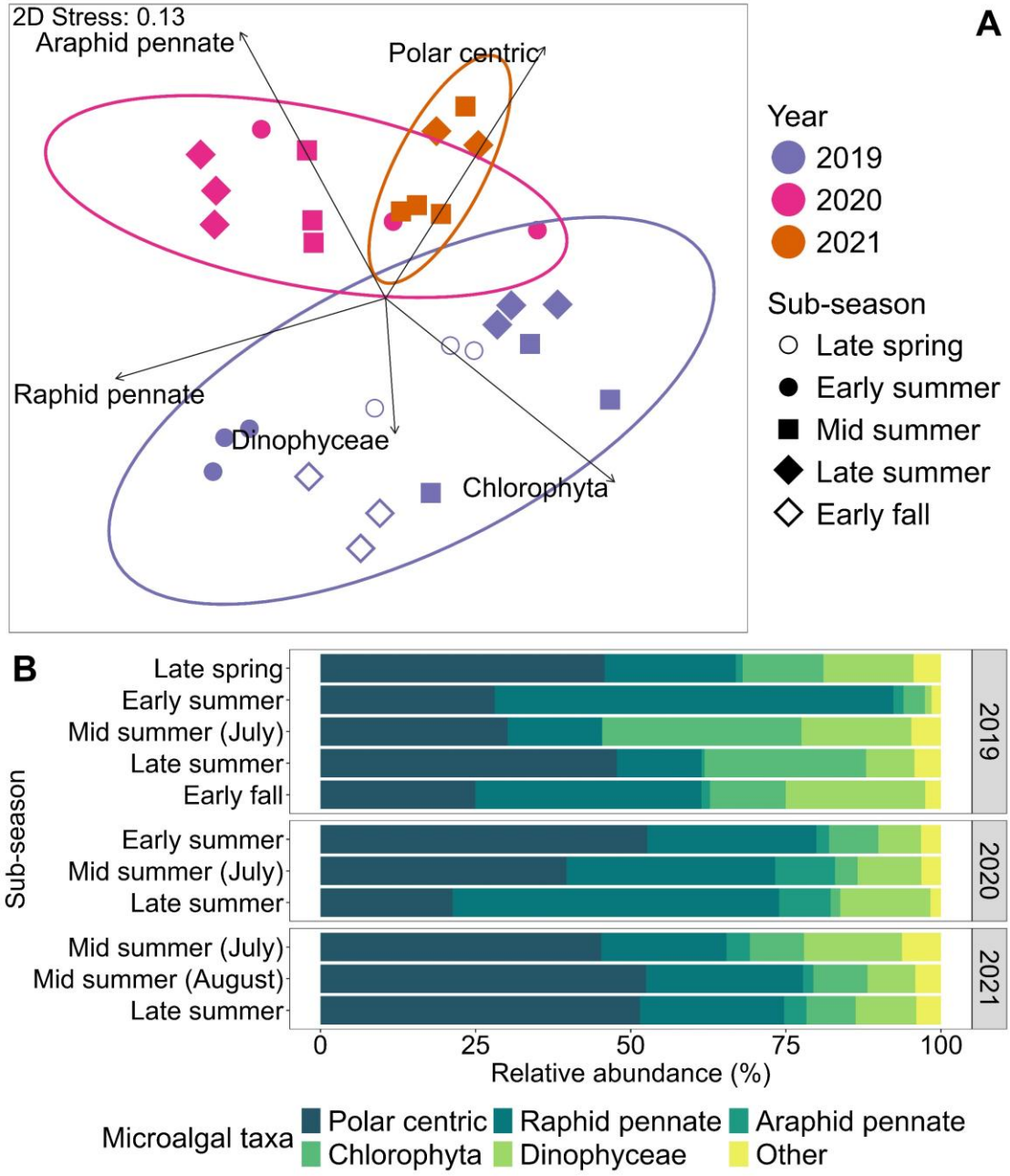
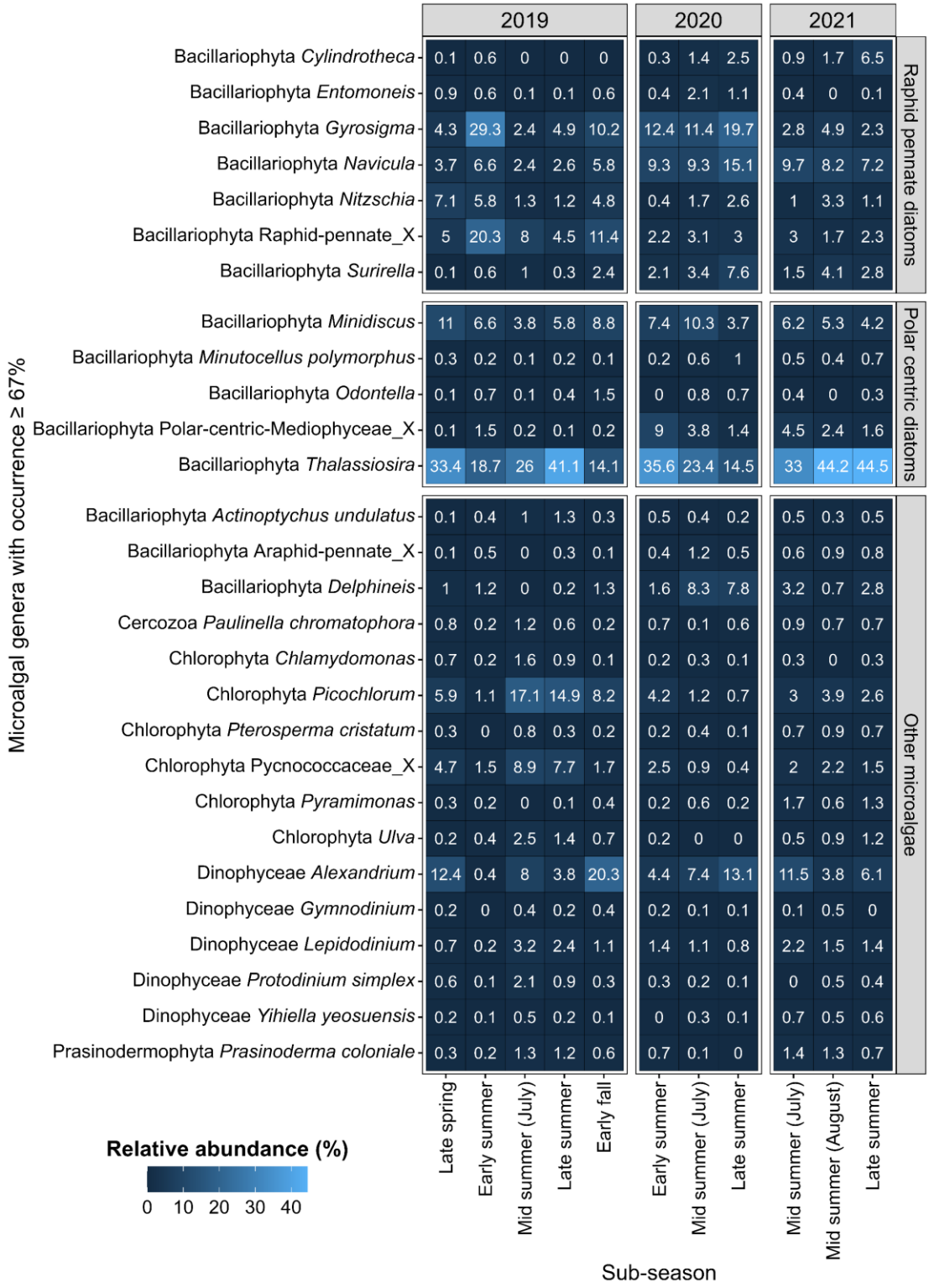

A
2D Stress: 0.13
Araphid pennate
Polar centric
Raphid pennate
Dinophyceae
Chlorophyta
Year
2019
2020
2021
Sub-season
Late spring
Early summer
Mid summer
Late summer
Early fall
B
Sub-season
Late spring
Early summer
Mid summer (July)
Late summer
Early fall
Early summer
Mid summer (July)
Late summer
Mid summer (July)
Mid summer (August)
Late summer
2019
2020
2021
0
25
50
75
100
Relative abundance (%)
Microalgal taxa
Polar centric
Raphid pennate
Araphid pennate
Chlorophyta
Dinophyceae
Other

Figure 7. Heatmap showing the temporal variability in the average relative abundances (percentages of microalgal 18S rRNA gene reads) of the most frequently detected microalgal genera, intra-seasonally in the summer, inter-seasonally during the spring–summer and summer–fall transitions, and interannually in 2019–2021, in surficial sediments (across the top 1 cm) of a Bay of Fundy mudflat. To estimate relative abundances, surficial depth samples were averaged within each sediment core (see Methods section for details), and the 2–3 replicate sediment cores collected in a given year and sub-season (refer to Table S1A for sampling dates) were averaged. n = 3 in 2019 and 2020, and n = 2 in 2021. Species names are indicated for monospecific genera in sediments. The designation "X" indicates taxonomically unresolved groups. The heatmap includes all microalgal genera with a percent occurrence ≥67% across 30 sediment cores. This included 28 groups with average relative abundances ranging from 0.2% to 35%, but excluded 43 groups (average relative abundances of 0.001% to 0.9%).


2019
2020
2021
Raphid pennate diatoms
Polar centric diatoms
Other microalgae
Microalgal genera with occurrence ≥ 67%
Bacillariophyta Cylindrotheca 0.1 0.6 0 0 0 0.3 1.4 2.5 0.9 1.7 6.5
Bacillariophyta Entomoneis 0.9 0.6 0.1 0.1 0.6 0.4 2.1 1.1 0.4 0 0.1
Bacillariophyta Gyrosigma 4.3 29.3 2.4 4.9 10.2 12.4 11.4 19.7 2.8 4.9 2.3
Bacillariophyta Navicula 3.7 6.6 2.4 2.6 5.8 9.3 9.3 15.1 9.7 8.2 7.2
Bacillariophyta Nitzschia 7.1 5.8 1.3 1.2 4.8 0.4 1.7 2.6 1 3.3 1.1
Bacillariophyta Raphid-pennate_X 5 20.3 8 4.5 11.4 2.2 3.1 3 3 1.7 2.3
Bacillariophyta Surirella 0.1 0.6 1 0.3 2.4 2.1 3.4 7.6 1.5 4.1 2.8
Bacillariophyta Minidiscus 11 6.6 3.8 5.8 8.8 7.4 10.3 3.7 6.2 5.3 4.2
Bacillariophyta Minutocellus polymorphus 0.3 0.2 0.1 0.2 0.1 0.2 0.6 1 0.5 0.4 0.7
Bacillariophyta Odontella 0.1 0.7 0.1 0.4 1.5 0 0.8 0.7 0.4 0 0.3
Bacillariophyta Polar-centric-Mediophyceae_X 0.1 1.5 0.2 0.1 0.2 9 3.8 1.4 4.5 2.4 1.6
Bacillariophyta Thalassiosira 33.4 18.7 26 41.1 14.1 35.6 23.4 14.5 33 44.2 44.5
Bacillariophyta Actinoptychus undulatus 0.1 0.4 1 1.3 0.3 0.5 0.4 0.2 0.5 0.3 0.5
Bacillariophyta Araphid-pennate_X 0.1 0.5 0 0.3 0.1 0.4 1.2 0.5 0.6 0.9 0.8
Bacillariophyta Delphineis 1 1.2 0 0.2 1.3 1.6 8.3 7.8 3.2 0.7 2.8
Cercozoa Paulinella chromatophora 0.8 0.2 1.2 0.6 0.2 0.7 0.1 0.6 0.9 0.7 0.7
Chlorophyta Chlamydomonas 0.7 0.2 1.6 0.9 0.1 0.2 0.3 0.1 0.3 0 0.3
Chlorophyta Picochlorum 5.9 1.1 17.1 14.9 8.2 4.2 1.2 0.7 3 3.9 2.6
Chlorophyta Pterosperma cristatum 0.3 0 0.8 0.3 0.2 0.2 0.4 0.1 0.7 0.9 0.7
Chlorophyta Pycnococcaceae_X 4.7 1.5 8.9 7.7 1.7 2.5 0.9 0.4 2 2.2 1.5
Chlorophyta Pyramimonas 0.3 0.2 0 0.1 0.4 0.2 0.6 0.2 1.7 0.6 1.3
Chlorophyta Ulva 0.2 0.4 2.5 1.4 0.7 0.2 0 0 0.5 0.9 1.2
Dinophyceae Alexandrium 12.4 0.4 8 3.8 20.3 4.4 7.4 13.1 11.5 3.8 6.1
Dinophyceae Gymnodinium 0.2 0 0.4 0.2 0.4 0.2 0.1 0.1 0.1 0.5 0
Dinophyceae Lepidodinium 0.7 0.2 3.2 2.4 1.1 1.4 1.1 0.8 2.2 1.5 1.4
Dinophyceae Protodinium simplex 0.6 0.1 2.1 0.9 0.3 0.3 0.2 0.1 0 0.5 0.4
Dinophyceae Yihiella yeosuensis 0.2 0.1 0.5 0.2 0.1 0 0.3 0.1 0.7 0.5 0.6
Prasinodermophyta Prasinoderma coloniale 0.3 0.2 1.3 1.2 0.6 0.7 0.1 0 1.4 1.3 0.7
Late spring
Early summer
Mid summer (July)
Late summer
Early fall
Early summer
Mid summer (July)
Late summer
Mid summer (July)
Mid summer (August)
Late summer
Sub-season
Relative abundance (%)
0 10 20 30 40

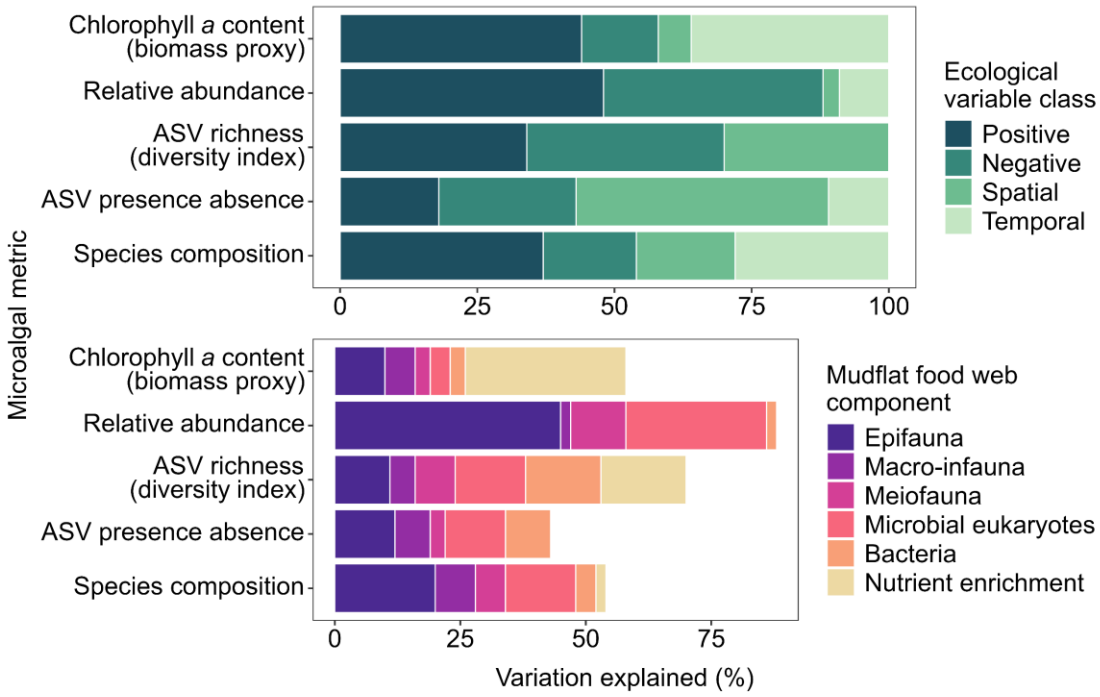
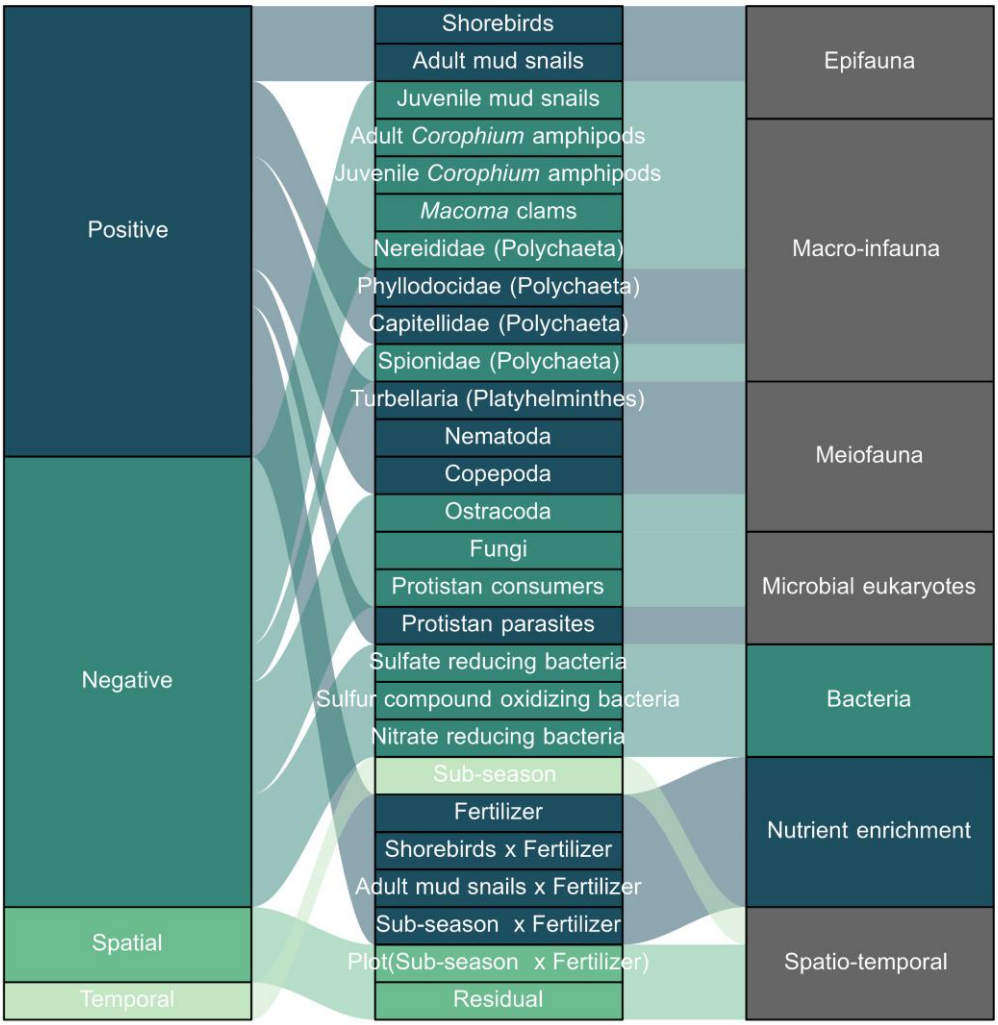


Figure 8. Relative contributions of biotic covariates and experimental factors to the variability of microalgal response variables, in surficial biofilms (top 2 mm) of a Bay of Fundy mudflat, in summer 2020. Five microalgal response variables were examined in univariate and multivariate analyses (see Table S8A–E for PERMANCOVA results). In Figure 8A above, covariates and factors are grouped into ecological variable classes (top panel) and mudflat food web components (bottom panel), with the contributions of these groups shown. Note that the combined contribution of all mudflat food web components is equal to that of the 'positive' and 'negative' ecological variables. For numerical representations of the percent contributions, refer to Table S10. In Figure 8B below, the composition of variables in each group is shown. Chlorophyll *a* content, µg of chlorophyll *a* per g of sediment. Relative abundance, percent of 18S rRNA gene reads of microalgae among microbial eukaryotes. ASV richness, number of microalgal amplicon sequence variants. ASV presence absence, presence or absence of microalgal amplicon sequence variants. Species composition, percentages of 18S rRNA gene reads for species of microalgae.



The 27 variables in PERMANCOVA models (centre) and 2 variable groupings (left and right)

Figure 8B. Alluvial flow diagram showing how variables in PERMANCOVA models were grouped into ecological classes (left) and mudflat food web components (right). While the bacterial and nutrient enrichment groups consist exclusively of 'negative' and 'positive' variables, respectively, other food web components had mixed compositions. See main figure caption above.

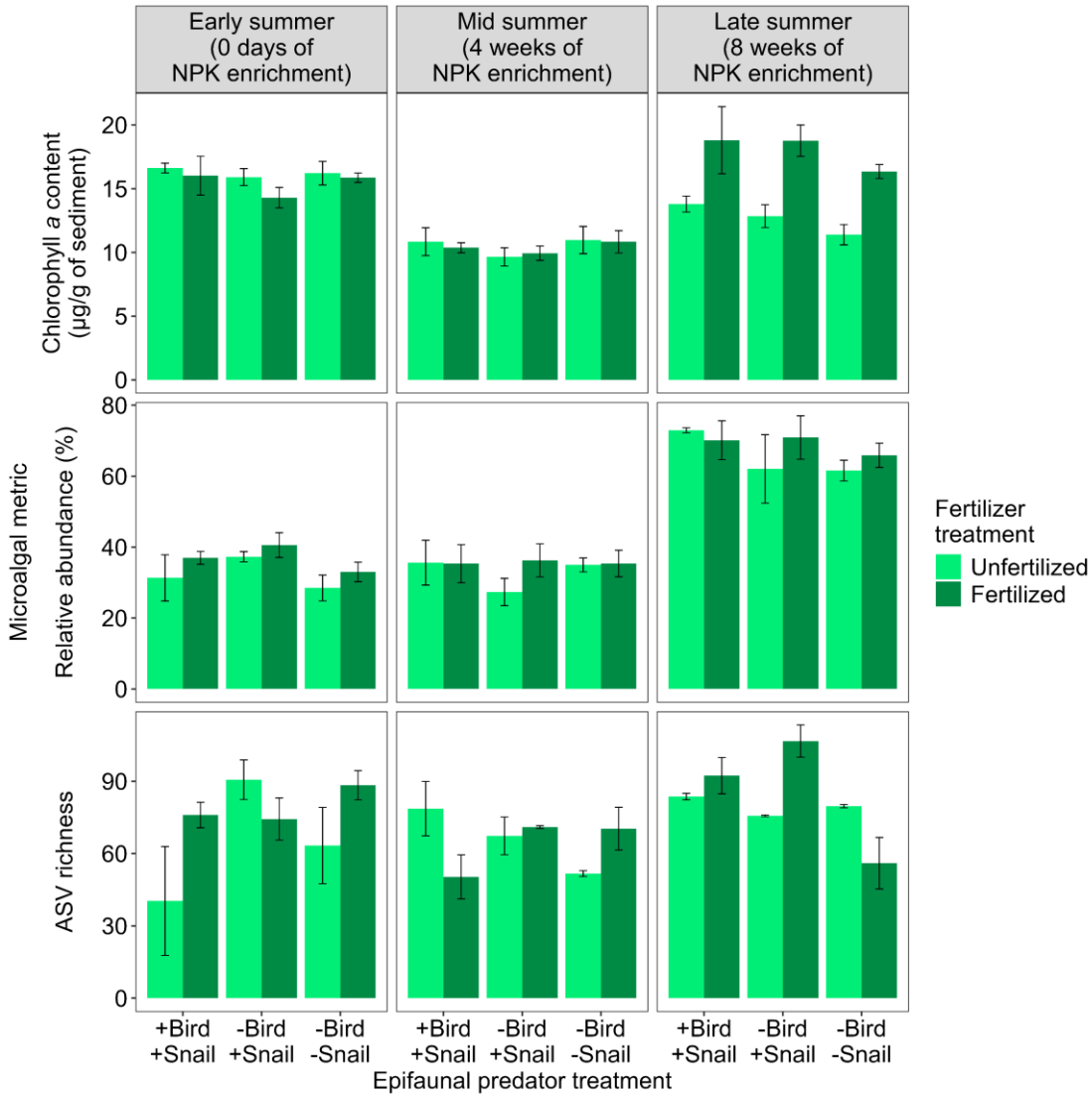


Figure 9. Variability (mean ± SE) in the chlorophyll *a* content, relative abundance and ASV richness of benthic microalgae, across epifaunal predator exclusion and nutrient enrichment treatments, in surficial biofilms (top 2 mm) of a Bay of Fundy mudflat, in summer 2020. n = 3. Note that epifaunal predators were entirely or mostly absent from the mudflat in early and mid-summer. Epifaunal predator treatments included a no-exclosure control (+Bird, +Snail), bird exclosure (−Bird, +Snail), and bird and snail exclosure (−Bird, −Snail). NPK, nitrogen, phosphorus and potassium. Chlorophyll *a* content, a microalgal biomass proxy. Relative abundance, percent of 18S rRNA gene reads of microalgae among microbial eukaryotes. ASV richness (an alpha diversity index), number of microalgal amplicon sequence variants.

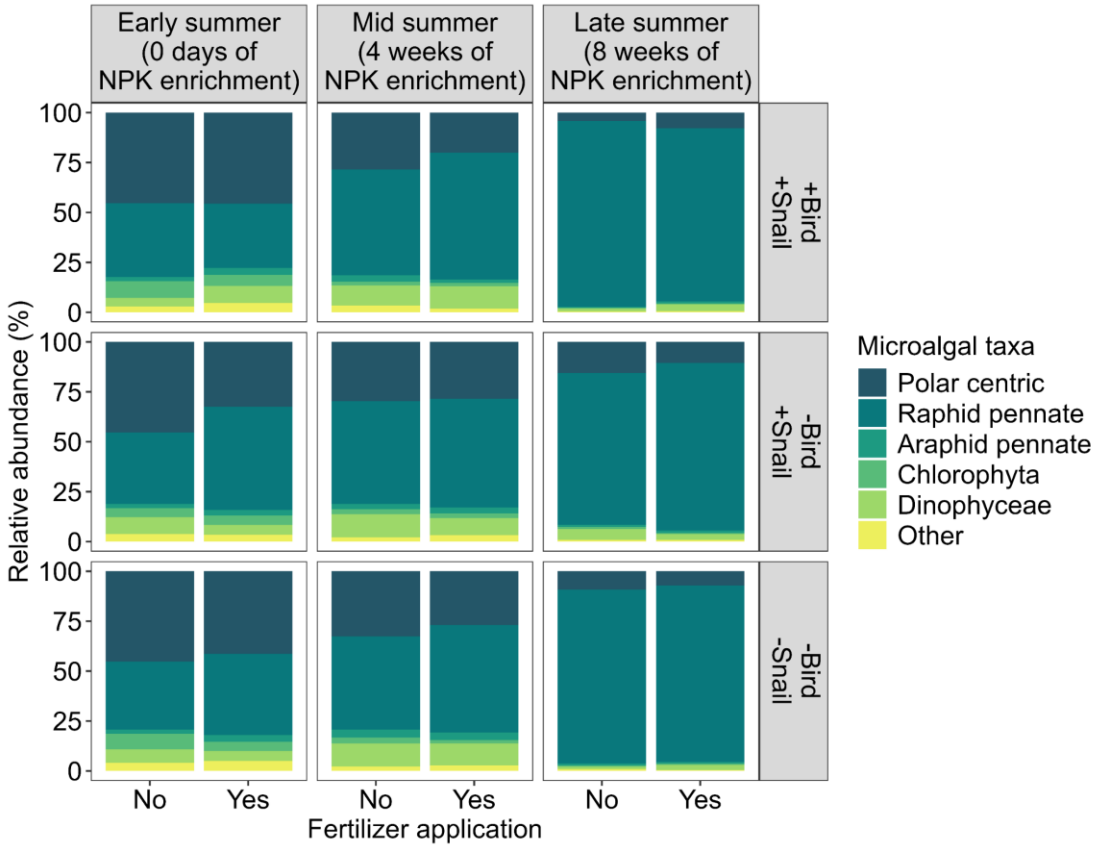


Figure 10. Variability in the composition of microalgal assemblages, across epifaunal predator exclusion and nutrient enrichment treatments, in surficial biofilms (top 2 mm) of a Bay of Fundy mudflat, in summer 2020. n = 3. Average relative abundances of microalgal lineage are based on percentages of microalgal 18S rRNA gene reads. Note that epifaunal predators were entirely or mostly absent from the mudflat in early and mid-summer. Epifaunal predator treatments included a no-exclosure control (+Bird, +Snail), bird exclosure (−Bird, +Snail), and bird and snail exclosure (−Bird, −Snail). NPK, nitrogen, phosphorus and potassium. Polar centric Mediophyceae, raphid pennates and araphid pennates are lineages of Bacillariophyta (diatoms). Chlorophyta, green algae. Dinophyceae, dinoflagellates. Refer to Figure 2A–C for "Other" microalgal lineages

Figure 11. Heatmap showing the strength and direction of relationships between the most abundant microalgal genera and biotic covariates associated with the significant variability in microalgal species composition (n = 54), pooling over sub-seasons as well as epifaunal predator exclusion and nutrient enrichment treatments, in surficial biofilms (top 2 mm) of a Bay of Fundy mudflat, in summer 2020. See Table S8E for the PERMANCOVA results on the variability in microalgal species composition. Correlation coefficients were omitted for non-significant relationships. Species names are indicated for monospecific genera in sediments. The designation “X” indicates taxonomically unresolved groups.

Raphid pennate diatoms
Polar centric diatoms
Other microalgae
Epifauna
Infauna
Microbes
Shorebird
Adult mud snail
Juvenile mud snail
Adult Corophium amphipod
Juvenile Corophium amphipod
Macoma clam
Phyllodocidae (Polychaeta)
Capitellidae (Polychaeta)
Copepoda
Ostracoda
Protist parasite
Sulfate reducing bacteria
Sulfur oxidizing bacteria
Nitrate reducing bacteria
Covariates significantly associated with microalgal species composition
Spearman's correlation
0.8
0.4
0.0
-0.4
Bacillariophyta Amphora
Bacillariophyta Cylindrotheca
Bacillariophyta Dickieia ulvacea
Bacillariophyta Entomoneis
Bacillariophyta Gyrosigma
Bacillariophyta Navicula
Bacillariophyta Nitzschia
Bacillariophyta Surirella
Bacillariophyta Raphid-pennate_X
Bacillariophyta Minidiscus
Bacillariophyta Minutocellus polymorphus
Bacillariophyta Skeletonema marinoi
Bacillariophyta Thalassiosira
Bacillariophyta Polar-centric-Mediophyceae_X
Bacillariophyta Actinoptychus
Bacillariophyta Delphineis
Bacillariophyta Paralia
Bacillariophyta Araphid-pennate_X
Cercozoa Gymnochlora stellata
Cercozoa Paulinella chromatophora
Chlorophyta Picochlorum
Chlorophyta Pyramimonas
Chlorophyta Ulva
Chlorophyta Pycnococcaceae_X
Dinophyceae Alexandrium
Dinophyceae Lepidodinium
Dinophyceae Prorocentrum nanum
Prasinodermophyta Prasinoderma coloniale
Microalgal genera with relative abundance ≥ 0.1%

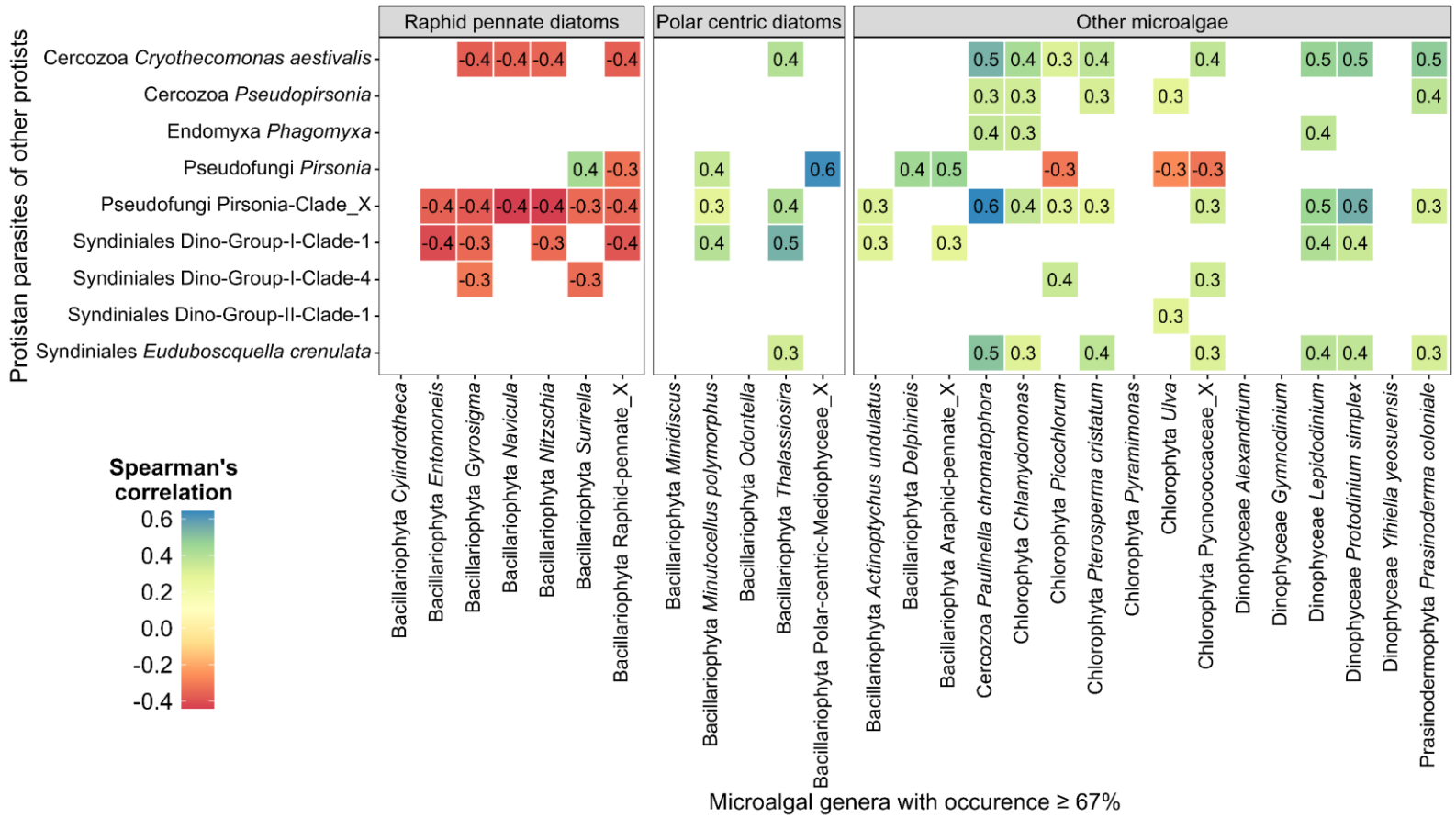


Figure 12. Heatmap showing the strength and direction of relationships between the most frequently detected microalgal genera and protistan parasite lineages (n = 30), averaging across sediment depths and pooling over sub-seasons, in surficial sediments (across the top 1 cm) of a Bay of Fundy mudflat, in 2019–2021. Refer to Table S1A for an overview of samples collected in late spring to early fall 2019, and the summers of 2020 and 2021. Correlation coefficients were omitted for non-significant relationships. Species names are indicated for monospecific genera in sediments. The designation "X" indicates taxonomically unresolved group.

References

Adam-Beyer, N., Deusner, C., Schmidt, M. & Perner, M. (2025). Microbial hydrogen oxidation potential in seasonally hypoxic Baltic Sea sediments. *Front Microbiol*, 16, 1565157.

Addo-Bediako, A., Chown, S.L. & Gaston, K.J. (2000). Thermal tolerance, climatic variability and latitude. *Proc R Soc Lond B Biol Sci*, 267, 739–745.

Adl, S.M., Bass, D., Lane, C.E., Lukeš, J., Schoch, C.L., Smirnov, A., *et al.* (2019). Revisions to the classification, nomenclature, and diversity of eukaryotes. *Journal of Eukaryotic Microbiology*, 66, 4–119.

Aikins, S. & Kikuchi, E. (2002). Grazing pressure by amphipods on microalgae in Gamo Lagoon, Japan. *Mar Ecol Prog Ser*, 245, 171–179.

Amend, A., Burgaud, G., Cunliffe, M., Edgcomb, V.P., Ettinger, C.L., Gutiérrez, M.H., *et al.* (2019). Fungi in the marine environment: Open questions and unsolved problems. *mBio*, 10.

Amir, A., Daniel, M., Navas-Molina, J., Kopylova, E., Morton, J., Xu, Z.Z., *et al.* (2017). Deblur rapidly resolves single-nucleotide community sequence patterns. *American Society for Microbiology*, 2, e00191-16.

Anderson, D.M., Alpermann, T.J., Cembella, A.D., Collos, Y., Masseret, E. & Montresor, M. (2012). The globally distributed genus *Alexandrium*: Multifaceted roles in marine ecosystems and impacts on human health. *Harmful Algae*, 14, 10–35.

Anderson, D.M., Keafer, B.A., Kleindinst, J.L., McGillicuddy, D.J., Martin, J.L., Norton, K., *et al.* (2014). *Alexandrium fundyense* cysts in the Gulf of Maine: Long-term time series of abundance and distribution, and linkages to past and future blooms. *Deep Sea Res 2 Top Stud Oceanogr*, 103, 6–26.

Anderson, D.M., Stock, C.A., Keafer, B.A., Bronzino Nelson, A., McGillicuddy, D.J., Keller, M., *et al.* (2005). *Alexandrium fundyense* cyst dynamics in the Gulf of Maine. *Deep Sea Res 2 Top Stud Oceanogr*, 52, 2522–2542.

Anderson, M.J. (2017). Permutational Multivariate Analysis of Variance (PERMANOVA). *Wiley StatsRef: Statistics Reference Online*, 1–15.

Anderson, M.J., Gorley, R.N. & Clarke, K.R. (2008). *PERMANOVA+ for PRIMER: Guide to software and statistical methods*. PRIMER-E, Plymouth, UK.

Anderson, S.R. & Harvey, E.L. (2020). Temporal variability and ecological interactions of parasitic marine Syndiniales in coastal protist communities. *mSphere*, 5, e00209-20.

Andres, B.A., Smith, P.A., Morrison, R.I.G., Gratto-Trevor, C.L., Brown, S.C. & Friis, C.A. (2012). Population estimates of North American shorebirds, 2012. *Wader Study Group Bulletin*, 119, 178–194.

Archer, A.W. (2013). World's highest tides: Hypertidal coastal systems in North America, South America and Europe. *Sediment Geol*, 284–285, 1–25.

Archibald, J.M., Simpson, A.G.B., Slamovits, C.H., Margulis, L., Melkonian, M., Chapman, D.J., *et al.* (Eds.). (2017). *Handbook of the protists*. 2nd edn. Springer, Cham, Switzerland.

Armbrust, E.V., Berges, J.A., Bowler, C., Green, B.R., Martinez, D., Putnam, N.H., *et al.* (2004). The genome of the diatom *Thalassiosira Pseudonana*: Ecology, evolution, and metabolism. *Science (1979)*, 306, 79–86.

Armitage, A.R. & Fong, P. (2004). Upward cascading effects of nutrients: Shifts in a benthic microalgal community and a negative herbivore response. *Oecologia*, 139, 560–567.

Arsenieff, L., Le Gall, F., Rigaut-Jalabert, F., Mahé, F., Sarno, D., Gouhier, L., *et al.* (2020). Diversity and dynamics of relevant nanoplanktonic diatoms in the Western English Channel. *ISME J*, 14, 1966–1981.

Azovsky, A.I., Saburova, M.A., Chertoprood, E.S. & Polikarpov, I.G. (2005). Selective feeding of littoral harpacticoids on diatom algae: Hungry gourmands? *Mar Biol*, 148, 327–337.

Barnett, A., Méléder, V., Blommaert, L., Lepetit, B., Gaudin, P., Vyverman, W., *et al.* (2015). Growth form defines physiological photoprotective capacity in intertidal benthic diatoms. *ISME J*, 9, 32–45.

Barnett, A., Méléder, V., Dupuy, C. & Lavaud, J. (2020). The vertical migratory rhythm of intertidal microphytobenthos in sediment depends on the light photoperiod, intensity, and spectrum: Evidence for a positive effect of blue wavelengths. *Front Mar Sci*, 7, 212.

Barranguet, C., Kromkamp, J. & Peene, J. (1998). Factors controlling primary production and photosynthetic characteristics of intertidal microphytobenthos. *Mar Ecol Prog Ser*, 173, 117–126.

Barten, R., Kleisman, M., D'Ermo, G., Nijveen, H., Wijffels, R.H. & Barbosa, M.J. (2022). Short-term physiologic response of the green microalga *Picochlorum* sp. (BPE23) to supra-optimal temperature. *Sci Rep*, 12, 3290.

Bates, S.T., Clemente, J.C., Flores, G.E., Walters, W.A., Parfrey, L.W., Knight, R., *et al.* (2013). Global biogeography of highly diverse protistan communities in soil. *ISME J*, 7, 652–659.

Bazin, P., Jouenne, F., Friedl, T., Deton-Cabanillas, A.F., Le Roy, B. & Véron, B. (2014). Phytoplankton diversity and community composition along the estuarine gradient of a temperate macrotidal ecosystem: Combined morphological and molecular approaches. *PLoS One*, 9, e94110.

Benjamini, Y. & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *J R Stat Soc Series B Stat Methodol*, 57, 289–300.

Berdjeb, L., Parada, A., Needham, D.M. & Fuhrman, J.A. (2018). Short-term dynamics and interactions of marine protist communities during the spring–summer transition. *ISME J*, 12, 1907–1917.

Berney, C., Romac, S., Mahé, F., Santini, S., Siano, R. & Bass, D. (2013). Vampires in the oceans: Predatory cercozoan amoebae in marine habitats. *ISME J*, 7, 2387–2399.

Björn, L.O., Papageorgiou, G.C., Blankenship, R.E. & Govindjee. (2009). A viewpoint: Why chlorophyll *a*? *Photosynth Res*, 99, 85–98.

Black, K.S., Tolhurst, T.J., Paterson, D.M. & Hagerthey, S.E. (2002). Working with natural cohesive sediments. *Journal of Hydraulic Engineering*, 128, 2–8.

Blanchard, G.F., Guarini, J., Gros, P. & Richard, P. (1997). Seasonal effect on the relationship between the photosynthetic capacity of intertidal microphytobenthos and temperature. *J Phycol*, 33, 723–728.

Blanchard, G.F., Guarini, J.-M., Provot, L., Richard, P. & Sauriau, P.-G. (2000). Measurement of ingestion rate of *Hydrobia ulvae* (Pennant) on intertidal epipelic microalgae: The effect of mud snail density. *J Exp Mar Biol Ecol*, 255, 247–260.

Blommaert, L., Lavaud, J., Vyverman, W. & Sabbe, K. (2018). Behavioural versus physiological photoprotection in epipelic and epipsammic benthic diatoms. *Eur J Phycol*, 53, 146–155.

Bolyen, E., Rideout, J.R., Dillon, M.R., Bokulich, N.A., Abnet, C.C., Al-Ghalith, G.A., *et al.* (2019). Reproducible, interactive, scalable and extensible microbiome data science using QIIME 2. *Nat Biotechnol*, 37, 852–857.

Bondoc, K.G. V., Heuschele, J., Gillard, J., Vyverman, W. & Pohnert, G. (2016). Selective silicate-directed motility in diatoms. *Nat Commun*, 7, 10540.

Bondoc, K.G. V., Lembke, C., Vyverman, W. & Pohnert, G. (2019). Selective chemoattraction of the benthic diatom *Seminavis robusta* to phosphate but not to inorganic nitrogen sources contributes to biofilm structuring. *Microbiologyopen*, 8, e00694.

Booty, J.M., Underwood, G.J.C., Parris, A., Davies, R.G. & Tolhurst, T.J. (2020). Shorebirds affect ecosystem functioning on an intertidal mudflat. *Front Mar Sci*, 7, 685.

Bourke, M.F., Marriott, P.J., Glud, R.N., Hasler-Sheetal, H., Kamalanathan, M., Beardall, J., *et al.* (2017). Metabolism in anoxic permeable sediments is dominated by eukaryotic dark fermentation. *Nat Geosci*, 10, 30–35.

Boyse, E., Robinson, K.P., Carr, I.M., Valsecchi, E., Beger, M. & Goodman, S.J. (2025). Inferring species interactions from co-occurrence networks with environmental DNA metabarcoding data in a coastal marine food web. *Mol Ecol*, 34, e17701.

Broman, E., Sachpazidou, V., Dopson, M. & Hylander, S. (2017). Diatoms dominate the eukaryotic metatranscriptome during spring in coastal 'dead zone' sediments. *Proceedings of the Royal Society B: Biological Sciences*, 284, 20171617.

Buchfink, B., Reuter, K. & Drost, H.-G. (2021). Sensitive protein alignments at tree-of-life scale using DIAMOND. *Nat Methods*, 18, 366–368.

Burki, F., Sandin, M.M. & Jamy, M. (2021). Diversity and ecology of protists revealed by metabarcoding. *Current Biology*, 31, R1267–R1280.

Butman, B., Aretxabaleta, A.L., Dickhudt, P.J., Dalyander, P.S., Sherwood, C.R., Anderson, D.M., *et al.* (2014). Investigating the importance of sediment resuspension in *Alexandrium fundyense* cyst population dynamics in the Gulf of Maine. *Deep Sea Research Part II: Topical Studies in Oceanography*, 103, 79–95.

Cahoon, L.B. (2002). The role of benthic microalgae in neritic ecosystems. In: *Oceanography and marine biology: An annual review* (eds. Ansell, A., Gibson, R.N. & Barnes, M.). CRC Press, London, UK, pp. 47–86.

Cahoon, L.B., Nearhoof, J.E. & Tilton, C.L. (1999). Sediment grain size effect on benthic microalgal biomass in shallow aquatic ecosystems. *Estuaries*, 22, 735.

Caron, D.A., Alexander, H., Allen, A.E., Archibald, J.M., Armbrust, E.V., Bachy, C., *et al.* (2017). Probing the evolution, ecology and physiology of marine protists using transcriptomics. *Nat Rev Microbiol*, 15, 6–20.

Caron, D.A., Countway, P.D., Jones, A.C., Kim, D.Y. & Schnetzer, A. (2012). Marine protistan diversity. *Ann Rev Mar Sci*, 4, 467–493.

Carpentier, A., Como, S., Dupuy, C., Lefrançois, C. & Feunteun, E. (2014). Feeding ecology of *Liza* spp. in a tidal flat: Evidence of the importance of primary production (biofilm) and associated meiofauna. *J Sea Res*, 92, 86–91.

Carstensen, J., Klais, R. & Cloern, J.E. (2015). Phytoplankton blooms in estuarine and coastal waters: Seasonal patterns and key species. *Estuar Coast Shelf Sci*, 162, 98–109.

Cartaxana, P., Cruz, S., Gameiro, C. & Kühl, M. (2016). Regulation of intertidal microphytobenthos photosynthesis over a diel emersion period is strongly affected by diatom migration patterns. *Front Microbiol*, 7, 872.

Cartaxana, P., Ruivo, M., Hubas, C., Davidson, I., Serôdio, J. & Jesus, B. (2011). Physiological versus behavioral photoprotection in intertidal epipelic and epipsammic benthic diatom communities. *J Exp Mar Biol Ecol*, 405, 120–127.

Cartaxana, P., Vieira, S., Ribeiro, L., Rocha, R.J.M., Cruz, S., Calado, R., *et al.* (2015). Effects of elevated temperature and CO_2 on intertidal microphytobenthos. *BMC Ecol*, 15, 10.

Catlett, D., Peacock, E.E., Taylor Crockford, E., Futrelle, J., Batchelder, S., Stevens, B.L.F., *et al.* (2023). Temperature dependence of parasitoid infection and abundance of a diatom revealed by automated imaging and classification. *Proceedings of the National Academy of Sciences*, 120.

Cavicchioli, R., Ripple, W.J., Timmis, K.N., Azam, F., Bakken, L.R., Baylis, M., *et al.* (2019). Scientists' warning to humanity: Microorganisms and climate change. *Nat Rev Microbiol*, 17, 569–586.

Chen, J., Deng, L., Pang, M., Li, Y., Xu, Z., Zhang, X., *et al.* (2024). Transcriptomic insights into the shift of trophic strategies in mixotrophic dinoflagellate *Lepidodinium* in the warming ocean. *ISME Communications*, 4, ycae087.

Chen, L., Weng, D., Du, C., Wang, J. & Cao, S. (2019). Contribution of frustules and mucilage trails to the mobility of diatom *Navicula* sp. *Sci Rep*, 9, 7342.

Chen, Z.L. & Lee, S.Y. (2022). Tidal flats as a significant carbon reservoir in global coastal ecosystems. *Front Mar Sci*, 9, 900896.

Cheverie, A. V., Hamilton, D.J., Coffin, M.R.S. & Barbeau, M.A. (2014). Effects of shorebird predation and snail abundance on an intertidal mudflat community. *J Sea Res*, 92, 102–114.

Christianen, M.J.A., Middelburg, J.J., Holthuijsen, S.J., Jouta, J., Compton, T.J., van der Heide, T., *et al.* (2017). Benthic primary producers are key to sustain the Wadden Sea food web: Stable carbon isotope analysis at landscape scale. *Ecology*, 98, 1498–1512.

Clark, K., Karsch-Mizrachi, I., Lipman, D.J., Ostell, J. & Sayers, E.W. (2016). GenBank. *Nucleic Acids Res*, 44, D67–D72.

Clarke, K.R. (1993). Non-parametric multivariate analyses of changes in community structure. *Australian Journal of Ecology*, 18, 117–143.

Clarke, K.R. & Gorley, R.N. (2006). *PRIMER v6: User manual/tutorial*. PRIMER-E, Plymouth, UK.

Clarke, L.J., Bestley, S., Bissett, A. & Deagle, B.E. (2019). A globally distributed Syndiniales parasite dominates the Southern Ocean micro-eukaryote community near the sea-ice edge. *ISME J*, 13, 734–737.

Van Colen, C., Underwood, G.J.C., Serôdio, J. & Paterson, D.M. (2014). Ecology of intertidal microbial biofilms: Mechanisms, patterns and future research needs. *J Sea Res*, 92, 2–5.

Comeau, A.M., Douglas, G.M. & Langille, M.G.I. (2017). Microbiome Helper: A custom and streamlined workflow for microbiome research. *mSystems*, 2, e00127-16.

Comeau, A.M., Li, W.K.W., Tremblay, J.-É., Carmack, E.C. & Lovejoy, C. (2011). Arctic Ocean microbial community structure before and after the 2007 record sea ice minimum. *PLoS One*, 6, e27492.

Conley, D.J., Paerl, H.W., Howarth, R.W., Boesch, D.F., Seitzinger, S.P., Havens, K.E., *et al.* (2009). Controlling eutrophication: Nitrogen and phosphorus. *Science (1979)*, 323, 1014–1015.

Connor, M.S., Teal, J.M. & Valiela, I. (1982). The effect of feeding by mud snails, *Ilyanassa obsoleta* (Say), on the structure and metabolism of a laboratory benthic algal community. *J Exp Mar Biol Ecol*, 65, 29–45.

Consalvey, M., Paterson, D.M. & Underwood, G.J.C. (2004). The ups and downs of life in a benthic biofilm: Migration of benthic diatioms. *Diatom Research*, 19, 181–202.

Cook, P.L.M., Revill, A.T., Butler, E.C. V. & Eyre, B.D. (2004). Carbon and nitrogen cycling on intertidal mudflats of a temperate Australian estuary. II. Nitrogen cycling. *Mar Ecol Prog Ser*, 280, 39–54.

Cordier, T., Angeles, I.B., Henry, N., Lejzerowicz, F., Berney, C., Morard, R., *et al.* (2022). Patterns of eukaryotic diversity from the surface to the deep-ocean sediment. *Sci Adv*, 8, 26–28.

Cornell, H. V. & Harrison, S.P. (2014). What are species pools and when are they important? *Annu Rev Ecol Evol Syst*, 45, 45–67.

Cranford, P.J. (1988). Behaviour and ecological importance of a mud snail (*Ilyanassa obsoleta*) population in a temperate macrotidal estuary. *Can J Zool*, 66, 459–466.

Crépeault, O., Otis, C., Pombert, J.F., Turmel, M. & Lemieux, C. (2024). Comparative plastome and mitogenome analyses indicate that the marine prasinophyte green algae *Pycnococcus provasolii* and *Pseudoscourfieldia marina* (Pseudoscourfieldiophyceae class nov., Chlorophyta) represent morphotypes of the same species. *J Phycol*.

Crump, B.C. & Bowen, J.L. (2024). The microbial ecology of estuarine ecosystems. *Ann Rev Mar Sci*, 16, 335–360.

Daborn, G.R., Amos, C.L., Brylinsky, M., Christian, H.A., Drapeau, G., Faas, R.W., *et al.* (1993). An ecological cascade effect: Migratory birds affect stability of intertidal sediments. *Limnol Oceanogr*, 38, 225–231.

Daggers, T.D., Herman, P.M.J. & van der Wal, D. (2020a). Seasonal and spatial variability in patchiness of microphytobenthos on intertidal flats from Sentinel-2 satellite imagery. *Front Mar Sci*, 7, 392.

Daggers, T.D., van Oevelen, D., Herman, P.M.J., Boschker, H.T.S. & van der Wal, D. (2020b). Spatial variability in macrofaunal diet composition and grazing pressure on microphytobenthos in intertidal areas. *Limnol Oceanogr*, 65, 2819–2834.

Dahlin, L.R., Gerritsen, A.T., Henard, C.A., Van Wychen, S., Linger, J.G., Kunde, Y., *et al.* (2019). Development of a high-productivity, halophilic, thermotolerant microalga *Picochlorum renovo*. *Commun Biol*, 2, 388.

Davidson, N.C. & Finlayson, C.M. (2019). Updating global coastal wetland areas presented in Davidson and Finlayson (2018). *Mar Freshw Res*, 70, 1195–1200.

Debroas, D., Domaizon, I., Humbert, J.-F., Jardillier, L., Lepère, C., Oudart, A., *et al.* (2017). Overview of freshwater microbial eukaryotes diversity: A first analysis of publicly available metabarcoding data. *FEMS Microbiol Ecol*, 93, fix023.

Decho, A.W. (2000). Microbial biofilms in intertidal systems: An overview. *Cont Shelf Res*, 20, 1257–1273.

Deiner, K., Bik, H.M., Mächler, E., Seymour, M., Lacoursière-Roussel, A., Altermatt, F., *et al.* (2017). Environmental DNA metabarcoding: Transforming how we survey animal and plant communities. *Mol Ecol*, 26, 5872–5895.

Demers, S., Therriault, J., Bourget, E. & Bah, A. (1987). Resuspension in the shallow sublittoral zone of a macrotidal estuarine environment: Wind influence. *Limnol Oceanogr*, 32, 327–339.

Desplanque, C. & Mossman, D.J. (2004). Tides and their seminal impact on the geology, geography, history, and socio-economics of the Bay of Fundy, eastern Canada. *Atlantic Geology*, 40, 1–130.

D'Hondt, A.-S., Stock, W., Blommaert, L., Moens, T. & Sabbe, K. (2018). Nematodes stimulate biomass accumulation in a multispecies diatom biofilm. *Mar Environ Res*, 140, 78–89.

Doucette, G.J. & Fryxell, G.A. (1985). *Thalassiosira antarctica* (Bacillariophyceae): Vegetative and resting stage ultrastructure of an ice-related marine diatom. *Polar Biol*, 4, 107–112.

Drebes, G., Kühn, S.F., Gmelch, A. & Schnepf, E. (1996). *Cryothecomonas aestivalis* sp. nov., a colourless nanoflagellate feeding on the marine centric diatom *Guinardia delicatula* (Cleve) Hasle. *Helgoländer Meeresuntersuchungen*, 50, 497–515.

Du, G., Yan, H. & Dupuy, C. (2017). Microphytobenthos as an indicator of environmental quality status in intertidal flats: Case study of coastal ecosystem in Pertuis Charentais, France. *Estuar Coast Shelf Sci*, 196, 217–226.

Dyksma, S., Lenk, S., Sawicka, J.E. & Mußmann, M. (2018). Uncultured Gammaproteobacteria and Desulfobacteraceae account for major acetate assimilation in a coastal marine sediment. *Front Microbiol*, 9, 3124.

Edgar, L.A. & Pickett-Heaps, J.D. (1983). The mechanism of diatom locomotion. I. An ultrastructural study of the motility. *Prog Nucl Energy 6 Biol Sci*, 218, 331–343.

Ellegaard, M. & Ribeiro, S. (2018). The long-term persistence of phytoplankton resting stages in aquatic 'seed banks.' *Biological Reviews*, 93, 166–183.

Epstein, S.S., Burkovsky, I. V. & Shiaris, M.P. (1992). Ciliate grazing on bacteria, flagellates, and microalgae in a temperate zone sandy tidal flat: Ingestion rates and food niche partitioning. *J Exp Mar Biol Ecol*, 165, 103–123.

Escalas, A., Hale, L., Voordeckers, J.W., Yang, Y., Firestone, M.K., Alvarez-Cohen, L., *et al.* (2019). Microbial functional diversity: From concepts to applications. *Ecol Evol*, 9, 12000–12016.

Falkowski, P.G., Fenchel, T. & Delong, E.F. (2008). The microbial engines that drive earth's biogeochemical cycles. *Science (1979)*, 320, 1034–1039.

Fawley, M.W., Qin, M. & Yun, Y. (1999). The relationship between *Pseudoscourfieldia marina* and *Pycnococcus provasolii* (Prasinophyceae, Chlorophyta): Evidence from 18S rDNA sequence data. *J Phycol*, 35, 838–843.

Fenchel, T. & Straarup, B.J. (1971). Vertical distribution of photosynthetic pigments and the penetration of light in marine sediments. *Oikos*, 22, 172–182.

Ferrier, M., Martin, J.L. & Rooney-Varga, J.N. (2002). Stimulation of *Alexandrium fundyense* growth by bacterial assemblages from the Bay of Fundy. *J Appl Microbiol*, 92, 706–716.

Fiore-Donno, A.M., Richter-Heitmann, T., Degrune, F., Dumack, K., Regan, K.M., Marhan, S., *et al.* (2019). Functional traits and spatio-temporal structure of a major group of soil protists (Rhizaria: Cercozoa) in a temperate grassland. *Front Microbiol*, 10, 1332.

Foflonker, F., Ananyev, G., Qiu, H., Morrison, A., Palenik, B., Dismukes, G.C., *et al.* (2016). The unexpected extremophile: Tolerance to fluctuating salinity in the green alga *Picochlorum*. *Algal Res*, 16, 465–472.

Foflonker, F., Mollegard, D., Ong, M., Yoon, H.S. & Bhattacharya, D. (2018). Genomic analysis of *Picochlorum* species reveals how microalgae may adapt to variable environments. *Mol Biol Evol*, 35, 2702–2711.

Foflonker, F., Price, D.C., Qiu, H., Palenik, B., Wang, S. & Bhattacharya, D. (2015). Genome of the halotolerant green alga *Picochlorum* sp. reveals strategies for thriving under fluctuating environmental conditions. *Environ Microbiol*, 17, 412–426.

Forster, D., Dunthorn, M., Mahé, F., Dolan, J.R., Audic, S., Bass, D., *et al.* (2016). Benthic protists: The under-charted majority. *FEMS Microbiol Ecol*, 92, fiw120.

Fukai, Y., Matsuno, K., Fujiwara, A., Suzuki, K., Richlen, M.L., Fachon, E., *et al.* (2021). Impact of sea-ice dynamics on the spatial distribution of diatom resting stages in sediments of the Pacific Arctic region. *J Geophys Res Oceans*, 126, e2021JC017223.

Gabr, A., Grossman, A.R. & Bhattacharya, D. (2020). *Paulinella*, a model for understanding plastid primary endosymbiosis. *J Phycol*, 56, 837–843.

Galván, K., Fleeger, J.W. & Fry, B. (2008). Stable isotope addition reveals dietary importance of phytoplankton and microphytobenthos to saltmarsh infauna. *Mar Ecol Prog Ser*, 359, 37–49.

Garner, R.E., Kraemer, S.A., Onana, V.E., Huot, Y., Gregory-Eaves, I. & Walsh, D.A. (2022). Protist diversity and metabolic strategy in freshwater lakes are shaped by trophic state and watershed land use on a continental scale. *mSystems*, 7, e00316-22.

Gates, C., Ananyev, G., Foflonker, F., Bhattacharya, D. & Dismukes, G.C. (2024). Exceptional quantum efficiency powers biomass production in halotolerant algae *Picochlorum* sp. *Photosynth Res*, 162, 439–457.

Gerdol, V. & Hughes, R.G. (1994a). Effect of *Corophium volutator* on the abundance of benthic diatoms, bacteria and sediment stability in two estuaries in southeastern England. *Mar Ecol Prog Ser*, 114, 109–115.

Gerdol, V. & Hughes, R.G. (1994b). Feeding behaviour and diet of *Corophium volutator* in an estuary in southeastern England. *Mar Ecol Prog Ser*, 114, 103–108.

Gerwing, T.G., Allen Gerwing, A.M., Drolet, D., Barbeau, M.A. & Hamilton, D.J. (2015a). Spatiotemporal variation in biotic and abiotic features of eight intertidal mudflats in the upper Bay of Fundy, Canada. *Northeast Nat (Steuben)*, 22, 1.

Gerwing, T.G., Allen Gerwing, A.M., Hamilton, D.J. & Barbeau, M.A. (2015b). Apparent redox potential discontinuity (aRPD) depth as a relative measure of sediment oxygen content and habitat quality. *International Journal of Sediment Research*, 30, 74–80.

Gerwing, T.G., Barbeau, M.A., Hamilton, D.J., Allen Gerwing, A.M., Sinclair, J., Campbell, L., *et al.* (2020). Assessment of sediment penetrability as an integrated in situ measure of intertidal soft-sediment conditions. *Mar Ecol Prog Ser*, 648, 67–78.

Gerwing, T.G., Kim, J.-H., Hamilton, D.J., Barbeau, M.A. & Addison, J.A. (2016). Diet reconstruction using next-generation sequencing increases the known ecosystem usage by a shorebird. *Auk*, 133, 168–177.

Glud, R.N. (2008). Oxygen dynamics of marine sediments. *Marine Biology Research*, 4, 243–289.

Guarini, J.-M., Blanchard, G.F., Bacher, C., Gros, P., Riera, P., Richard, P., *et al.* (1998). Dynamics of spatial patterns of microphytobenthic biomass: Inferences from a geostatistical analysis of two comprehensive surveys in Marennes-Oléron Bay (France). *Mar Ecol Prog Ser*, 166, 131–141.

Guillard, R.R.L., Keller, M.D., O'Kelly, C.J. & Floyd, G.L. (1991). *Pycnococcus provasolii* gen. et sp. nov., a coccoid prasinoxanthin-containing phytoplankter from the Western North Atlantic and Gulf of Mexico. *J Phycol*, 27, 39–47.

Guillou, L., Bachar, D., Audic, S., Bass, D., Berney, C., Bittner, L., *et al.* (2013). The Protist Ribosomal Reference database (PR2): A catalog of unicellular eukaryote Small Sub-Unit rRNA sequences with curated taxonomy. *Nucleic Acids Res*, 41, D597–D604.

Hagerthey, S.E., Defew, E.C. & Paterson, D.M. (2002). Influence of *Corophium volutator* and *Hydrobia ulvae* on intertidal benthic diatom assemblages under different nutrient and temperature regimes. *Mar Ecol Prog Ser*, 245, 47–59.

Hamilton, D.J., Diamond, A.W. & Wells, P.G. (2006). Shorebirds, snails, and the amphipod (*Corophium volutator*) in the upper Bay of Fundy: Top–down vs. bottom–up factors, and the influence of compensatory interactions on mudflat ecology. *Hydrobiologia*, 567, 285–306.

Happey-Wood, C.M. & Jones, P. (1988). Rhythms of vertical migration and motility in intertidal benthic diatoms with particular reference to *Pleurosigma angulatum*. *Diatom Research*, 3, 83–93.

Hargrave, B.T., Prouse, N.J., Phillips, G.A. & Neame, P.A. (1983). Primary production and respiration in pelagic and benthic communities at two intertidal sites in the upper Bay of Fundy. *Canadian Journal of Fisheries and Aquatic Sciences*, 40, s229–s243.

Hasegawa, T., Miyashita, H., Kawachi, M., Ikemoto, H., Kurano, N., Miyachi, S., *et al.* (1996). *Prasinoderma coloniale* gen. et sp. nov., a new pelagic coccoid prasinophyte from the western Pacific Ocean. *Phycologia*, 35, 170–176.

Hattenrath-Lehmann, T.K. & Gobler, C.J. (2017). Identification of unique microbiomes associated with harmful algal blooms caused by *Alexandrium fundyense* and *Dinophysis acuminata*. *Harmful Algae*, 68, 17–30.

Haubois, A.G., Sylvestre, F., Guarini, J.-M., Richard, P. & Blanchard, G.F. (2005). Spatio-temporal structure of the epipelic diatom assemblage from an intertidal mudflat in Marennes-Oléron Bay, France. *Estuar Coast Shelf Sci*, 64, 385–394.

Henley, W.J., Hironaka, J.L., Guillou, L., Buchheim, M.A., Buchheim, J.A., Fawley, M.W., *et al.* (2004). Phylogenetic analysis of the '*Nannochloris*-like' algae and diagnoses of *Picochlorum oklahomensis* gen. et sp. nov. (Trebouxiophyceae, Chlorophyta). *Phycologia*, 43, 641–652.

Herman, P.M.J., Middelburg, J.J., Widdows, J., Lucas, C.H. & Heip, C.H.R. (2000). Stable isotopes as trophic tracers: Combining field sampling and manipulative labelling of food resources for macrobenthos. *Mar Ecol Prog Ser*, 204, 79–92.

Hicklin, P.W. (1987). The migration of shorebirds in the Bay of Fundy. *Wilson Bull*, 99, 540–570.

Hicklin, P.W. & Smith, P.C. (1979). The diets of five species of migrant shorebirds in the Bay of Fundy. *Proc N S Inst Sci*, 29, 483–488.

Hinga, K.R. (2002). Effects of pH on coastal marine phytoplankton. *Mar Ecol Prog Ser*, 238, 281–300.

Hoffman, L. (1999). Marine cyanobacteria in tropical regions: Diversity and ecology. *Eur J Phycol*, 34, 371–379.

Hope, J.A., Paterson, D.M. & Thrush, S.F. (2020). The role of microphytobenthos in soft-sediment ecological networks and their contribution to the delivery of multiple ecosystem services. *Journal of Ecology*, 108, 815–830.

Hoppenrath, M., Beszteri, B., Drebes, G., Halliger, H., Van Beusekom, J.E.E., Janisch, S., *et al.* (2007). *Thalassiosira* species (Bacillariophyceae, Thalassiosirales) in the North Sea at Helgoland (German Bight) and Sylt (North Frisian Wadden Sea) – A first approach to assessing diversity. *Eur J Phycol*, 42, 271–288.

Hubas, C., Sachidhanandam, C., Rybarczyk, H., Lubarsky, H. V., Rigaux, A., Moens, T., *et al.* (2010). Bacterivorous nematodes stimulate microbial growth and exopolymer production in marine sediment microcosms. *Mar Ecol Prog Ser*, 419, 85–94.

Huggins, K., Frenette, J. & Arts, M.T. (2004). Nutritional quality of biofilms with respect to light regime in Lake Saint-Pierre (Québec, Canada). *Freshw Biol*, 49, 945–959.

Hutchins, D.A. & Fu, F. (2017). Microorganisms and ocean global change. *Nat Microbiol*, 2, 17058.

Hyde, K.D., Jones, E.B.G., Leaño, E., Pointing, S.B., Poonyth, A.D. & Vrijmoed, L.L.P. (1998). Role of fungi in marine ecosystems. *Biodivers Conserv*, 7, 1147–1161.

Ishida, K., Nakayama, T. & Hara, Y. (1996). Taxonomic studies on the Chlorarachniophyta. II. Generic delimitation of the chlorarachniophytes and description of *Gymnochlora stellata* gen. et sp. nov. and *Lotharella* gen. nov. *Phycological Res*, 44, 37–45.

Itakura, S., Imai, I. & Itoh, K. (1997). "Seed bank" of coastal planktonic diatoms in bottom sediments of Hiroshima Bay, Seto Inland Sea, Japan. *Mar Biol*, 128, 497–508.

Jacobs, P., Pitarch, J., Kromkamp, J.C. & Philippart, C.J.M. (2021). Assessing biomass and primary production of microphytobenthos in depositional coastal systems using spectral information. *PLoS One*, 16, e0246012.

Jauffrais, T., Drouet, S., Turpin, V., Méléder, V., Jesus, B., Cognie, B., *et al.* (2015). Growth and biochemical composition of a microphytobenthic diatom (*Entomoneis paludosa*) exposed to shorebird (Calidris alpina) droppings. *J Exp Mar Biol Ecol*, 469, 83–92.

Jesus, B., Brotas, V., Ribeiro, L., Mendes, C.R., Cartaxana, P. & Paterson, D.M. (2009). Adaptations of microphytobenthos assemblages to sediment type and tidal position. *Cont Shelf Res*, 29, 1624–1634.

Jesus, B., Jauffrais, T., Trampe, E., Méléder, V., Ribeiro, L., Bernhard, J.M., *et al.* (2023). Microscale imaging sheds light on species-specific strategies for photo-regulation and photo-acclimation of microphytobenthic diatoms. *Environ Microbiol*, 25, 3087–3103.

de Jonge, V.N., de Boer, W.F., de Jong, D.J. & Brauer, V.S. (2012). Long-term mean annual microphytobenthos chlorophyll *a* variation correlates with air temperature. *Mar Ecol Prog Ser*, 468, 43–56.

de Jorge, V.N. & van Beusekom, J.E.E. (1995). Wind- and tide-induced resuspension of sediment and microphytobenthos from tidal flats in the Ems estuary. *Limnol Oceanogr*, 40, 776–778.

Jousset, A., Bienhold, C., Chatzinotas, A., Gallien, L., Gobet, A., Kurm, V., *et al.* (2017). Where less may be more: How the rare biosphere pulls ecosystems strings. *ISME Journal*.

Jumars, P.A., Dorgan, K.M. & Lindsay, S.M. (2015). Diet of worms emended: An update of polychaete feeding guilds. *Ann Rev Mar Sci*, 7, 497–520.

Juneja, A., Ceballos, R. & Murthy, G. (2013). Effects of environmental factors and nutrient availability on the biochemical composition of algae for biofuels production: A review. *Energies (Basel)*, 6, 4607–4638.

Kaczmarska, I., Lovejoy, C., Potvin, M. & Macgillivary, M. (2009). Morphological and molecular characteristics of selected species of *Minidiscus* (Bacillariophyta, Thalassiosiraceae). *Eur J Phycol*, 44, 461–475.

Kalu, E.I. (2020). Spatial and temporal structure of microbial eukaryote communities in two mudflats in the upper Bay of Fundy. Honours thesis. University of New Brunswick, Fredericton.

Kalu, E.I., Reyes-Prieto, A. & Barbeau, M.A. (2023). Community dynamics of microbial eukaryotes in intertidal mudflats in the hypertidal Bay of Fundy. *ISME Communications*, 3.

Kamp, A., De Beer, D., Nitsch, J.L., Lavik, G. & Stief, P. (2011). Diatoms respire nitrate to survive dark and anoxic conditions. *Proc Natl Acad Sci U S A*, 108, 5649–5654.

Kamp, A., Høgslund, S., Risgaard-Petersen, N. & Stief, P. (2015). Nitrate storage and dissimilatory nitrate reduction by eukaryotic microbes. *Front Microbiol*, 6, 1492.

Kamp, A., Stief, P., Knappe, J. & de Beer, D. (2013). Response of the ubiquitous pelagic diatom *Thalassiosira weissflogii* to darkness and anoxia. *PLoS One*, 8, e82605.

Kanaya, G., Takagi, S. & Kikuchi, E. (2008). Dietary contribution of the microphytobenthos to infaunal deposit feeders in an estuarine mudflat in Japan. *Mar Biol*, 155, 543–553.

Kang, C.K., Kim, J.B., Lee, K.S., Kim, J. Bin, Lee, P.Y. & Hong, J.S. (2003). Trophic importance of benthic microalgae to macrozoobenthos in coastal bay systems in Korea: Dual stable C and N isotope analyses. *Mar Ecol Prog Ser*, 259, 79–92.

Keeling, P.J. (2017). Chlorarachniophytes. In: *Handbook of the protists* (eds. Archibald, J.M., Simpson, A.G.B. & Slamovits, C.H.). Springer, Cham, Switzerland, pp. 765–781.

Kelly, D.P. & Wood, A.P. (2006). The chemolithotrophic prokaryotes. In: *The prokaryotes* (eds. Dworkin, M., Falkow, S., Rosenberg, E., Schleifer, K.-H. & Stackebrandt, E.). Springer, New York, USA, pp. 441–456.

Kholssi, R., Lougraimzi, H. & Moreno-Garrido, I. (2023). Effects of global environmental change on microalgal photosynthesis, growth and their distribution. *Mar Environ Res*, 184, 105877.

Knight, R., Vrbanac, A., Taylor, B.C., Aksenov, A., Callewaert, C., Debelius, J., *et al.* (2018). Best practices for analysing microbiomes. *Nat Rev Microbiol*, 16, 410–422.

Koedooder, C., Stock, W., Willems, A., Mangelinckx, S., De Troch, M., Vyverman, W., *et al.* (2019). Diatom-bacteria interactions modulate the composition and productivity of benthic diatom biofilms. *Front Microbiol*, 10, 1255.

Koh, C.-H., Khim, J.S., Araki, H., Yamanishi, H., Mogi, H. & Koga, K. (2006). Tidal resuspension of microphytobenthic chlorophyll *a* in a Nanaura mudflat, Saga, Ariake Sea, Japan: Flood–ebb and spring–neap variations. *Mar Ecol Prog Ser*, 312, 85–100.

Kostecki, C., Roussel, J.M., Desroy, N., Roussel, G., Lanshere, J., Le Bris, H., *et al.* (2012). Trophic ecology of juvenile flatfish in a coastal nursery ground: Contributions of intertidal primary production and freshwater particulate organic matter. *Mar Ecol Prog Ser*, 449, 221–232.

Krasovec, M., Vancaester, E., Rombauts, S., Bucchini, F., Yau, S., Hemon, C., *et al.* (2018). Genome analyses of the microalga *Picochlorum* provide insights into the evolution of thermotolerance in the green lineage. *Genome Biol Evol*, 10, 2347–2365.

Krishnan, A., Likhogrud, M., Cano, M., Edmundson, S., Melanson, J.B., Huesemann, M., *et al.* (2021). *Picochlorum celeri* as a model system for robust outdoor algal growth in seawater. *Sci Rep*, 11, 11649.

Kühl, M., Lassen, C. & Jorgensen, B.B. (1994). Light penetration and light intensity in sandy marine sediments measured with irradiance and scalar irradiance fiber-optic microprobes. *Mar Ecol Prog Ser*, 105, 139–148.

Kuwae, T., Beninger, P.G., Decottignies, P., Mathot, K.J., Lund, D.R. & Elner, R.W. (2008). Biofilm grazing in a higher vertebrate: The Western Sandpiper, Calidris mauri. *Ecology*, 89, 599–606.

Kuwae, T., Miyoshi, E., Hosokawa, S., Ichimi, K., Hosoya, J., Amano, T., *et al.* (2012). Variable and complex food web structures revealed by exploring missing trophic links between birds and biofilm. *Ecol Lett*, 15, 347–356.

Kuypers, M.M.M., Marchant, H.K. & Kartal, B. (2018). The microbial nitrogen-cycling network. *Nat Rev Microbiol*, 16, 263–276.

Leander, B.S. (2020). Predatory protists. *Current Biology*, 30, R510–R516.

Leblanc, K., Quéguiner, B., Diaz, F., Cornet, V., Michel-Rodriguez, M., Durrieu de Madron, X., *et al.* (2018). Nanoplanktonic diatoms are globally overlooked but play a role in spring blooms and carbon export. *Nat Commun*, 9, 953.

Lewis, J., Harris, A.S.D., Jones, K.J. & Edmonds, R.L. (1999). Long-term survival of marine planktonic diatoms and dinoflagellates in stored sediment samples. *J Plankton Res*, 21, 343–354.

Li, L., Wang, S., Wang, H., Sahu, S.K., Marin, B., Li, H., *et al.* (2020). The genome of *Prasinoderma coloniale* unveils the existence of a third phylum within green plants. *Nat Ecol Evol*, 4, 1220–1231.

Lichtenberg, M., Cartaxana, P. & Kühl, M. (2020). Vertical migration optimizes photosynthetic efficiency of motile cyanobacteria in a coastal microbial mat. *Front Mar Sci*, 7, 359.

Lin, Q., Zhuo, W.-H., Wang, X.-W., Chen, C.-P., Gao, Y.-H. & Liang, J.-R. (2018). Effects of fundamental nutrient stresses on the lipid accumulation profiles in two diatom species *Thalassiosira weissflogii* and *Chaetoceros muelleri*. *Bioprocess Biosyst Eng*, 41, 1213–1224.

Lin, W.J., Wu, J. & Lin, H.J. (2020). Contribution of unvegetated tidal flats to coastal carbon flux. *Glob Chang Biol*, 26, 3443–3454.

Liu, K., Liu, S., Cui, Z., Zhao, Y. & Chen, N. (2024). Rich diversity and active spatial–temporal dynamics of *Thalassiosira* species revealed by time-series metabarcoding analysis. *ISME Communications*, 4, ycad009.

Liu, K., Ng, H., Zhang, S. & Liu, H. (2021). Effects of temperature on a mixotrophic dinoflagellate (*Lepidodinium* sp.) under different nutritional strategies. *Mar Ecol Prog Ser*, 678, 37–49.

Liu, K., Ng, H.Y.-T., Gao, Z. & Liu, H. (2022). Selective feeding of a mixotrophic dinoflagellate (*Lepidodinium* sp.) in response to experimental warming and inorganic nutrient imbalance. *Front Microbiol*, 13, 805306.

Louca, S., Parfrey, L.W. & Doebeli, M. (2016). Decoupling function and taxonomy in the global ocean microbiome. *Science (1979)*, 353, 1272–1277.

Lundholm, N., Hansen, P.J. & Kotaki, Y. (2004). Effect of pH on growth and domoic acid production by potentially toxic diatoms of the genera *Pseudo-nitzschia* and *Nitzschia*. *Mar Ecol Prog Ser*, 273, 1–15.

Lundkvist, M., Grue, M., Friend, P.L. & Flindt, M.R. (2007). The relative contributions of physical and microbiological factors to cohesive sediment stability. *Cont Shelf Res*, 27, 1143–1152.

MacIntyre, H.L., Geider, R.J. & Miller, D.C. (1996). Microphytobenthos: The ecological role of the "secret garden" of unvegetated, shallow-water marine habitats. I. Distribution, abundance and primary production. *Estuaries*, 19, 186–201.

Macorano, L. & Nowack, E.C.M. (2021). *Paulinella chromatophora*. *Current Biology*, 31, R1024–R1026.

Madsen, E.L. (2011). Microorganisms and their roles in fundamental biogeochemical cycles. *Curr Opin Biotechnol*, 22, 456–464.

Maillet, D. & Weber, J.-M. (2006). Performance-enhancing role of dietary fatty acids in a long-distance migrant shorebird: The semipalmated sandpiper. *Journal of Experimental Biology*, 209, 2686–2695.

Maillet, D. & Weber, J.-M. (2007). Relationship between n-3 PUFA content and energy metabolism in the flight muscles of a migrating shorebird: Evidence for natural doping. *Journal of Experimental Biology*, 210, 413–420.

Malviya, S., Scalco, E., Audic, S., Vincent, F., Veluchamy, A., Poulain, J., *et al.* (2016). Insights into global diatom distribution and diversity in the world's ocean. *Proc Natl Acad Sci U S A*, 113, E1516–E1525.

Martin, J.L., LeGresley, M.M. & Hanke, A.R. (2014). Thirty years – *Alexandrium fundyense* cyst, bloom dynamics and shellfish toxicity in the Bay of Fundy, eastern Canada. *Deep Sea Research Part II: Topical Studies in Oceanography*, 103, 27–39.

Martinez-Garcia, M., Brazel, D., Poulton, N.J., Swan, B.K., Gomez, M.L., Masland, D., *et al.* (2012). Unveiling in situ interactions between marine protists and bacteria through single cell sequencing. *ISME J*, 6, 703–707.

Massana, R. (2024). The nitroplast: A nitrogen-fixing organelle. *Science (1979)*, 384, 160–161.

Massana, R., Gobet, A., Audic, S., Bass, D., Bittner, L., Boutte, C., *et al.* (2015). Marine protist diversity in European coastal waters and sediments as revealed by high-throughput sequencing. *Environ Microbiol*, 17, 4035–4049.

Mathot, K.J., Lund, D.R. & Elner, R.W. (2010). Sediment in stomach contents of Western Sandpipers and Dunlin provide evidence of biofilm feeding. *Waterbirds*, 33, 300–306.

Mathot, K.J., Piersma, T. & Elner, R.W. (2018). Shorebirds as integrators and indicators of mudflat ecology. In: *Mudflat Ecology. Aquatic Ecology Series* (ed. Beninger, P.G.). Springer, Cham, Switzerland, pp. 309–338.

Matrai, P., Thompson, B. & Keller, M. (2005). Circannual excystment of resting cysts of *Alexandrium* spp. from eastern Gulf of Maine populations. *Deep Sea Research Part II: Topical Studies in Oceanography*, 52, 2560–2568.

McGillicuddy, D.J., Anderson, D.M., Lynch, D.R. & Townsend, D.W. (2005). Mechanisms regulating large-scale seasonal fluctuations in *Alexandrium fundyense* populations in the Gulf of Maine: Results from a physical–biological model. *Deep Sea Research Part II: Topical Studies in Oceanography*, 52, 2698–2714.

McQuoid, M.R., Godhe, A. & Nordberg, K. (2002). Viability of phytoplankton resting stages in the sediments of a coastal Swedish fjord. *Eur J Phycol*, 37, 191–201.

McQuoid, M.R. & Hobson, L.A. (1996). Diatom resting stages. *J Phycol*, 32, 889–902.

Méléder, V., Rincé, Y., Barillé, L., Gaudin, P. & Rosa, P. (2007). Spatiotemporal changes in microphytobenthos assemblages in a macrotidal flat (Bourgneuf Bay, France). *J Phycol*, 43, 1177–1190.

Middelburg, J.J. (2018). Reviews and syntheses: To the bottom of carbon processing at the seafloor. *Biogeosciences*, 15, 413–427.

Middelburg, J.J., Barranguet, C., Boschker, H.T.S., Herman, P.M.J., Moens, T. & Heip, C.H.R. (2000). The fate of intertidal microphytobenthos carbon: An in situ ^{13}C-labeling study. *Limnol Oceanogr*, 45, 1224–1234.

Miller, D.C., Geider, R.J. & MacIntyre, H.L. (1996). Microphytobenthos: The ecological role of the "secret garden" of unvegetated, shallow-water marine habitats. II. Role in sediment stability and shallow-water food webs. *Estuaries*, 19, 202–212.

Mitbavkar, S. & Anil, A. (2002). Diatoms of the microphytobenthic community: Population structure in a tropical intertidal sand flat. *Mar Biol*, 140, 41–57.

Mitbavkar, S. & Anil, A.C. (2006). Diatoms of the microphytobenthic community in a tropical intertidal sand flat influenced by monsoons: Spatial and temporal variations. *Mar Biol*, 148, 693–709.

Moens, T., Luyten, C., Middelburg, J.J., Herman, P.M.J. & Vincx, M. (2002). Tracing organic matter sources of estuarine tidal flat nematodes with stable carbon isotopes. *Mar Ecol Prog Ser*, 234, 127–137.

Moestrup, Ø. & Throndsen, J. (1988). Light and electron microscopical studies on *Pseudoscourfieldia marina*, a primitive scaly green flagellate (Prasinophyceae) with posterior flagella. *Canadian Journal of Botany*, 66, 1415–1434.

Mogle, M.J. (2021). Dynamics of intertidal biofilm in relation to Semipalmated Sandpiper (*Calidris pusilla*) migratory stopover and assessments of benthic invertebrate and biofilm nutritional content in the upper Bay of Fundy, Canada. Master's thesis. Mount Allison University, Sackville.

Montagna, P.A., Blanchard, G.F. & Dinet, A. (1995). Effect of production and biomass of intertidal microphytobenthos on meiofaunal grazing rates. *J Exp Mar Biol Ecol*, 185, 149–165.

Montresor, M., Di Prisco, C., Sarno, D., Margiotta, F. & Zingone, A. (2013). Diversity and germination patterns of diatom resting stages at a coastal Mediterranean site. *Mar Ecol Prog Ser*, 484, 79–95.

Morales, M., Aflalo, C. & Bernard, O. (2021). Microalgal lipids: A review of lipids potential and quantification for 95 phytoplankton species. *Biomass Bioenergy*, 150, 106108.

Murray, N.J., Phinn, S.R., DeWitt, M., Ferrari, R., Johnston, R., Lyons, M.B., *et al.* (2019). The global distribution and trajectory of tidal flats. *Nature*, 565, 222–225.

Neima, S.G. (2016). Stopover ecology of semipalmated sandpipers (*Calidris pusilla*) during fall migration through the upper Bay of Fundy, Canada. Master's thesis. Mount Allison University, Sackville.

Ng, W.H.A., Liu, H. & Zhang, S. (2017). Diel variation of grazing of the dinoflagellate *Lepidodinium* sp. and ciliate *Euplotes* sp. on algal prey: The effect of prey cell properties. *J Plankton Res*, 39, 450–462.

Nguyen, N.H., Song, Z., Bates, S.T., Branco, S., Tedersoo, L., Menke, J., *et al.* (2016). FUNGuild: An open annotation tool for parsing fungal community datasets by ecological guild. *Fungal Ecol*, 20, 241–248.

Nowack, E.C.M. & Grossman, A.R. (2012). Trafficking of protein into the recently established photosynthetic organelles of *Paulinella chromatophora*. *Proceedings of the National Academy of Sciences*, 109, 5340–5345.

van Oevelen, D., Soetaert, K., Middelburg, J.J., Herman, P.M.J., Moodley, L., Hamels, I., *et al.* (2006). Carbon flows through a benthic food web: Integrating biomass, isotope and tracer data. *J Mar Res*, 64, 453–482.

Oliverio, A.M., Geisen, S., Delgado-Baquerizo, M., Maestre, F.T., Turner, B.L. & Fierer, N. (2020). The global-scale distributions of soil protists and their contributions to belowground systems. *Sci Adv*, 6, eaax8787.

Orr, R.J.S., Stüken, A., Rundberget, T., Eikrem, W. & Jakobsen, K.S. (2011). Improved phylogenetic resolution of toxic and non-toxic *Alexandrium* strains using a concatenated rDNA approach. *Harmful Algae*, 10, 676–688.

Pan, J., Cuadrado, D.G. & Bournod, C.N. (2017). Diatom-driven recolonization of microbial mat-dominated siliciclastic tidal flat sediments. *FEMS Microbiol Ecol*, 93, fix111.

Park, J.S., Jung, S.W., Lee, S.D., Yun, S.M. & Lee, J.H. (2016). Species diversity of the genus *Thalassiosira* (Thalassiosirales, Bacillariophyta) in South Korea and its biogeographical distribution in the world. *Phycologia*, 55, 403–423.

Parker, M.S., Mock, T. & Armbrust, E.V. (2008). Genomic insights into marine microalgae. *Annu Rev Genet*, 42, 619–645.

Pascal, P.-Y., Dupuy, C., Richard, P., Rzeznik-Orignac, J. & Niquil, N. (2008). Bacterivory of a mudflat nematode community under different environmental conditions. *Mar Biol*, 154, 671–682.

Pascoal, F., Costa, R. & Magalhães, C. (2021). The microbial rare biosphere: Current concepts, methods and ecological principles. *FEMS Microbiol Ecol*, 97, fiaa227.

Pawlowski, J., Christen, R., Lecroq, B., Bachar, D., Shahbazkia, H.R., Amaral-Zettler, L., *et al.* (2011). Eukaryotic richness in the abyss: Insights from pyrotag sequencing. *PLoS One*, 6, e18169.

Peacock, E.E., Olson, R.J. & Sosik, H.M. (2014). Parasitic infection of the diatom *Guinardia delicatula*, a recurrent and ecologically important phenomenon on the New England Shelf. *Mar Ecol Prog Ser*, 503, 1–10.

Peer, D.L., Linkletter, L.E. & Hicklin, P.W. (1986). Life history and reproductive biology of *Corophium volutator* (Crustacea: Amphipoda) and the influence of shorebird predation on population structure in Chignecto Bay, Bay of Fundy, Canada. *Netherlands Journal of Sea Research*, 20, 359–373.

Pelusi, A., Ambrosino, L., Miralto, M., Chiusano, M.L., Rogato, A., Ferrante, M.I., *et al.* (2023). Gene expression during the formation of resting spores induced by nitrogen starvation in the marine diatom *Chaetoceros socialis*. *BMC Genomics*, 24, 106.

Petro, C., Starnawski, P., Schramm, A. & Kjeldsen, K.U. (2017). Microbial community assembly in marine sediments. *Aquatic Microbial Ecology*, 79, 177–195.

Pierella Karlusich, J.J., Ibarbalz, F.M. & Bowler, C. (2020). Phytoplankton in the Tara Ocean. *Ann Rev Mar Sci*, 12, 233–265.

Pinckney, J., Piceno, Y. & Lovell, C.R. (1994). Short-term changes in the vertical distribution of benthic microalgal biomass in intertidal muddy sediments. *Diatom Research*, 9, 143–153.

Pinckney, J.L., Carman, K.R., Lumsden, S.E. & Hymel, S.N. (2003). Microalgal-meiofaunal trophic relationships in muddy intertidal estuarine sediments. *Aquatic Microbial Ecology*, 31, 99–108.

Piredda, R., Claverie, J.M., Decelle, J., de Vargas, C., Dunthorn, M., Edvardsen, B., *et al.* (2018). Diatom diversity through HTS-metabarcoding in coastal European seas. *Sci Rep*, 8, 18059.

Plugge, C.M., Zhang, W., Scholten, J.C.M. & Stams, A.J.M. (2011). Metabolic flexibility of sulfate-reducing bacteria. *Front Microbiol*, 2, 81.

Post, D.M. (2002). Using stable isotopes to estimate trophic position: Models, methods, and assumptions. *Ecology*, 83, 703–718.

Poulsen, N.C., Spector, I., Spurck, T.P., Schultz, T.F. & Wetherbee, R. (1999). Diatom gliding is the result of an actin-myosin motility system. *Cell Motil Cytoskeleton*, 44, 23–33.

Prins, A., Deleris, P., Hubas, C. & Jesus, B. (2020). Effect of light intensity and light quality on diatom behavioral and physiological photoprotection. *Front Mar Sci*, 7, 203.

Prokina, K.I., Yubuki, N., Tikhonenkov, D. V., Ciobanu, M.C., López-García, P. & Moreira, D. (2024). Refurbishing the marine parasitoid order Pirsoniales with newly (re)described marine and freshwater free-living predators. *Journal of Eukaryotic Microbiology*, 71, e13061.

Prosser, J.I. (2015). Dispersing misconceptions and identifying opportunities for the use of “omics” in soil microbial ecology. *Nat Rev Microbiol*, 13, 439–446.

Prouse, N.J., Gordon Jr., D.C., Hargrave, B.T., Bird, C.J. & McLachlan, J. (1984). Primary production: Organic matter supply to ecosystems in the Bay of Fundy. In: *Update on the marine environmental consequences of tidal power development in the upper reaches of the Bay of Fundy* (eds. Gordon Jr., D.C. & Dadswell, M.J.). Canadian Technical Report of Fisheries and Aquatic Sciences No. 1256, Dartmouth, Canada, pp. 65–95.

Quast, C., Pruesse, E., Yilmaz, P., Gerken, J., Schweer, T., Yarza, P., *et al.* (2013). The SILVA ribosomal RNA gene database project: Improved data processing and web-based tools. *Nucleic Acids Res*, 41, D590–D596.

Quinn, J.T. & Hamilton, D.J. (2012). Variation in diet of Semipalmated Sandpipers *Calidris pusilla* during stopover in the upper Bay of Fundy, Canada. *Can J Zool*, 90, 1181–1190.

Rabalais, N.N., Turner, R.E., Díaz, R.J. & Justić, D. (2009). Global change and eutrophication of coastal waters. *ICES Journal of Marine Science*, 66, 1528–1537.

Reyes-Prieto, A., Weber, A.P.M. & Bhattacharya, D. (2007). The origin and establishment of the plastid in algae and plants. *Annu Rev Genet*, 41, 147–168.

Ribeiro, L., Brotas, V., Mascarell, G. & Couté, A. (2003). Taxonomic survey of the microphytobenthic communities of two Tagus estuary mudflats. *Acta Oecologica*, 24, S117–S123.

Ribeiro, L., Brotas, V., Rincé, Y. & Jesus, B. (2013). Structure and diversity of intertidal benthic diatom assemblages in contrasting shores: A case study from the Tagus estuary. *J Phycol*, 49, 258–270.

Richards, T.A., Jones, M.D.M., Leonard, G. & Bass, D. (2012). Marine fungi: Their ecology and molecular diversity. *Ann Rev Mar Sci*, 4, 495–522.

Riera, P., Richard, P., Grémare, A. & Blanchard, G. (1996). Food source of intertidal nematodes in the Bay of Marennes-Oléron (France), as determined by dual stable isotope analysis. *Mar Ecol Prog Ser*, 142, 303–309.

Risgaard-Petersen, N., Nicolaisen, M.H., Revsbech, N.P. & Lomstein, B.A. (2004). Competition between ammonia-oxidizing bacteria and benthic microalgae. *Appl Environ Microbiol*, 70, 5528–5537.

Robar, N.D.P. & Hamilton, D.J. (2007). A method for estimating habitat use by shorebirds using footprints. *Waterbirds*, 30, 116–120.

Robitaille, M.C. (2021). The effects of top-down and bottom-up forces on the nutritional quality of biofilm at an intertidal mudflat in the Bay of Fundy. Honours thesis. Mount Allison University, Sackville.

Rosenberg, K. V., Dokter, A.M., Blancher, P.J., Sauer, J.R., Smith, A.C., Smith, P.A., *et al.* (2019). Decline of the North American avifauna. *Science (1979)*, 366, 120–124.

Roux, P., Schapira, M., Mertens, K.N., André, C., Terre-Terrillon, A., Schmitt, A., *et al.* (2023). When phytoplankton do not bloom: The case of the dinoflagellate *Lepidodinium chlorophorum* in southern Brittany (France) assessed by environmental DNA. *Prog Oceanogr*, 212, 102999.

Ruppert, K.M., Kline, R.J. & Rahman, M.S. (2019). Past, present, and future perspectives of environmental DNA (eDNA) metabarcoding: A systematic review in methods, monitoring, and applications of global eDNA. *Glob Ecol Conserv*, 17, e00547.

Sahan, E., Sabbe, K., Creach, V., Hernandez-Raquet, G., Vyverman, W., Stal, L.J., *et al.* (2007). Community structure and seasonal dynamics of diatom biofilms and associated grazers in intertidal mudflats. *Aquatic Microbial Ecology*, 47, 253–266.

Salazar, G. & Sunagawa, S. (2017). Marine microbial diversity. *Current Biology*, 27, R489–R494.

Sanyal, A., Larsson, J., van Wirdum, F., Andrén, T., Moros, M., Lönn, M., *et al.* (2022). Not dead yet: Diatom resting spores can survive in nature for several millennia. *Am J Bot*, 109, 67–82.

Savelli, R., Dupuy, C., Barillé, L., Lerouxel, A., Guizien, K., Philippe, A., *et al.* (2018). On biotic and abiotic drivers of the microphytobenthos seasonal cycle in a temperate intertidal mudflat: A modelling study. *Biogeosciences*, 15, 7243–7271.

Schnurr, P.J., Drever, M.C., Elner, R.W., Harper, J. & Arts, M.T. (2020). Peak abundance of fatty acids from intertidal biofilm in relation to the breeding migration of shorebirds. *Front Mar Sci*, 7, 63.

Schnurr, P.J., Drever, M.C., Kling, H.J., Elner, R.W. & Arts, M.T. (2019). Seasonal changes in fatty acid composition of estuarine intertidal biofilm: Implications for western sandpiper migration. *Estuar Coast Shelf Sci*, 224, 94–107.

Scholz, B. & Liebezeit, G. (2012a). Growth responses of 25 benthic marine Wadden Sea diatoms isolated from the Solthörn tidal flat (southern North Sea) in relation to varying culture conditions. *Diatom Research*, 27, 65–73.

Scholz, B. & Liebezeit, G. (2012b). Microphytobenthic dynamics in a Wadden Sea intertidal flat – Part I: Seasonal and spatial variation of diatom communities in relation to macronutrient supply. *Eur J Phycol*, 47, 105–119.

Scholz, B. & Liebezeit, G. (2012c). Microphytobenthic dynamics in a Wadden Sea intertidal flat – Part II: Seasonal and spatial variability of non-diatom community components in relation to abiotic parameters. *Eur J Phycol*, 47, 120–137.

Scholz, B. & Liebezeit, G. (2012d). Screening for competition effects and allelochemicals in benthic marine diatoms and cyanobacteria isolated from an intertidal flat (southern North Sea). *Phycologia*, 51, 432–450.

Scholz, B. & Liebezeit, G. (2013). Biochemical characterisation and fatty acid profiles of 25 benthic marine diatoms isolated from the Solthörn tidal flat (southern North Sea). *J Appl Phycol*, 25, 453–465.

Schratzberger, M., Holterman, M., van Oevelen, D. & Helder, J. (2019). A worm's world: Ecological flexibility pays off for free-living nematodes in sediments and soils. *Bioscience*, 69, 867–876.

Schwinghamer, P., Tan, F.C. & Gordon Jr., D.C. (1983). Stable carbon isotope studies on the Pecks Cove mudflat ecosystem in the Cumberland Basin, Bay of Fundy. *Canadian Journal of Fisheries and Aquatic Sciences*, 40, s262–s272.

Searle, S.R., Casella, G. & McCulloch, C.E. (1992). *Variance components*. Wiley Series in Probability and Statistics. Wiley, New York, USA.

Seckbach, J. & Oren, A. (2007). Oxygenic photosynthetic microorganisms in extreme environments: Possibilities and limitations. In: *Algae and cyanobacteria in extreme environments. Cellular origin, life in extreme habitats and astrobiology*, Cellular Origin, Life in Extreme Habitats and Astrobiology (ed. Seckbach, J.). Springer, Dordrecht, Netherlands, pp. 5–25.

Shahraki, M., Fry, B., Krumme, U. & Rixen, T. (2014). Microphytobenthos sustain fish food webs in intertidal arid habitats: A comparison between mangrove-lined and un-vegetated creeks in the Persian Gulf. *Estuar Coast Shelf Sci*, 149, 203–212.

Sheahan, D.C., Arnold, E.E., Hamilton, D.J., Grecian, L.A. & Barbeau, M.A. (2009). Spatial and temporal variation in the population dynamics of the intertidal amphipod *Corophium Volutator* in the upper Bay of Fundy, Canada. *Journal of Crustacean Biology*, 29, 491–506.

Sherr, E.B. & Sherr, B.F. (2002). Significance of predation by protists in aquatic microbial food webs. *Antonie Van Leeuwenhoek*, 81, 293–308.

Shiozaki, T., Hirose, Y., Hamasaki, K., Kaneko, R., Ishikawa, K. & Harada, N. (2019). Eukaryotic phytoplankton contributing to a seasonal bloom and carbon export revealed by tracking sequence variants in the western North Pacific. *Front Microbiol*, 10, 2722.

Shoaf, W.T. & Lium, B.W. (1976). Improved extraction of chlorophyll *a* and *b* from algae using dimethyl sulfoxide. *Limnol Oceanogr*, 21, 926–928.

Sieracki, M.E., Poulton, N.J., Jaillon, O., Wincker, P., de Vargas, C., Rubinat-Ripoll, L., *et al.* (2019). Single cell genomics yields a wide diversity of small planktonic protists across major ocean ecosystems. *Sci Rep*, 9, 6025.

Singer, D., Seppey, C.V.W., Lentendu, G., Dunthorn, M., Bass, D., Belbahri, L., *et al.* (2021). Protist taxonomic and functional diversity in soil, freshwater and marine ecosystems. *Environ Int*, 146, 106262.

Smith, D., Hughes, R.G. & Cox, E.J. (1996). Predation of epipelic diatoms by the amphipod *Corophium volutator* and the polychaete *Nereis diversicolor*. *Mar Ecol Prog Ser*, 145, 53–61.

Smith, P.A., Smith, A.C., Andres, B., Francis, C.M., Harrington, B., Friis, C., *et al.* (2023). Accelerating declines of North America's shorebirds signal the need for urgent conservation action. *Ornithological Applications*, 125, 1–14.

Stal, L.J. (2003). Microphytobenthos, their extracellular polymeric substances, and the morphogenesis of intertidal sediments. *Geomicrobiol J*, 20, 463–478.

Stenow, R., Robertson, E.K., Kourtchenko, O., Whitehouse, M.J., Pinder, M.I.M., Benvenuto, G., *et al.* (2024). Resting cells of *Skeletonema marinoi* assimilate organic compounds and respire by dissimilatory nitrate reduction to ammonium in dark, anoxic conditions. *Environ Microbiol*, 26, e16625.

Sterrenburg, F.A.S. (1994). Studies on the Genera *Gyrosigma* and *Pleurosigma* (Bacillariophyceae). *Proceedings of the Academy of Natural Sciences of Philadelphia*, 145, 217–236.

Sterrenburg, F.A.S. & Underwood, G.J.C. (1997). Studies on the genera *Gyrosigma* and *Pleurosigma* (Bacillariophyceae). The marine "*Gyrosigma spenceri*" records: Gyrosigma limosum Sterrenburg et Underwood nov. sp. *Proc Acad Nat Sci Phila*, 148, 165–169.

Sugie, K. & Kuma, K. (2008). Resting spore formation in the marine diatom *Thalassiosira nordenskioeldii* under iron- and nitrogen-limited conditions. *J Plankton Res*, 30, 1245–1255.

Takai, N., Mishima, Y., Yorozu, A. & Hoshika, A. (2002). Carbon sources for demersal fish in the western Seto Inland Sea, Japan, examined by δ13C and δ15N analyses. *Limnol Oceanogr*, 47, 730–741.

Tamames, J. & Puente-Sánchez, F. (2019). SqueezeMeta, a highly portable, fully automatic metagenomic analysis pipeline. *Front Microbiol*, 9.

Taraldsvik, M. & Myklestad, S. (2000). The effect of pH on growth rate, biochemical composition and extracellular carbohydrate production of the marine diatom *Skeletonema costatum*. *Eur J Phycol*, 35, 189–194.

Taş, N., de Jong, A.E.E., Li, Y., Trubl, G., Xue, Y. & Dove, N.C. (2021). Metagenomic tools in microbial ecology research. *Curr Opin Biotechnol*, 67, 184–191.

Thaler, M. & Kaczmarska, I. (2009). *Gyrosigma orbitum* sp. nov. (Bacillariophyta) from a salt marsh in the Bay of Fundy, eastern Canada. *Botanica Marina*, 52, 60–68.

Thiel, V., Tank, M. & Bryant, D.A. (2018). Diversity of chlorophototrophic bacteria revealed in the omics era. *Annu Rev Plant Biol*, 69, 21–49.

Thornton, D.C.O., Dong, L.F., Underwood, G.J.C. & Nedwell, D.B. (2002). Factors affecting microphytobenthic biomass, species composition and production in the Colne Estuary (UK). *Aquatic Microbial Ecology*, 27, 285–300.

Tillmann, U. (2004). Interactions between planktonic microalgae and protozoan grazers. *Journal of Eukaryotic Microbiology*, 51, 156–168.

Townsend, D.W., Pettigrew, N.R. & Thomas, A.C. (2005). On the nature of *Alexandrium fundyense* blooms in the Gulf of Maine. *Deep Sea Research Part II: Topical Studies in Oceanography*, 52, 2603–2630.

Trites, M., Kaczmarska, I., Ehrman, J.M., Hicklin, P.W. & Ollerhead, J. (2005). Diatoms from two macro-tidal mudflats in Chignecto Bay, upper Bay of Fundy, New Brunswick, Canada. *Hydrobiologia*, 544, 299–319.

Trobajo, R., Rovira, L., Ector, L., Wetzel, C.E., Kelly, M. & Mann, D.G. (2013). Morphology and identity of some ecologically important small *Nitzschia* species. *Diatom Research*, 28, 37–59.

De Troch, M., Houthoofd, L., Chepurnov, V. & Vanreusel, A. (2006). Does sediment grain size affect diatom grazing by harpacticoid copepods? *Mar Environ Res*, 61, 265–277.

Tweedley, J.R., Warwick, R.M. & Potter, I.C. (2016). The contrasting ecology of temperate macrotidal and microtidal estuaries. *Oceanography and Marine Biology: An Annual Review*, 54, 73–171.

Ubertini, M., Lefebvre, S., Gangnery, A., Grangeré, K., Le Gendre, R. & Orvain, F. (2012). Spatial variability of benthic-pelagic coupling in an estuary ecosystem: Consequences for microphytobenthos resuspension phenomenon. *PLoS One*, 7, e44155.

Underwood, G., Phillips, J. & Saunders, K. (1998). Distribution of estuarine benthic diatom species along salinity and nutrient gradients. *Eur J Phycol*, 33, 173–183.

Underwood, G. & Provot, L. (2000). Determining the environmental preferences of four estuarine epipelic diatom taxa: Growth across a range of salinity, nitrate and ammonium conditions. *Eur J Phycol*, 35, 173–182.

Underwood, G.J.C. (1994). Seasonal and spatial variation in epipelic diatom assemblages in the Severn Estuary. *Diatom Research*, 9, 451–472.

Underwood, G.J.C. & Kromkamp, J. (1999). Primary production by phytoplankton and microphytobenthos in estuaries. *Adv Ecol Res*, 29, 93–153.

Underwood, G.J.C. & Paterson, D.M. (1993). Recovery of intertidal benthic diatoms after biocide treatment and associated sediment dynamics. *Journal of the Marine Biological Association of the United Kingdom*, 73, 25–45.

Underwood, G.J.C. & Paterson, D.M. (2003). The importance of extracellular carbohydrate production by marine epipelic diatoms. *Adv Bot Res*, 40, 183–240.

Vanbaelinghem, L. (2021). The response of mudflat microphytobenthos to bottom-up and top-down factors in the Bay of Fundy. Honours thesis. Mount Allison University, Sackville.

de Vargas, C., Audic, S., Henry, N., Decelle, J., Mahé, F., Logares, R., *et al.* (2015). Eukaryotic plankton diversity in the sunlit ocean. *Science (1979)*, 348, 1261605.

Velo-Suárez, L., Brosnahan, M.L., Anderson, D.M. & McGillicuddy, D.J. (2013). A quantitative assessment of the role of the parasite *Amoebophrya* in the termination of *Alexandrium fundyense* blooms within a small coastal embayment. *PLoS One*, 8, e81150.

Vieira, S., Ribeiro, L., Marques da Silva, J. & Cartaxana, P. (2013). Effects of short-term changes in sediment temperature on the photosynthesis of two intertidal microphytobenthos communities. *Estuar Coast Shelf Sci*, 119, 112–118.

Virta, L., Gammal, J., Järnström, M., Bernard, G., Soininen, J., Norkko, J., *et al.* (2019). The diversity of benthic diatoms affects ecosystem productivity in heterogeneous coastal environments. *Ecology*, 100, e02765.

Vogt, J.C., Albach, D.C. & Palinska, K.A. (2018). Cyanobacteria of the Wadden Sea: Seasonality and sediment influence on community composition. *Hydrobiologia*, 811, 103–117.

van der Wal, D., Wielemaker-van den Dool, A. & Herman, P.M.J. (2010). Spatial synchrony in intertidal benthic algal biomass in temperate coastal and estuarine ecosystems. *Ecosystems*, 13, 338–351.

Wang, G., Huang, L., Zhuang, S., Han, F., Huang, Q., Hao, M., *et al.* (2024). Resting cell formation in the marine diatom *Thalassiosira pseudonana*. *New Phytologist*, 243, 1347–1360.

Watanabe, M.M., Suda, S., Inouya, I., Sawaguchi, T. & Chihara, M. (1990). *Lepidodinium viride* gen. et sp. nov. (Gymnodiniales, Dinophyta), a green dinoflagellate with a chlorophyll *a*- and *b*-containing endosymbiont. *J Phycol*, 26, 741–751.

Watermann, F., Hillebrand, H., Gerdes, G., Krumbein, W.E. & Sommer, U. (1999). Competition between benthic cyanobacteria and diatoms as influenced by different grain sizes and temperatures. *Mar Ecol Prog Ser*, 187, 77–87.

Wells, M.L., Karlson, B., Wulff, A., Kudela, R., Trick, C., Asnaghi, V., *et al.* (2020). Future HAB science: Directions and challenges in a changing climate. *Harmful Algae*, 91, 101632.

Welschmeyer, N.A. (1994). Fluorometric analysis of chlorophyll *a* in the presence of chlorophyll *b* and pheopigments. *Limnol Oceanogr*, 39, 1985–1992.

Wetzel, C.E., Ector, L., Van de Vijver, B., Compère, P. & Mann, D.G. (2015). Morphology, typification and critical analysis of some ecologically important small naviculoid species (Bacillariophyta). *Fottea*, 15, 203–234.

Willis, A.D. (2019). Rarefaction, alpha diversity, and statistics. *Front Microbiol*, 10, 2407.

Yan, Y., Lin, T., Xie, W., Zhang, D., Jiang, Z., Han, Q., *et al.* (2023). Contrasting mechanisms determine the microeukaryotic and Syndiniales community assembly in a eutrophic bay. *Microb Ecol*, 86, 1575–1588.

Yang, Z., Ma, X., Wang, Q., Tian, X., Sun, J., Zhang, Z., *et al.* (2023). Phylotranscriptomics unveil a Paleoproterozoic-Mesoproterozoic origin and deep relationships of the Viridiplantae. *Nat Commun*, 14, 5542.

Yao, S., Lyu, S., An, Y., Lu, J., Gjermansen, C. & Schramm, A. (2019). Microalgae-bacteria symbiosis in microalgal growth and biofuel production: A review. *J Appl Microbiol*, 126, 359–368.

Yin, K., Zetsche, E.-M. & Harrison, P.J. (2016). Effects of sandy vs muddy sediments on the vertical distribution of microphytobenthos in intertidal flats of the Fraser River Estuary, Canada. *Environmental Science and Pollution Research*, 23, 14196–14209.

Zhou, J. & Ning, D. (2017). Stochastic community assembly: Does it matter in microbial ecology? *Microbiology and Molecular Biology Reviews*, 81, e00002-17.

Zulu, N.N., Zienkiewicz, K., Vollheyde, K. & Feussner, I. (2018). Current trends to comprehend lipid metabolism in diatoms. *Prog Lipid Res*, 70, 1–16.

Appendix A

Table S1. Study design components for the multi-year (2019–2021) investigation of microphytobenthos ecology in shallow sediments at the Grande Anse mudflat in the Bay of Fundy. (A) Sub-seasonal dates and sediment depths sampled across different years to characterize microalgal natural dynamics. The investigation of microalgal temporal dynamics involved averaging the depth samples within each core (see Methods section for details). (B) Types of data obtained in each year to address research questions. Classifications are based on astronomical seasons.

(A)

Year	Sub-season	Date	Replicate cores	Sediment depth
2019	Late spring	June 8	3	Top 0.5 cm 0.75 to 1.25 cm
	Early summer	July 20	3	
	Mid-summer (July)	July 31	3	
	Late summer	September 2	3	
	Early fall	October 14	3	
2020	Early summer	July 4–5	3	Top 0.2 cm 0.75 to 1.25 cm
	Mid-summer (July)	July 30	3	
	Late summer	August 28	3	
2021	Mid-summer (July)	July 29	2	Top 1 cm
	Mid-summer (August)	August 16	2	
	Late summer	August 28	2	

(B)

Data type	2019	2020	2021
Bacterial 16S rRNA gene amplicon sequences		Yes	
Eukaryotic 18S rRNA gene amplicon sequences	Yes	Yes	Yes
Metagenomes for cyanobacterial diversity and relative abundance			Yes
Chlorophyll *a* content		Yes	
Sediment physico-chemical properties	Yes		
Benthic fauna densities or equivalent		Yes	

Table S2. Description of covariates and factors used in PERMANCOVA (permutational multivariate analysis of covariance) models to investigate potential contributors, and quantify their respective contributions, to the variability in benthic microalgal assemblages in surficial biofilms of a Bay of Fundy mudflat in summer 2020.

Ecological covariate or factor	Description and unit of measure
Shorebirds	Habitat usage: Percent cover of footprints
Adult mud snails	Density: Number of individuals per m^2
Juvenile mud snails	Density: Number of individuals per m^2
Adult *Corophium* amphipods	Density: Number of individuals per m^2
Juvenile *Corophium* amphipods	Density: Number of individuals per m^2
Macoma clams	Density: Number of individuals per m^2
Nereididae (errant Polychaeta)	Density: Number of individuals per m^2
Phyllodocidae (errant Polychaeta)	Density: Number of individuals per m^2
Capitellidae (sessile Polychaeta)	Density: Number of individuals per m^2
Spionidae (sessile Polychaeta)	Density: Number of individuals per m^2
Turbellaria (Platyhelminthes)	Relative abundance: Percent of meiofaunal 18S rRNA gene reads
Nematoda	Relative abundance: Percent of meiofaunal 18S rRNA gene reads
Copepoda	Relative abundance: Percent of meiofaunal 18S rRNA gene reads
Ostracoda	Relative abundance: Percent of meiofaunal 18S rRNA gene reads
Fungi	Relative abundance: Percent of microbial eukaryotic 18S rRNA gene reads
Protistan consumers	Relative abundance: Percent of microbial eukaryotic 18S rRNA gene reads
Protistan parasites	Relative abundance: Percent of microbial eukaryotic 18S rRNA gene reads
Sulfate-reducing bacteria	Relative abundance: Percent of bacterial 16S rRNA gene reads
Sulfur compound-oxidizing bacteria	Relative abundance: Percent of bacterial 16S rRNA gene reads
Nitrate-reducing bacteria	Relative abundance: Percent of bacterial 16S rRNA gene reads
Sub-season	Categorical (3 levels): Early summer, mid-summer, and late summer
Fertilizer	Categorical (2 levels): Unfertilized and fertilized
Plot(nested in Sub-season and Fertilizer factors)	Unit of replication: 3 per combination of fertilizer and predator treatments (see Methods section for details)

Relative abundance was expressed as the number of reads of a taxonomic or functional group relative to the total number of reads for their corresponding biological community (see Table S9), in each sample. Covariate units were normalized prior to analyses.

Table S3. SIMPER results showing the contributions of microalgal species to the significant variation between depths (top 2–5 mm and 0.75–1.25 cm), within surficial sediments of a Bay of Fundy mudflat, both intra-seasonally (early, mid and late summer) and interannually in 2019 and 2020 (see Table 1 for significant depth-related effects). Only species contributing ≥1% of the overall dissimilarity between depths (together 85–90% of dissimilarity), are shown here. Species with an average dissimilarity/SD of dissimilarities ≥1 are consistent contributors to the dissimilarity in microalgal assemblages between depths.

(A) Early, mid and late summer 2019. Overall average dissimilarity between depths: 50.11%

Species	Average relative abundance (%)		Average diss.	Diss./SD	Contribution (%)
	Top 0.5 cm	0.75–1.25 cm			
Bacillariophyta *Thalassiosira* spp.	21.57	34.12	7.78	1.36	15.52
Bacillariophyta *Gyrosigma limosum*	12.40	9.65	6.72	1.14	13.41
Bacillariophyta Raphid-pennate_X	14.81	5.97	6.59	1.20	13.16
Chlorophyta *Picochlorum* spp.	13.34	8.76	5.61	1.35	11.20
Dinophyceae *Alexandrium fundyense*	2.40	5.71	2.91	0.96	5.80
Chlorophyta Pycnococcaceae_X	7.44	4.58	2.64	1.29	5.26
Bacillariophyta *Minidiscus spinulatus*	3.14	6.79	1.90	1.75	3.79
Bacillariophyta *Navicula* spp.	2.04	3.47	1.82	1.10	3.62
Bacillariophyta *Nitzschia* spp.	3.77	1.44	1.72	1.14	3.43
Dinophyceae *Lepidodinium* sp.	2.00	1.88	0.88	1.40	1.75
Chlorophyta *Ulva* spp.	2.09	0.72	0.82	1.20	1.64
Dinophyceae *Gonyaulax spinifera*	0.02	1.48	0.75	0.37	1.49
Bacillariophyta *Navicula phyllepta*	1.48	0.15	0.70	0.73	1.39
Dinophyceae *Protodinium simplex*	0.66	1.42	0.67	1.24	1.34
Bacillariophyta *Gyrosigma* spp.	1.35	1.01	0.60	1.35	1.20
Bacillariophyta *Surirella* spp.	1.12	0.20	0.51	0.66	1.01

(B) Early summer 2020. Overall average dissimilarity between depths: 45.91%

Species	Average relative abundance (%)		Avg. diss.	Diss./SD	Contribution (%)
	Top 0.2 cm	0.75–1.25 cm			
Bacillariophyta *Gyrosigma limosum*	20.89	3.87	9.44	0.91	20.56
Bacillariophyta *Thalassiosira* spp.	33.09	37.24	5.67	1.34	12.36
Bacillariophyta *Navicula cryptocephala*	10.29	2.39	4.75	0.94	10.34
Bacillariophyta Polar-centric-Mediophyceae_X	6.26	11.66	3.61	1.56	7.87
Dinophyceae *Alexandrium fundyense*	2.63	6.24	2.99	1.19	6.51
Chlorophyta *Picochlorum* spp.	4.79	3.60	2.94	1.11	6.40
Bacillariophyta *Minidiscus spinulatus*	4.36	6.95	2.02	1.35	4.40
Bacillariophyta *Navicula* spp.	1.77	3.80	1.73	1.10	3.77
Chlorophyta Pycnococcaceae_X	2.83	2.18	1.68	1.32	3.65
Bacillariophyta Raphid-pennate_X	1.02	3.33	1.25	1.62	2.72
Bacillariophyta *Surirella* spp.	1.43	2.79	1.18	1.13	2.57
Bacillariophyta *Delphineis* sp.	1.63	1.63	1.09	1.51	2.37
Bacillariophyta *Minidiscus comicus*	1.02	2.50	0.88	1.58	1.92
Dinophyceae *Lepidodinium* sp.	0.81	2.05	0.72	1.36	1.58
Prasinodermophyta *Prasinoderma coloniale*	0.88	0.59	0.54	1.29	1.18

(C) Mid-summer 2020. Overall average dissimilarity between depths: 49.89%

Species	Average relative abundance (%)		Avg. diss.	Diss./SD	Contribution (%)
	Top 0.2 cm	0.75–1.25 cm			
Bacillariophyta *Gyrosigma limosum*	21.80	0.71	10.55	3.38	21.14
Bacillariophyta *Delphineis* sp.	2.47	14.21	5.87	6.45	11.76
Bacillariophyta *Thalassiosira* spp.	17.07	27.40	5.16	2.59	10.35
Dinophyceae *Alexandrium fundyense*	7.79	7.03	2.45	1.39	4.91
Bacillariophyta *Minidiscus spinulatus*	4.22	8.67	2.22	3.00	4.45
Bacillariophyta *Navicula* spp.	6.02	2.27	1.87	1.91	3.75
Bacillariophyta *Navicula cryptocephala*	4.98	5.26	1.81	2.31	3.62
Bacillariophyta *Surirella* spp.	4.67	2.09	1.68	1.40	3.37
Bacillariophyta *Minidiscus comicus*	2.40	5.30	1.45	1.61	2.91
Bacillariophyta *Cylindrotheca* spp.	2.76	0.00	1.38	1.90	2.77
Bacillariophyta Polar-centric-Mediophyceae_X	3.07	4.62	1.19	2.53	2.38
Bacillariophyta *Entomoneis* spp.	3.06	1.22	1.17	1.72	2.34
Bacillariophyta Raphid-pennate_X	4.13	1.97	1.08	3.38	2.17
Bacillariophyta *Amphora* spp.	2.25	0.10	1.07	1.98	2.15
Bacillariophyta *Nitzschia apiculata*	2.14	0.26	0.99	1.47	1.98
Bacillariophyta *Paralia longispina*	1.68	1.53	0.74	1.38	1.48
Chlorophyta *Picochlorum* spp.	0.67	1.78	0.68	1.26	1.36
Dinophyceae *Prorocentrum nanum*	1.18	0.00	0.59	2.35	1.18
Bacillariophyta Araphid-pennate_X	0.70	1.63	0.52	1.49	1.04
Chlorophyta *Pyramimonas australis*	0.09	1.11	0.51	1.49	1.02

(D) Late summer 2020. Overall average dissimilarity between depths: 81.81%

Species	Average relative abundance (%)		Avg. diss.	Diss./SD	Contribution (%)
	Top 0.2 cm	0.75–1.25 cm			
Bacillariophyta *Gyrosigma limosum*	36.03	2.93	16.55	3.33	20.23
Dinophyceae *Alexandrium fundyense*	1.16	24.78	11.81	1.85	14.44
Bacillariophyta *Thalassiosira* spp.	2.95	24.82	10.94	9.36	13.37
Bacillariophyta *Navicula cryptocephala*	22.18	3.55	9.31	2.71	11.38
Bacillariophyta *Delphineis* sp.	0.39	15.25	7.43	17.11	9.08
Bacillariophyta *Surirella* spp.	13.21	2.05	5.58	4.81	6.82
Bacillariophyta *Cylindrotheca* spp.	3.81	0.27	1.77	3.38	2.16
Bacillariophyta *Minidiscus spinulatus*	0.65	3.87	1.61	1.49	1.96
Bacillariophyta *Nitzschia apiculata*	3.58	0.47	1.55	3.27	1.90
Bacillariophyta Raphid-pennate_X	4.49	1.44	1.53	2.96	1.87
Bacillariophyta Polar-centric-Mediophyceae_X	0.09	2.77	1.34	5.02	1.64
Bacillariophyta *Minidiscus comicus*	0.23	2.62	1.19	1.88	1.46
Bacillariophyta *Navicula* spp.	3.01	0.74	1.13	2.31	1.39
Bacillariophyta *Entomoneis* spp.	2.08	0.14	0.97	2.57	1.19
Bacillariophyta *Minutocellus polymorphus*	0.03	1.89	0.93	2.97	1.14

Table S4. SIMPER results showing the contributions of microalgal species to the significant inter-seasonal variability during the spring–summer and summer–fall transitions, and interannual variability in 2019–2021, in surficial sediments (across the top 1 cm) of a Bay of Fundy mudflat (see Table 3 for significant temporal contrasts). Only species contributing ≥1% of the overall dissimilarity between years or sub-seasons (together 82–87% of dissimilarity), are shown here. Species with an average dissimilarity/SD of dissimilarities ≥1 are consistent contributors to the dissimilarity in microalgal assemblages between years and sub-seasons.

(A) 2019 vs. 2020 (early, mid and late summer). Overall average dissimilarity between years: 53.86%

Species	Avg. relative abundance (%)		Avg. diss.	Diss./SD	Contribution (%)
	2019	2020			
Bacillariophyta *Gyrosigma limosum*	11.02	14.37	6.28	1.51	11.66
Bacillariophyta *Thalassiosira* spp.	27.85	23.76	5.93	1.37	11.00
Chlorophyta *Picochlorum* spp.	11.05	2.03	4.97	1.20	9.22
Bacillariophyta Raphid-pennate_X	10.39	2.73	4.20	1.02	7.81
Bacillariophyta *Navicula cryptocephala*	0.29	8.11	3.91	1.42	7.26
Dinophyceae *Alexandrium fundyense*	4.06	8.27	3.24	1.29	6.02
Bacillariophyta *Delphineis* sp.	0.48	5.93	2.76	1.76	5.12
Chlorophyta Pycnococcaceae_X	6.01	1.29	2.56	1.35	4.75
Bacillariophyta Polar-centric-Mediophyceae_X	0.62	4.74	2.08	1.26	3.87
Bacillariophyta *Surirella* spp.	0.66	4.37	1.90	1.41	3.53
Bacillariophyta *Navicula* spp.	2.75	2.94	1.30	1.53	2.41
Bacillariophyta *Minidiscus spinulatus*	4.96	4.79	1.27	1.62	2.36
Bacillariophyta *Nitzschia* spp.	2.61	0.42	1.17	0.99	2.17
Bacillariophyta *Minidiscus comicus*	0.44	2.35	0.95	1.43	1.77
Dinophyceae *Lepidodinium* sp.	1.94	1.10	0.75	1.61	1.40
Chlorophyta *Ulva* spp.	1.40	0.07	0.67	1.34	1.24
Bacillariophyta *Cylindrotheca* spp.	0.01	1.24	0.61	1.44	1.14
Bacillariophyta *Nitzschia apiculata*	0.00	1.10	0.55	1.18	1.02

(B) 2019 vs. 2021 (mid and late summer). Overall average dissimilarity between years: 42.59%

Species	Avg. relative abundance (%)		Avg. diss.	Diss./SD	Contribution (%)
	2019	2021			
Chlorophyta *Picochlorum* spp.	16.04	2.83	6.60	2.17	15.50
Bacillariophyta *Thalassiosira* spp.	32.90	37.80	4.85	1.37	11.38
Chlorophyta Pycnococcaceae_X	8.26	1.74	3.26	2.40	7.66
Bacillariophyta *Navicula cryptocephala*	0.32	5.69	2.68	3.03	6.30
Bacillariophyta Raphid-pennate_X	5.84	2.61	2.22	0.81	5.22
Dinophyceae *Alexandrium fundyense*	5.87	8.80	2.13	1.51	5.00
Bacillariophyta *Cylindrotheca closterium*	0.01	3.00	1.50	0.83	3.52
Bacillariophyta Polar-centric-Mediophyceae_X	0.16	3.05	1.44	1.84	3.39
Bacillariophyta *Delphineis* sp.	0.12	2.99	1.44	2.67	3.37
Bacillariophyta *Gyrosigma limosum*	2.90	2.46	1.08	1.94	2.53
Bacillariophyta *Navicula* spp.	1.15	2.37	1.02	1.04	2.41
Bacillariophyta *Surirella* spp.	0.67	2.16	0.88	1.82	2.07
Bacillariophyta *Minidiscus spinulatus*	4.44	3.96	0.82	1.34	1.92
Bacillariophyta *Paralia longispina*	0.04	1.60	0.78	1.58	1.83
Dinophyceae *Protodinium simplex*	1.53	0.24	0.65	1.62	1.52
Chlorophyta *Ulva* spp.	1.90	0.85	0.64	1.48	1.51
Dinophyceae *Gonyaulax spinifera*	1.11	0.26	0.63	0.55	1.49
Dinophyceae *Lepidodinium* sp.	2.83	1.81	0.57	1.31	1.34
Bacillariophyta *Navicula phyllepta*	1.03	0.21	0.49	0.96	1.15
Bacillariophyta *Nitzschia* spp.	1.17	0.64	0.48	1.01	1.13
Chlorophyta *Chlamydomonas bilatus*	1.13	0.24	0.45	1.14	1.05
Bacillariophyta *Minidiscus comicus*	0.35	1.23	0.44	1.70	1.03

(C) 2020 vs. 2021 (mid and late summer). Overall average dissimilarity between years: 43.36%

Species	Avg. relative abundance (%)		Avg. diss.	Diss./SD	Contribution (%)
	2020	2021			
Bacillariophyta *Thalassiosira* spp.	18.06	37.80	9.87	2.34	22.77
Bacillariophyta *Gyrosigma limosum*	15.37	2.46	6.45	1.94	14.88
Dinophyceae *Alexandrium fundyense*	10.19	8.80	2.71	1.51	6.26
Bacillariophyta *Delphineis* sp.	8.08	2.99	2.54	3.65	5.86
Bacillariophyta *Navicula cryptocephala*	8.99	5.69	2.40	1.31	5.55
Bacillariophyta *Surirella* spp.	5.51	2.16	1.88	1.68	4.34
Bacillariophyta *Cylindrotheca closterium*	0.26	3.00	1.49	0.87	3.44
Bacillariophyta *Minidiscus spinulatus*	4.35	3.96	1.13	1.50	2.60
Bacillariophyta *Navicula* spp.	3.01	2.37	1.08	1.62	2.48
Chlorophyta *Picochlorum* spp.	0.95	2.83	0.94	2.93	2.17
Bacillariophyta Polar-centric-Mediophyceae_X	2.63	3.05	0.84	1.34	1.94
Bacillariophyta *Minidiscus comicus*	2.64	1.23	0.81	1.32	1.87
Bacillariophyta Raphid-pennate_X	3.01	2.61	0.74	1.62	1.70
Bacillariophyta *Nitzschia apiculata*	1.61	0.20	0.72	2.15	1.65
Bacillariophyta *Entomoneis* spp.	1.62	0.25	0.69	1.84	1.58
Bacillariophyta *Paralia longispina*	1.19	1.60	0.62	1.42	1.44
Chlorophyta Pycnococcaceae_X	0.68	1.74	0.53	2.04	1.23
Bacillariophyta *Cylindrotheca* spp.	1.71	0.71	0.53	1.55	1.22
Prasinodermophyta *Prasinoderma coloniale*	0.03	1.08	0.53	2.62	1.21
Dinophyceae *Lepidodinium* sp.	0.94	1.81	0.46	1.65	1.06

(D) Late spring vs. early summer 2019. Overall average dissimilarity between sub-seasons: 50.69%

Species	Avg. relative abundance (%)		Avg. diss.	Diss./SD	Contribution (%)
	Late Spring	Early Summer			
Bacillariophyta *Gyrosigma limosum*	4.22	27.27	11.53	5.49	22.74
Bacillariophyta Raphid-pennate_X	4.97	19.49	7.26	2.10	14.33
Bacillariophyta *Thalassiosira* spp.	31.02	17.74	6.64	2.10	13.10
Dinophyceae *Alexandrium fundyense*	12.36	0.43	5.97	1.72	11.77
Chlorophyta *Picochlorum* spp.	5.85	1.09	2.38	2.04	4.70
Chlorophyta Pycnococcaceae_X	4.70	1.50	1.60	2.34	3.15
Bacillariophyta *Minidiscus spinulatus*	8.98	6.02	1.48	3.51	2.92
Bacillariophyta *Navicula* spp.	3.52	5.97	1.43	1.51	2.82
Bacillariophyta *Nitzschia* spp.	7.10	5.47	1.22	1.52	2.40
Bacillariophyta *Gyrosigma* spp.	0.05	2.03	0.99	3.21	1.95
Phaeophyceae *Fucus* sp.	1.61	0.02	0.80	0.73	1.58
Bacillariophyta Polar-centric-Mediophyceae_X	0.11	1.54	0.71	5.14	1.40
Bacillariophyta *Minidiscus comicus*	2.01	0.63	0.69	1.57	1.37
Bacillariophyta *Delphineis* sp.	1.01	1.19	0.66	3.49	1.31

(E) Late summer vs. early fall 2019. Overall average dissimilarity between sub-seasons: 50.46%

Species	Avg. relative abundance (%)		Avg. diss.	Diss./SD	Contribution (%)
	Late Summer	Early Fall			
Bacillariophyta *Thalassiosira* spp.	40.23	13.57	13.33	8.02	26.41
Dinophyceae *Alexandrium fundyense*	3.76	19.04	7.64	2.06	15.14
Bacillariophyta Raphid-pennate_X	4.13	11.17	3.53	1.16	6.99
Chlorophyta *Picochlorum* spp.	14.94	8.22	3.46	1.50	6.85
Bacillariophyta *Gyrosigma limosum*	3.62	9.56	3.08	1.26	6.10
Chlorophyta Pycnococcaceae_X	7.66	1.71	2.97	5.90	5.89
Bacillariophyta *Nitzschia* spp.	1.09	4.71	1.81	2.38	3.58
Bacillariophyta *Minidiscus spinulatus*	5.24	5.77	1.40	1.37	2.77
Bacillariophyta *Navicula phyllepta*	1.74	3.77	1.36	1.42	2.70
Bacillariophyta *Minidiscus comicus*	0.52	3.01	1.25	2.93	2.47
Bacillariophyta *Surirella* spp.	0.33	2.38	1.02	1.31	2.03
Dinophyceae *Lepidodinium* sp.	2.45	1.06	0.69	3.68	1.38
Bacillariophyta *Odontella* spp.	0.42	1.49	0.64	1.04	1.28
Dinophyceae *Alexandrium ostenfeldii*	0.00	1.27	0.64	0.98	1.26
Bacillariophyta *Delphineis* sp.	0.23	1.27	0.55	1.41	1.10
Bacillariophyta *Navicula* spp.	0.63	1.60	0.52	1.32	1.03

Table S5. PERMANCOVA results for spatio-temporal variability in microalgal assemblages, in relation to covariability in environmental variables, in surficial sediments (top 5 mm and 0.75–1.25 cm) of a Bay of Fundy mudflat, from early summer to early fall 2019. Among the environmental covariates, temperature varies seasonally, redox potential (a proxy for oxygen content) and pH decrease with sediment depth, and penetrability is a proxy for sediment water content. Four response variables of microalgae (A–D) were examined in univariate and multivariate analyses. Significant *P* values are indicated in bold. Number of unique permutations = 994–999. Estimates and percentages of variation accounted for by each source of variation are provided. Negative estimates were replaced with zeros (Searle *et al.* 1992). Spearman's rank correlation coefficients are shown for significant associations between a univariate response variable and a covariate. Environmental covariates accounted for 8–22% of microalgal variability, across response variables, from early summer to early fall.

(A) Relative abundance: Percent of 18S rRNA gene reads of microalgae among microbial eukaryotes

Source of variation	df	MS	Pseudo-*F*	*P*	Components of variation		Spearman's ρ
					Estimate	%	
Temperature	1	75.39	0.77	0.519	0.00	0.00	
Redox potential	1	243.56	2.43	0.137	6.56	1.45	
pH	1	558.02	8.19	**0.027**	22.47	4.97	0.21
Penetrability	1	265.84	2.45	0.161	7.70	1.70	
Sub-season	3	795.20	7.60	**0.022**	340.99	75.40	
Depth	1	107.67	5.13	0.100	7.83	1.73	
Sub-season × Depth	3	38.97	1.86	0.278	6.40	1.42	
Plot(Sub-season)	7	68.18	3.73	0.108	42.05	9.30	
Depth × Plot(Sub-season)	4	18.27			18.27	4.04	

Univariate analysis.

(B) ASV richness, an alpha diversity index: Number of microalgal amplicon sequence variants

Source of variation	df	MS	Pseudo-*F*	*P*	Components of variation		Spearman's ρ
					Estimate	%	
Temperature	1	1028.50	6.26	**0.026**	38.39	13.77	0.46
Redox potential	1	13.72	0.20	0.918	0.00	0.00	
pH	1	595.30	5.88	**0.040**	22.67	8.13	0.13
Penetrability	1	55.29	0.46	0.734	0.00	0.00	
Sub-season	3	399.04	2.65	0.174	126.93	45.54	
Depth	1	39.09	1.05	0.366	0.15	0.05	
Sub-season × Depth	3	7.58	0.20	0.871	0.00	0.00	
Plot(Sub-season)	7	101.24	3.00	0.115	56.90	20.42	
Depth × Plot(Sub-season)	4	33.70			33.70	12.09	

Univariate analysis.

(C) ASV presence absence: Presence or absence of microalgal amplicon sequence variants

Source of variation	df	MS	Pseudo-*F*	*P*	Components of variation	
					Estimate	%
Temperature	1	2035.40	1.80	0.088	49.54	2.67
Redox potential	1	677.74	0.84	0.605	0.00	0.00
pH	1	4618.10	5.37	**0.002**	172.40	9.28
Penetrability	1	1080.00	1.13	0.364	8.59	0.46
Sub-season	3	1869.20	1.73	0.072	477.29	25.70
Depth	1	3860.60	4.78	**0.013**	275.94	14.86
Sub-season × Depth	3	871.04	1.08	0.459	22.71	1.22
Plot(Sub-season)	7	859.57	1.07	0.422	46.68	2.51
Depth × Plot(Sub-season)	4	804.17			804.17	43.30

Multivariate analysis.

(D) Species composition: Percentages of 18S rRNA gene reads for species of microalgae

Source of variation	df	MS	Pseudo-*F*	*P*	Components of variation	
					Estimate	%
Temperature	1	2985.70	2.50	**0.048**	86.66	3.88
Redox potential	1	1300.70	1.28	0.282	15.22	0.68
pH	1	4441.40	5.53	**0.003**	166.89	7.46
Penetrability	1	2034.50	1.78	0.129	48.66	2.18
Sub-season	3	2528.30	2.25	0.072	763.26	34.14
Depth	1	3713.60	6.99	**0.006**	287.62	12.86
Sub-season × Depth	3	839.41	1.58	0.173	109.74	4.91
Plot(Sub-season)	7	803.19	1.56	0.138	242.71	10.86
Depth × Plot(Sub-season)	4	515.10			515.10	23.04

Multivariate analysis.

Table S6. PERMANCOVA results for spatio-temporal variability in microalgal assemblages, in relation to covariability in environmental variables, in surficial sediments (top 5 mm and 0.75–1.25 cm) of a Bay of Fundy mudflat, in early, mid and late summer 2019. Among the environmental covariates, temperature varies seasonally, redox potential (a proxy for oxygen content) and pH decrease with sediment depth, and penetrability is a proxy for sediment water content. Four response variables of microalgae (A–D) were examined in univariate and multivariate analyses. Significant *P* values are indicated in bold. Number of unique permutations = 996–999. Estimates and percentages of variation accounted for by each source of variation are provided. Negative estimates were replaced with zeros (Searle *et al.* 1992). Spearman's rank correlation coefficients are shown for significant associations between a univariate response variable and a covariate. Environmental covariates typically accounted for 15–19% of microalgal variability, across response variables, except with microalgal ASV richness where they accounted for 58% of the variation in the summer.

(A) Relative abundance: Percent of 18S rRNA gene reads of microalgae among microbial eukaryotes

Source of variation	df	MS	Pseudo-*F*	*P*	Components of variation		Spearman's ρ
					Estimate	%	
Temperature	1	1098.80	6.81	**0.039**	56.29	14.44	0.28
Redox potential	1	67.25	0.56	0.662	0.00	0.00	
pH	1	390.85	4.16	0.078	18.12	4.65	
Penetrability	1	6.31	0.18	0.908	0.00	0.00	
Sub-season	2	601.54	4.18	0.117	222.87	57.16	
Depth	1	43.69	2.02	0.270	2.66	0.68	
Sub-season × Depth	2	47.32	2.06	0.316	8.84	2.27	
Plot(Sub-season)	5	73.10	4.04	0.215	63.06	16.17	
Depth × Plot(Sub-season)	2	18.09			18.09	4.64	

Univariate analysis.

(B) ASV richness, an alpha diversity index: Number of microalgal amplicon sequence variants

Source of variation	df	MS	Pseudo-*F*	*P*	Components of variation		Spearman's *ρ*
					Estimate	%	
Temperature	1	1918.00	9.35	**0.018**	104.15	43.64	0.61
Redox potential	1	15.40	0.37	0.790	0.00	0.00	
pH	1	375.84	3.19	0.120	16.17	6.77	
Penetrability	1	400.36	2.17	0.192	17.88	7.49	
Sub-season	2	56.51	0.57	0.639	0.00	0.00	
Depth	1	19.16	0.35	0.610	0.00	0.00	
Sub-season × Depth	2	9.26	0.17	0.860	0.00	0.00	
Plot(Sub-season)	5	94.25	1.82	0.405	48.74	20.42	
Depth × Plot(Sub-season)	2	51.73			51.73	21.67	

Univariate analysis.

(C) ASV presence absence: Presence or absence of microalgal amplicon sequence variants

Source of variation	df	MS	Pseudo-*F*	*P*	Components of variation	
					Estimate	%
Temperature	1	1814.40	1.60	0.224	61.02	3.20
Redox potential	1	744.56	0.98	0.488	0.00	0.00
pH	1	3661.70	3.91	**0.036**	174.23	9.13
Penetrability	1	1435.00	1.38	0.352	46.72	2.45
Sub-season	2	2267.40	1.96	0.153	705.10	36.93
Depth	1	2057.30	2.73	0.131	157.22	8.24
Sub-season × Depth	2	627.01	0.83	0.593	0.00	0.00
Plot(Sub-season)	5	763.20	1.01	0.510	12.65	0.66
Depth × Plot(Sub-season)	2	752.17			752.17	39.40

Multivariate analysis.

(D) Species composition: Percentages of 18S rRNA gene reads for species of microalgae

Source of variation	df	MS	Pseudo-*F*	*P*	Components of variation	
					Estimate	%
Temperature	1	3579.50	2.52	0.101	144.46	7.17
Redox potential	1	704.35	0.75	0.624	0.00	0.00
pH	1	3735.40	4.13	**0.030**	175.77	8.72
Penetrability	1	1301.90	1.10	0.424	11.96	0.59
Sub-season	2	2331.50	1.88	0.230	604.68	29.99
Depth	1	2157.40	5.25	**0.035**	210.54	10.44
Sub-season × Depth	2	708.31	1.69	0.301	105.10	5.21
Plot(Sub-season)	5	715.77	1.84	0.232	373.82	18.54
Depth × Plot(Sub-season)	2	389.68			389.68	19.33

Multivariate analysis.

Table S7. RELATE results showing correlations between fluorescence-based chlorophyll *a* content and environmental DNA metabarcoding measurements of benthic microalgal assemblages in surficial biofilms of a Bay of Fundy mudflat in summer 2020. Significant *P* values are indicated in bold and represent a significant correlation between chlorophyll *a* content and a metabarcoding variable. Number of samples for each compassion = 54. Number of unique permutations = 999.

Metabarcoding variable	Spearman's ρ	*P*
Relative abundance	0.056	0.090
ASV richness	0.181	**0.002**
Species composition	0.133	**0.003**

Statistical analysis method: I used the PRIMER v6 statistical package (Clarke & Gorley 2006) to investigate associations between fluorescence-based chlorophyll *a* content measurements and environmental DNA metabarcoding estimations of the benthic microalgal relative abundance (among microbial eukaryotes), ASV (amplicon sequence variant) richness, and species composition, in surficial mudflat biofilms. I used Euclidean distances to create resemblance matrices for the univariate chlorophyll *a* content, relative abundance and ASV richness, and Bray-Curtis similarities for the multivariate microalgal species composition. I then performed RELATE tests using these resemblance matrices. For this analysis, I used 54 biofilm samples from the top 2 mm of mudflat sediments, collected in early, mid and late summer 2020, including from plots subjected to epifaunal predator exclusion and/or nutrient enrichment treatments.

Table S8. PERMANCOVA results for spatio-temporal variability in microalgal assemblages, in relation to covariability in biotic and nutrient-related variables, and across epifaunal predator exclusion and nutrient enrichment treatments, in surficial biofilms (top 2 mm) of a Bay of Fundy mudflat, in summer 2020. Biotic covariates, spanning from shorebirds to nitrate reducing bacteria, represent various levels of the mudflat food web, and are reflective of predator-prey, parasite-host and resource availability interactions with microalgae (see Table S9 for classification of covariates). Five response variables of microalgae (A–E) were examined in univariate and multivariate analyses. Significant *P* values are indicated in bold. Number of unique permutations = 993–999. Estimates and percentages of variation accounted for by each source of variation are provided. Negative estimates were replaced with zeros (Searle *et al.* 1992). Shading indicates top (i.e. above-average) determinants of microalgal variability (i.e., components of variation estimate ≥4%); these 13 variables together accounted for 80–96% of microalgal variability. Spearman's rank correlation coefficients are shown for significant associations between a univariate response variable and a covariate. Biotic and nutrient-related variables specifically accounted for 43–88% of microalgal variability, across response variables, in the summer. Juv., juvenile.

(A) Chlorophyll *a* content, a proxy for microalgal biomass: µg of chlorophyll *a* per g of sediment

Source of variation	df	MS	*Pseudo-F*	*P*	Components of variation		Spearman's ρ
					Estimate	%	
Shorebirds	1	26.78	12.50	**0.002**	0.46	1.24	0.29
Adult mud snails	1	58.62	19.78	**0.001**	1.68	4.57	0.24
Juv. mud snails	1	54.72	18.19	**0.001**	1.57	4.28	-0.28
Adult *Corophium*	1	23.32	9.87	**0.002**	0.42	1.14	-0.41
Juv. *Corophium*	1	0.11	0.04	0.837	0.00	0.00	
Macoma clams	1	21.29	10.47	**0.004**	0.44	1.20	-0.05
Nereididae (Polychaeta)	1	42.02	17.49	**0.001**	0.88	2.38	-0.48
Phyllodocidae (Polychaeta)	1	4.34	1.70	0.212	0.04	0.10	
Capitellidae (Polychaeta)	1	19.86	6.47	**0.020**	0.47	1.28	0.42
Spionidae (Polychaeta)	1	0.02	0.01	0.926	0.00	0.00	
Turbellaria (Platyhelminthes)	1	13.06	6.80	**0.015**	0.26	0.71	0.22
Nematoda	1	37.75	15.62	**0.004**	0.97	2.62	0.20
Copepoda	1	1.86	0.81	0.394	0.00	0.00	
Ostracoda	1	2.97	1.37	0.255	0.03	0.09	
Fungi	1	27.46	11.92	**0.005**	1.04	2.83	-0.60
Protistan consumers	1	1.42	0.58	0.455	0.00	0.00	
Protistan parasites	1	17.24	6.13	**0.025**	0.45	1.23	0.08
Sulfate reducing bacteria	1	20.23	10.40	**0.003**	0.68	1.86	-0.12
Sulfur compound oxidizing bacteria	1	0.45	0.22	0.628	0.00	0.00	
Nitrate reducing bacteria	1	7.63	3.24	0.088	0.24	0.65	
Sub-season	2	25.47	10.21	**0.002**	13.05	35.45	
Fertilizer	1	17.91	5.75	**0.034**	1.17	3.18	
Shorebirds × Fertilizer	1	26.61	14.73	**0.003**	1.36	3.70	
Adult mud snails × Fertilizer	1	22.69	10.20	**0.004**	3.66	9.94	
Sub-season × Fertilizer	2	21.60	8.56	**0.008**	5.69	15.46	
Plot(Sub-season × Fertilizer)	12	2.64	1.70	0.186	0.69	1.88	
Residual	14	1.55			1.55	4.22	

Univariate analysis.

(B) Relative abundance: Percent of 18S rRNA gene reads of microalgae among microbial eukaryotes

Source of variation	df	MS	*Pseudo-F*	*P*	Components of variation		Spearman's ρ
					Estimate	%	
Shorebirds	1	4337.20	309.58	**0.001**	80.06	14.72	0.58
Adult mud snails	1	5331.80	429.66	**0.001**	160.45	29.50	0.74
Juv. mud snails	1	129.70	10.49	**0.007**	3.63	0.67	0.36
Adult *Corophium*	1	34.93	2.76	0.110	0.44	0.08	
Juv. *Corophium*	1	297.28	26.63	**0.001**	5.60	1.03	-0.23
Macoma clams	1	13.15	0.90	0.339	0.00	0.00	
Nereididae (Polychaeta)	1	21.39	1.72	0.188	0.20	0.04	
Phyllodocidae (Polychaeta)	1	145.83	12.64	**0.006**	2.65	0.49	0.01
Capitellidae (Polychaeta)	1	18.91	1.70	0.213	0.25	0.05	
Spionidae (Polychaeta)	1	42.30	3.04	0.085	0.62	0.11	
Turbellaria (Platyhelminthes)	1	116.98	7.61	**0.014**	2.39	0.44	-0.33
Nematoda	1	0.25	0.02	0.883	0.00	0.00	
Copepoda	1	205.38	15.65	**0.002**	4.23	0.78	0.24
Ostracoda	1	1301.70	93.51	**0.001**	51.47	9.46	0.12
Fungi	1	638.02	48.97	**0.001**	25.85	4.75	-0.45
Protistan consumers	1	2227.50	183.22	**0.001**	120.28	22.11	-0.87
Protistan parasites	1	337.13	28.64	**0.001**	10.19	1.87	-0.55
Sulfate reducing bacteria	1	8.75	0.57	0.452	0.00	0.00	
Sulfur compound oxidizing bacteria	1	219.37	14.97	**0.003**	6.06	1.11	0.10
Nitrate reducing bacteria	1	63.23	4.96	**0.046**	2.28	0.42	-0.45
Sub-season	2	99.21	8.35	**0.007**	49.61	9.12	
Fertilizer	1	0.44	0.30	0.588	0.00	0.00	
Shorebirds × Fertilizer	1	0.00	0.00	0.992	0.00	0.00	
Adult mud snails × Fertilizer	1	0.54	0.04	0.850	0.00	0.00	
Sub-season × Fertilizer	2	10.47	0.89	0.423	0.00	0.00	
Plot(Sub-season × Fertilizer)	12	11.01	0.62	0.785	0.00	0.00	
Residual	14	17.61			17.61	3.24	

Univariate analysis. This response variable provides insights into the dynamics of photosynthetic microbial eukaryotes, relative to non-photosynthetic lineages. For example, a significant positive association suggests that a mudflat organism stimulates the abundance of microalgae more it does for non-photosynthetic microbial eukaryotes, or decreases the abundance microalgae less than it does for non-photosynthetic microbial eukaryotes. Similarly, a significant negative association suggests that a mudflat organism preys on microalgae more than non-photosynthetic microbial eukaryotes, or increases the abundance of microalgae less than it does for non-photosynthetic microbial eukaryotes, either directly or indirectly. This response variable also helps to characterize the dynamics of the chlorophyll *a* content, for example, by indicating that while microalgal biomass may be highly temporally variable, relative to non-photosynthetic microbial eukaryotes, it may be a lot less variable.

(C) ASV richness, an alpha diversity index: Number of microalgal amplicon sequence variants.

Source of variation	df	MS	*Pseudo-F*	*P*	Components of variation		Spearman's ρ
					Estimate	%	
Shorebirds	1	1421.10	6.50	**0.031**	22.27	3.03	0.43
Adult mud snails	1	2103.70	8.21	**0.010**	56.40	7.68	0.32
Juv. mud snails	1	106.80	0.52	0.556	0.00	0.00	
Adult *Corophium*	1	1373.50	6.14	**0.021**	22.96	3.13	-0.27
Juv. *Corophium*	1	127.34	0.56	0.478	0.00	0.00	
Macoma clams	1	196.34	0.91	0.388	0.00	0.00	
Nereididae (Polychaeta)	1	251.16	1.12	0.301	0.59	0.08	
Phyllodocidae (Polychaeta)	1	63.35	0.28	0.589	0.00	0.00	
Capitellidae (Polychaeta)	1	829.12	3.19	0.087	16.46	2.24	
Spionidae (Polychaeta)	1	124.88	0.57	0.489	0.00	0.00	
Turbellaria (Platyhelminthes)	1	721.44	3.38	0.092	11.96	1.63	
Nematoda	1	95.30	0.42	0.516	0.00	0.00	
Copepoda	1	938.45	4.23	0.068	15.76	2.15	
Ostracoda	1	952.67	4.35	**0.047**	29.32	4.00	-0.07
Fungi	1	520.50	2.34	0.140	12.33	1.68	
Protistan consumers	1	1873.60	8.30	**0.012**	89.47	12.19	-0.54
Protistan parasites	1	18.26	0.14	0.812	0.00	0.00	
Sulfate reducing bacteria	1	849.62	3.97	0.075	23.79	3.24	
Sulfur compound oxidizing bacteria	1	3007.40	13.91	**0.001**	82.68	11.27	-0.04
Nitrate reducing bacteria	1	285.36	1.28	0.286	2.80	0.38	
Sub-season	2	200.86	0.89	0.460	0.00	0.00	
Fertilizer	1	6.54	0.17	0.672	0.00	0.00	
Shorebirds × Fertilizer	1	667.71	3.17	0.074	25.10	3.42	
Adult mud snails × Fertilizer	1	785.58	3.56	0.075	101.00	13.76	
Sub-season × Fertilizer	2	29.96	0.13	0.851	0.00	0.00	
Plot(Sub-season × Fertilizer)	12	230.04	1.12	0.404	15.97	2.18	
Residual	14	205.01			205.01	27.94	

Univariate analysis.

(D) ASV presence absence: Presence or absence of microalgal amplicon sequence variants

Source of variation	df	MS	*Pseudo-F*	*P*	Components of variation	
					Estimate	%
Shorebirds	1	3393.50	5.15	**0.001**	50.65	3.44
Adult mud snails	1	3849.20	4.55	**0.001**	92.16	6.27
Juv. mud snails	1	1960.20	2.33	**0.011**	35.29	2.40
Adult *Corophium*	1	2884.70	4.09	**0.001**	43.52	2.96
Juv. *Corophium*	1	959.03	1.27	0.257	3.95	0.27
Macoma clams	1	845.86	1.33	0.174	4.82	0.33
Nereididae (Polychaeta)	1	1265.40	1.77	0.087	12.20	0.83
Phyllodocidae (Polychaeta)	1	761.31	1.02	0.418	0.33	0.02
Capitellidae (Polychaeta)	1	1930.20	2.25	**0.015**	31.17	2.12
Spionidae (Polychaeta)	1	594.05	0.90	0.487	0.00	0.00
Turbellaria (Platyhelminthes)	1	979.39	1.60	0.121	8.66	0.59
Nematoda	1	1138.00	1.59	0.115	11.52	0.78
Copepoda	1	874.76	1.27	0.252	4.08	0.28
Ostracoda	1	1265.00	1.91	**0.038**	24.11	1.64
Fungi	1	1779.20	2.57	**0.013**	44.94	3.06
Protistan consumers	1	2065.90	2.86	**0.006**	72.91	4.96
Protistan parasites	1	2692.20	3.33	**0.001**	59.68	4.06
Sulfate reducing bacteria	1	1000.80	1.62	0.106	14.38	0.98
Sulfur compound oxidizing bacteria	1	3641.80	5.72	**0.001**	89.03	6.05
Nitrate reducing bacteria	1	1345.40	1.91	**0.039**	29.04	1.97
Sub-season	2	1011.40	1.38	0.141	158.25	10.76
Fertilizer	1	388.11	0.54	0.845	0.00	0.00
Shorebirds × Fertilizer	1	481.92	0.82	0.606	0.00	0.00
Adult mud snails × Fertilizer	1	555.24	0.82	0.610	0.00	0.00
Sub-season × Fertilizer	2	598.20	0.81	0.662	0.00	0.00
Plot(Sub-season × Fertilizer)	12	763.08	1.43	**0.009**	146.44	9.96
Residual	14	533.48			533.48	36.28

Multivariate analysis.

(E) Species composition: Percentages of 18S rRNA gene reads for species of microalgae

Source of variation	df	MS	*Pseudo-F*	*P*	Components of variation	
					Estimate	%
Shorebirds	1	6463.40	17.71	**0.001**	112.93	5.23
Adult mud snails	1	9052.80	16.65	**0.001**	257.36	11.91
Juv. mud snails	1	2402.40	4.40	**0.002**	57.05	2.64
Adult *Corophium*	1	2649.00	6.34	**0.001**	44.55	2.06
Juvenile *Corophium*	1	1207.10	2.53	**0.045**	14.29	0.66
Macoma clams	1	1168.50	3.45	**0.008**	19.01	0.88
Nereididae (Polychaeta)	1	943.58	2.21	0.083	11.41	0.53
Phyllodocidae (Polychaeta)	1	1490.40	3.22	**0.022**	20.26	0.94
Capitellidae (Polychaeta)	1	2610.40	4.65	**0.001**	57.58	2.67
Spionidae (Polychaeta)	1	412.47	1.12	0.389	0.96	0.04
Turbellaria (Platyhelminthes)	1	413.24	1.33	0.246	2.40	0.11
Nematoda	1	637.06	1.48	0.206	5.64	0.26
Copepoda	1	1495.20	3.74	**0.004**	24.10	1.12
Ostracoda	1	2551.50	6.93	**0.001**	87.25	4.04
Fungi	1	765.38	1.90	0.111	14.96	0.69
Protistan consumers	1	847.34	1.93	0.119	22.22	1.03
Protistan parasites	1	9211.50	17.77	**0.001**	271.81	12.58
Sulfate reducing bacteria	1	754.30	2.38	**0.047**	16.37	0.76
Sulfur compound oxidizing bacteria	1	1207.90	3.56	**0.016**	25.72	1.19
Nitrate reducing bacteria	1	1640.40	3.95	**0.003**	55.37	2.56
Sub-season	2	1531.20	3.41	**0.002**	614.58	28.45
Fertilizer	1	559.50	1.04	0.404	1.79	0.08
Shorebirds × Fertilizer	1	210.67	0.74	0.595	0.00	0.00
Adult mud snails × Fertilizer	1	570.78	1.48	0.226	33.32	1.54
Sub-season × Fertilizer	2	267.76	0.59	0.818	0.00	0.00
Plot(Sub-season × Fertilizer)	12	483.54	2.17	**0.001**	166.35	7.70
Residual	14	222.74			222.74	10.31

Multivariate analysis. This response variable provides insights into potential drivers of species-specific dynamics in microalgal assemblages. For example, a significant association implies that a mudflat organism interacts differently with various groups of species, like the case of juvenile mud snails which has a positive association with raphid pennate diatoms, negative association with polar centric diatoms and green algae, and no significant association with dinoflagellates and other diatoms (see Figures 11 and 12 for genus-specific correlations with biotic covariates and protistan parasite lineages, respectively).

Table S9. Classification of biotic, nutrient-related and spatio-temporal variables based on their relationships and net-impacts on microalgal assemblages in surficial biofilms (top 2 mm) of a Bay of Fundy mudflat, in summer 2020, using PERMANCOVA results (see Table S8A–E).

Source of variation	Biological community	Ecological variable class
Shorebirds	Epifauna	Positive
Adult mud snails	Epifauna	Positive
Juvenile mud snails	Epifauna	Negative
Adult *Corophium* amphipods	Macro-infauna	Negative
Juvenile *Corophium* amphipods	Macro-infauna	Negative
Macoma clams	Macro-infauna	Negative
Nereididae (Polychaeta)	Macro-infauna	Negative
Phyllodocidae (Polychaeta)	Macro-infauna	Positive
Capitellidae (Polychaeta)	Macro-infauna	Positive
Spionidae (Polychaeta)	Macro-infauna	Negative
Turbellaria (Platyhelminthes)	Meiofauna	Positive
Nematoda	Meiofauna	Positive
Copepoda	Meiofauna	Positive
Ostracoda	Meiofauna	Negative
Fungi	Microbial eukaryotes	Negative
Protistan consumers	Microbial eukaryotes	Negative
Protistan parasites	Microbial eukaryotes	Positive
Sulfate-reducing bacteria	Bacteria	Negative
Sulfur compound-oxidizing bacteria	Bacteria	Negative
Nitrate-reducing bacteria	Bacteria	Negative
Sub-season		Temporal
Fertilizer		Positive
Shorebirds × Fertilizer		Positive
Adult mud snails × Fertilizer		Positive
Sub-season × Fertilizer		Positive
Plot(Sub-season × Fertilizer)		Spatial
Residual		Spatial

Refer to the Methods section for a comprehensive classification guide. Positive ecological variables, indicated by positive correlations with microalgae, include bottom-up control through trophic interactions, facilitation (such as habitat amelioration), and parasitism. In contrast, negative ecological variables, indicated by negative correlations with microalgae, encompass top-down control through predation, competition, and habitat deterioration. Habitat deterioration refers to activities by mudflat organisms that degrade habitat quality, making it less suitable for microalgal growth or survival. In essence, positive variables generally enhance while negative variables generally reduce the survival, growth, fitness and diversity of microalgae. The term “protistan parasites” refers to protists that parasitize other protists; similarly, the term “protistan consumers” denotes protists that consume other protists.

Table S10. Summary of PERMANCOVA results for spatio-temporal variability in microalgal assemblages, in relation to covariability in biotic and nutrient-related variables, and across epifaunal predator exclusion and nutrient enrichment treatments, in surficial biofilms (top 2 mm) of a Bay of Fundy mudflat, in summer 2020. Biotic covariates, spanning from shorebirds to nitrate reducing bacteria, represent various levels of the mudflat food web, as well as different types of ecological relationships, and are reflective of predator-prey, parasite-host and resource availability interactions with microalgae (see Table S9 for classification of covariates). Five response variables of microalgae were examined in univariate and multivariate analyses (see Table S8A–E for PERMANCOVA results). The percentages of microalgal variability accounted for different types of ecological relationships and different biological communities are provided. Negative estimates were replaced with zeros (Searle *et al.* 1992). ASV, amplicon sequence variant. Species comp., species composition.

Source of variation	Estimate of components of variation (%) of microalgae				
	Chlorophyll *a* content	Relative abundance	ASV richness	ASV presence absence	Species comp.
Ecological relationship					
Positive	44	48	34	18	37
Negative	14	40	36	25	17
Spatial	6	3	30	46	18
Temporal	36	9	0	11	28
Biological community or fertilizer treatment					
Epifauna	10	45	11	12	20
Macro-infauna	6	2	5	7	8
Meiofauna	3	11	8	3	6
Microbial eukaryotes	4	28	14	12	14
Bacteria	3	2	15	9	4
Nutrient enrichment	32	0	17	0	2

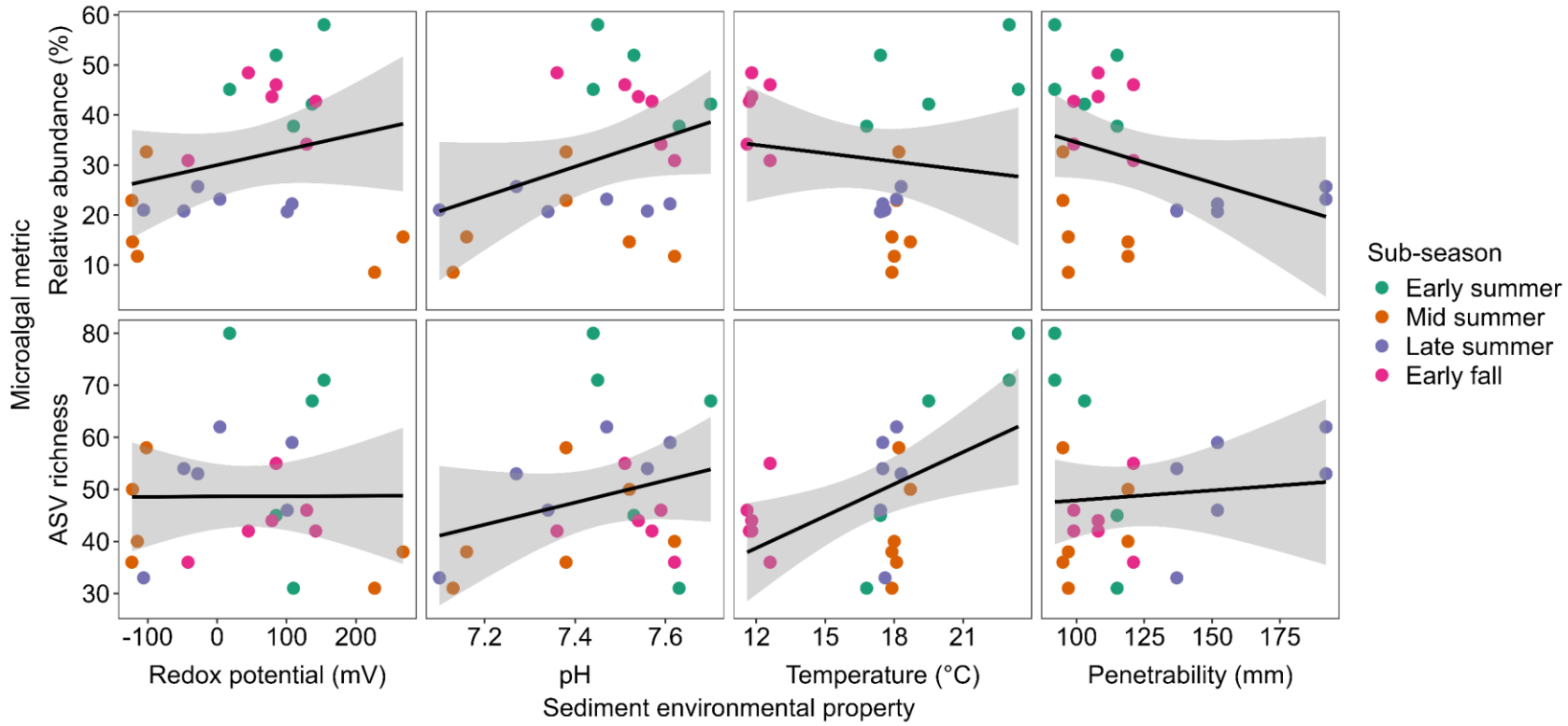


Figure S1. Scatterplots showing the relationship between various metrics of microalgal assemblages, and environmental covariates (n = 23), pooling over sediment depths (top 5 mm and 0.75–1.25 cm) and sub-seasons, in surficial sediments of a Bay of Fundy mudflat, in 2019. Each interaction is represented by a general trendline, accompanied by a 95% confidence interval, indicating the general strength and direction of the relationship. Refer to Table S5A–D for significant correlations.

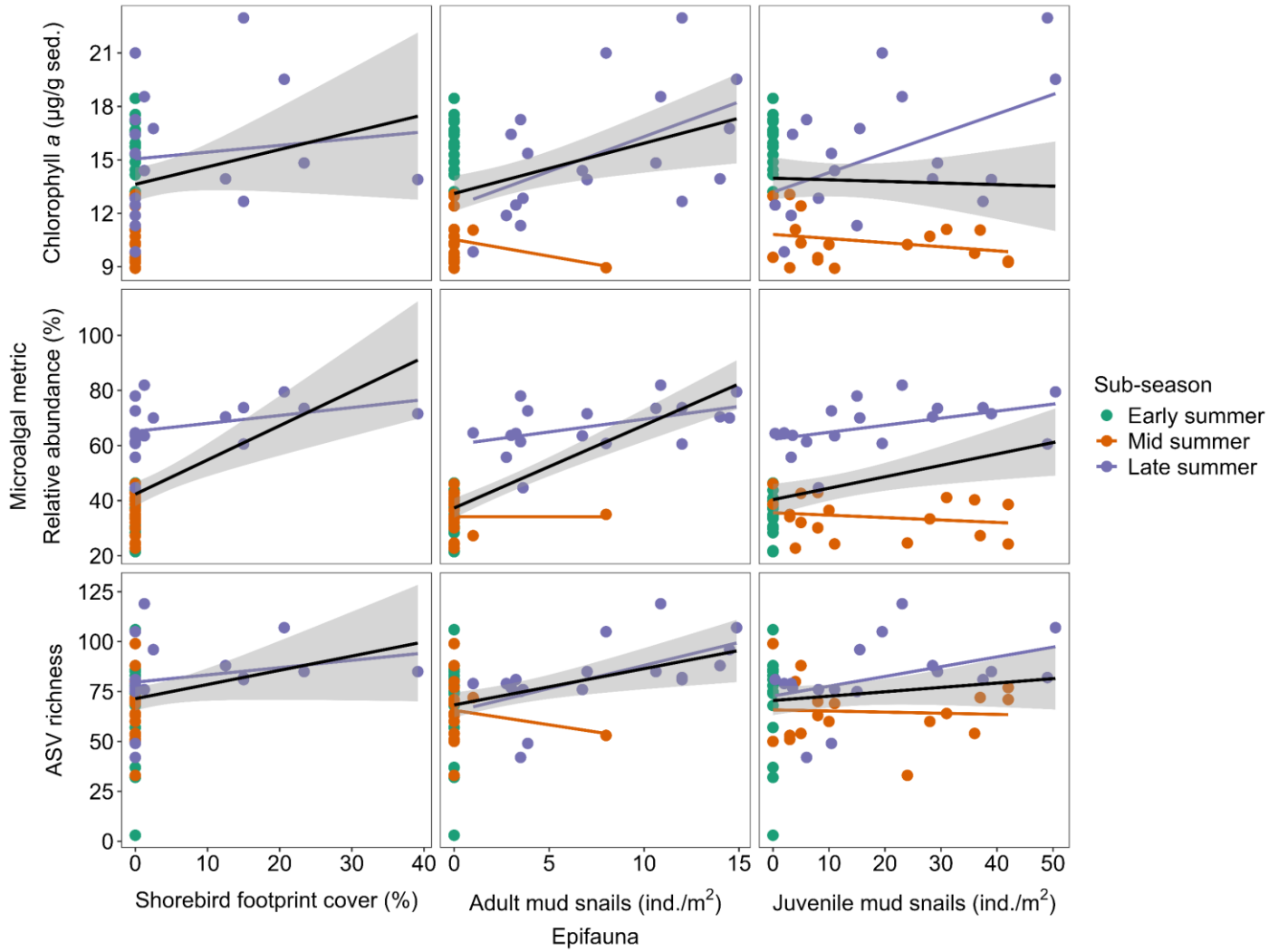


Figure S2A. Relationship between microalgae and epifauna. See figure caption below.

Figure S2. Scatterplots showing the relationship between various metrics of microalgal assemblages, and biotic covariates (n = 54), pooling over sub-seasons, in surficial biofilms (top 2 mm) of a Bay of Fundy mudflat, in summer 2020. Microalgal relationships with (A) epifauna, (B) macro-infauna (amphipods and clams), (C) macro-infauna (polychaete worms), (D) meiofauna, (E) microbial eukaryotes, and (F) bacteria are shown. Each interaction is represented by a general trendline, accompanied by a 95% confidence interval, indicating the general strength and direction of the relationship. A separate trendline is also shown for each sub-season. Refer to Table S8A–E for significant correlations.

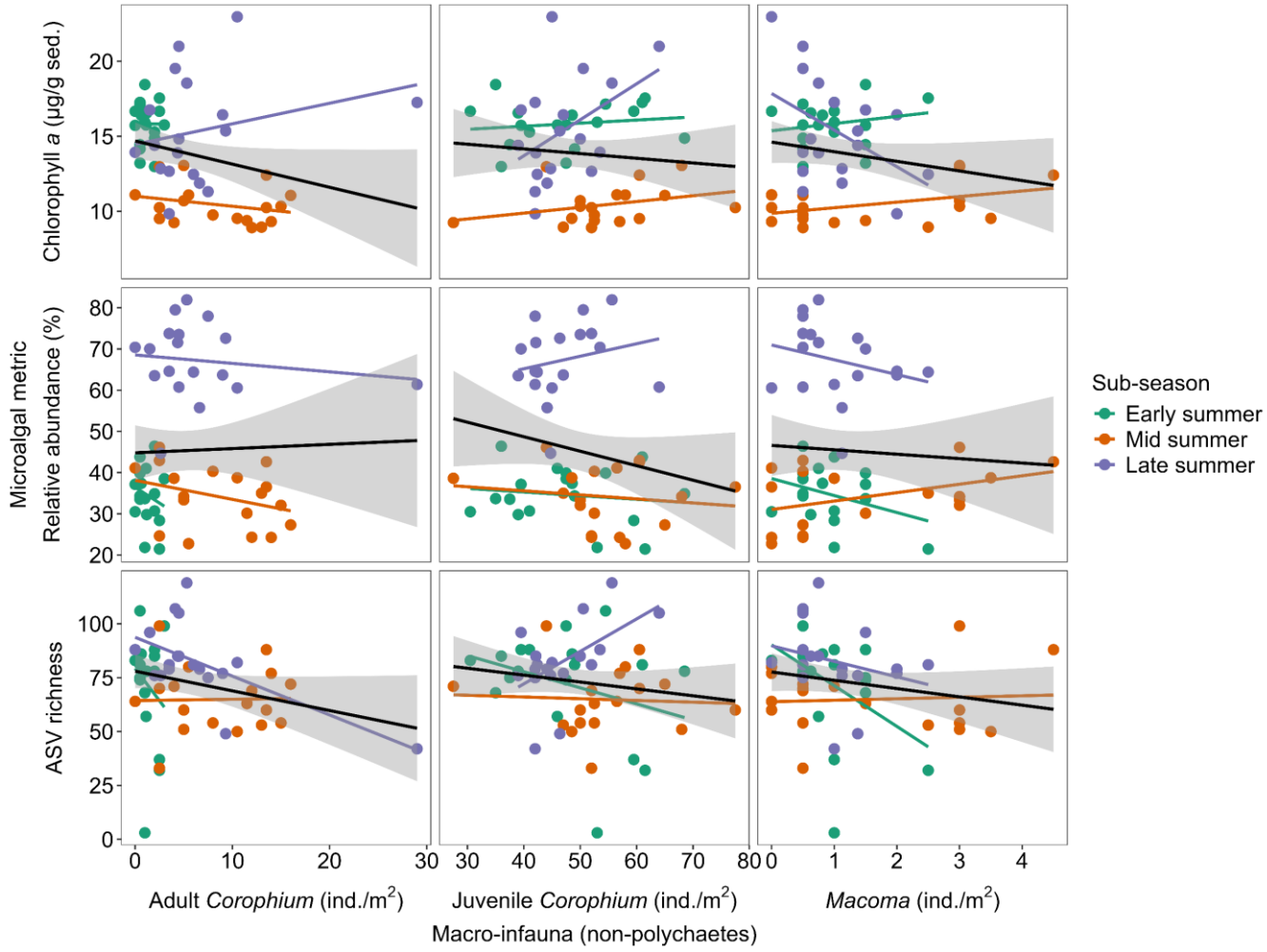


Figure S2B. Relationship between microalgae and macro-infauna (amphipods and clams). See figure caption above.

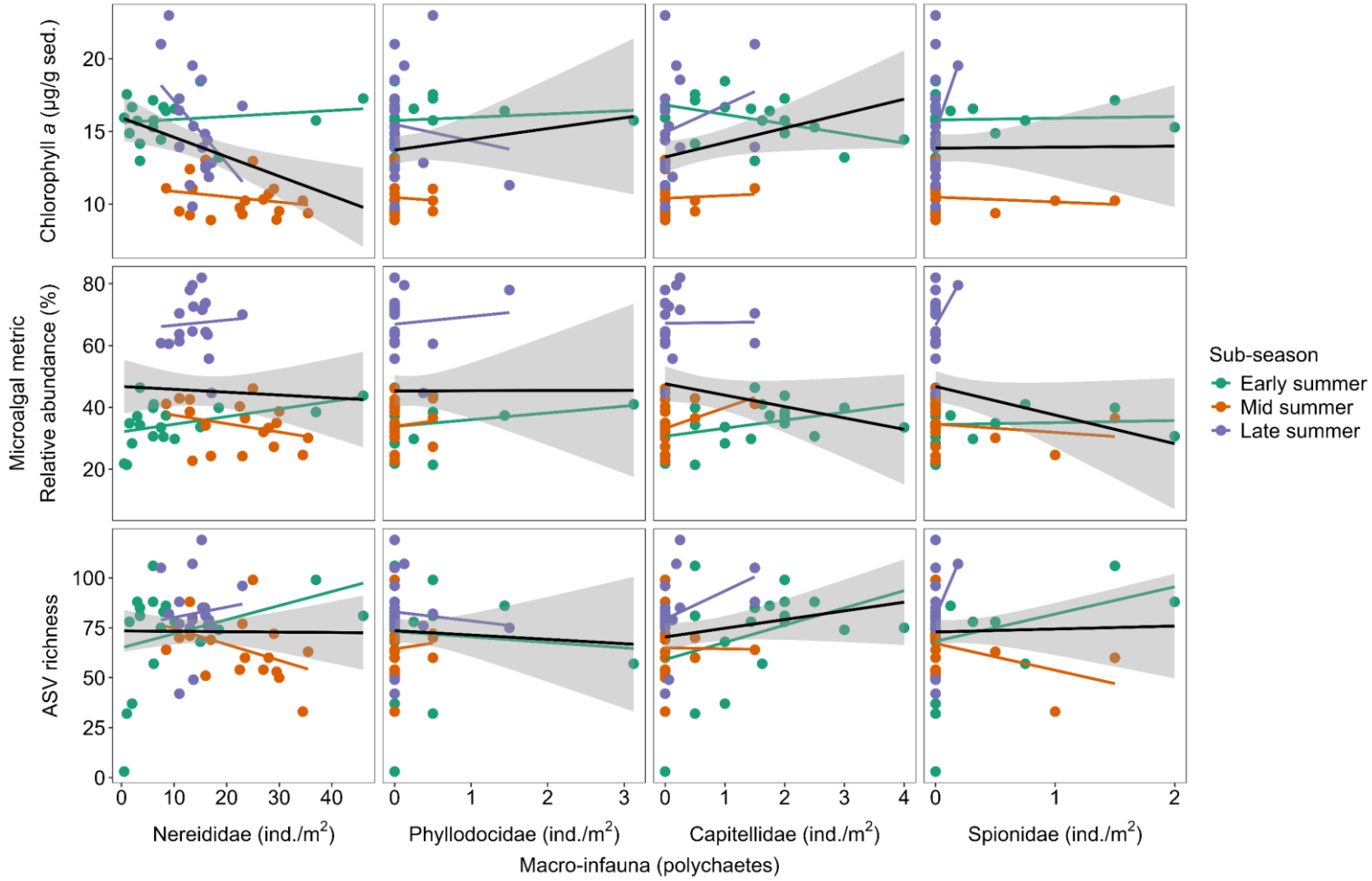


Figure S2C. Relationship between microalgae and macro-infauna (polychaetes). See figure caption above.

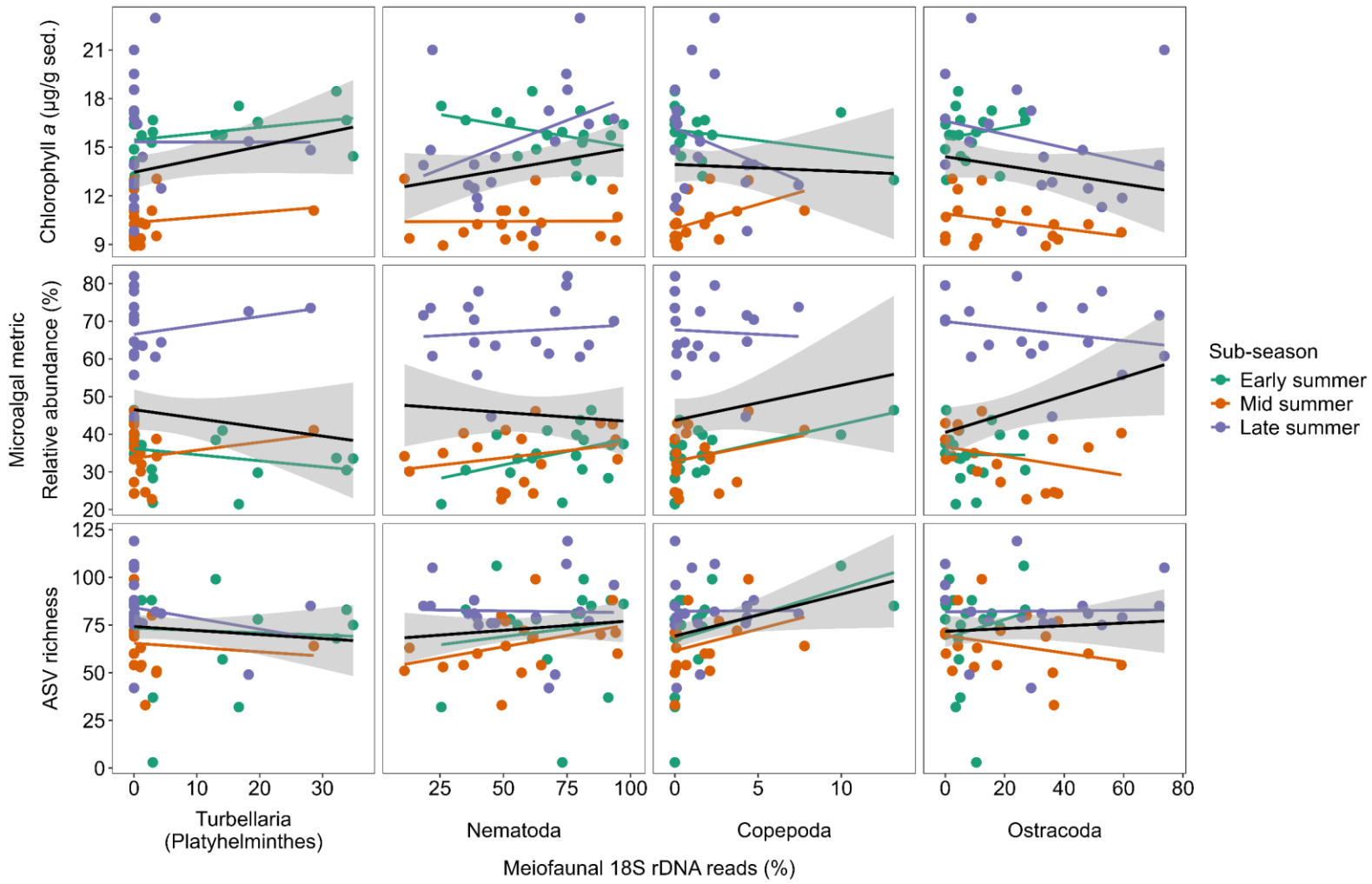


Figure S2D. Relationship between microalgae and meiofauna. See figure caption above.

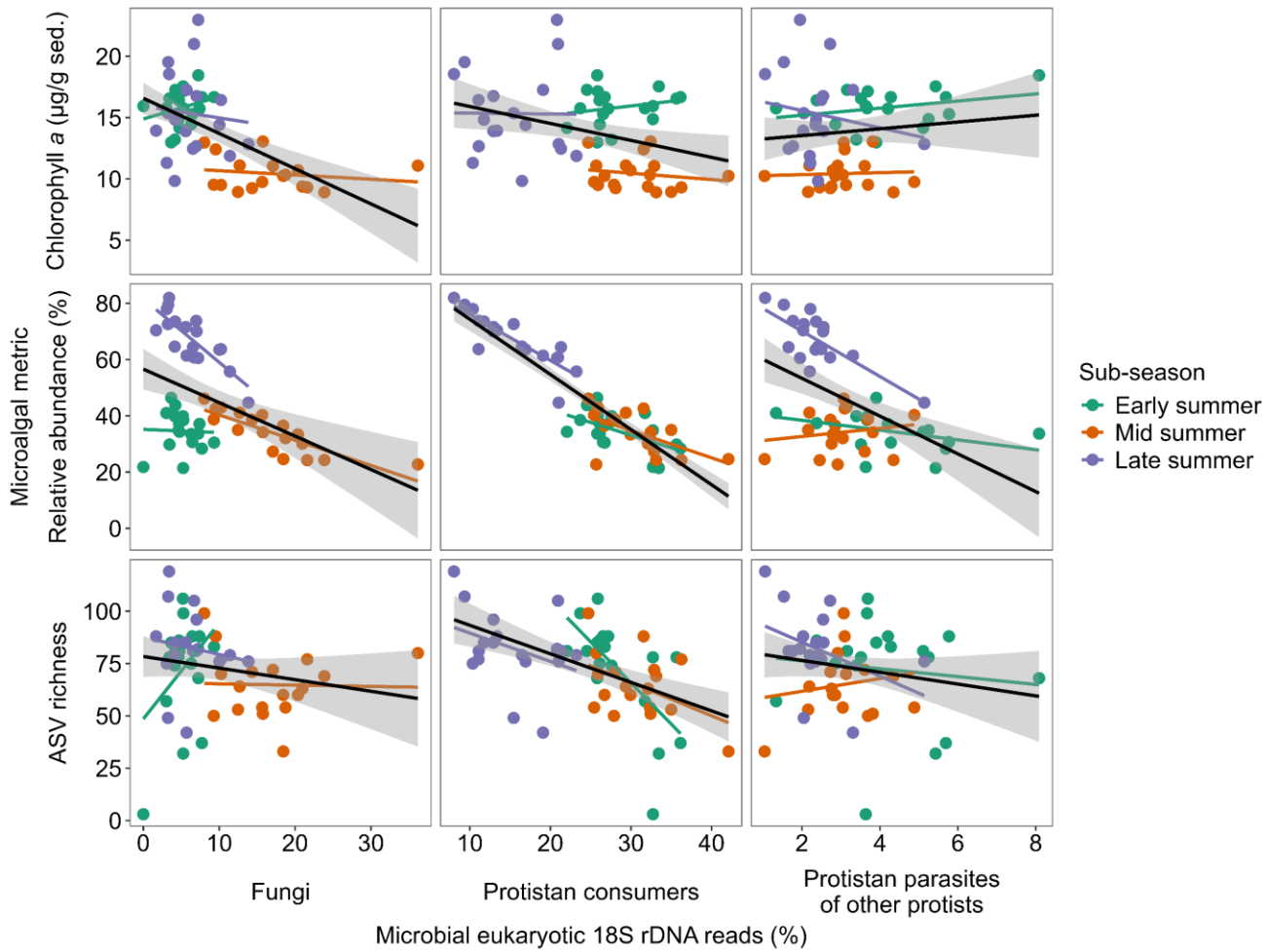


Figure S2E. Relationship between microalgae and microbial eukaryotes. See figure caption above. Note that microbial eukaryotes were defined as encompassing all eukaryotic diversity except animals and land plants, and are predominantly unicellular; while protists were defined as encompassing all microbial eukaryotic diversity except fungi.

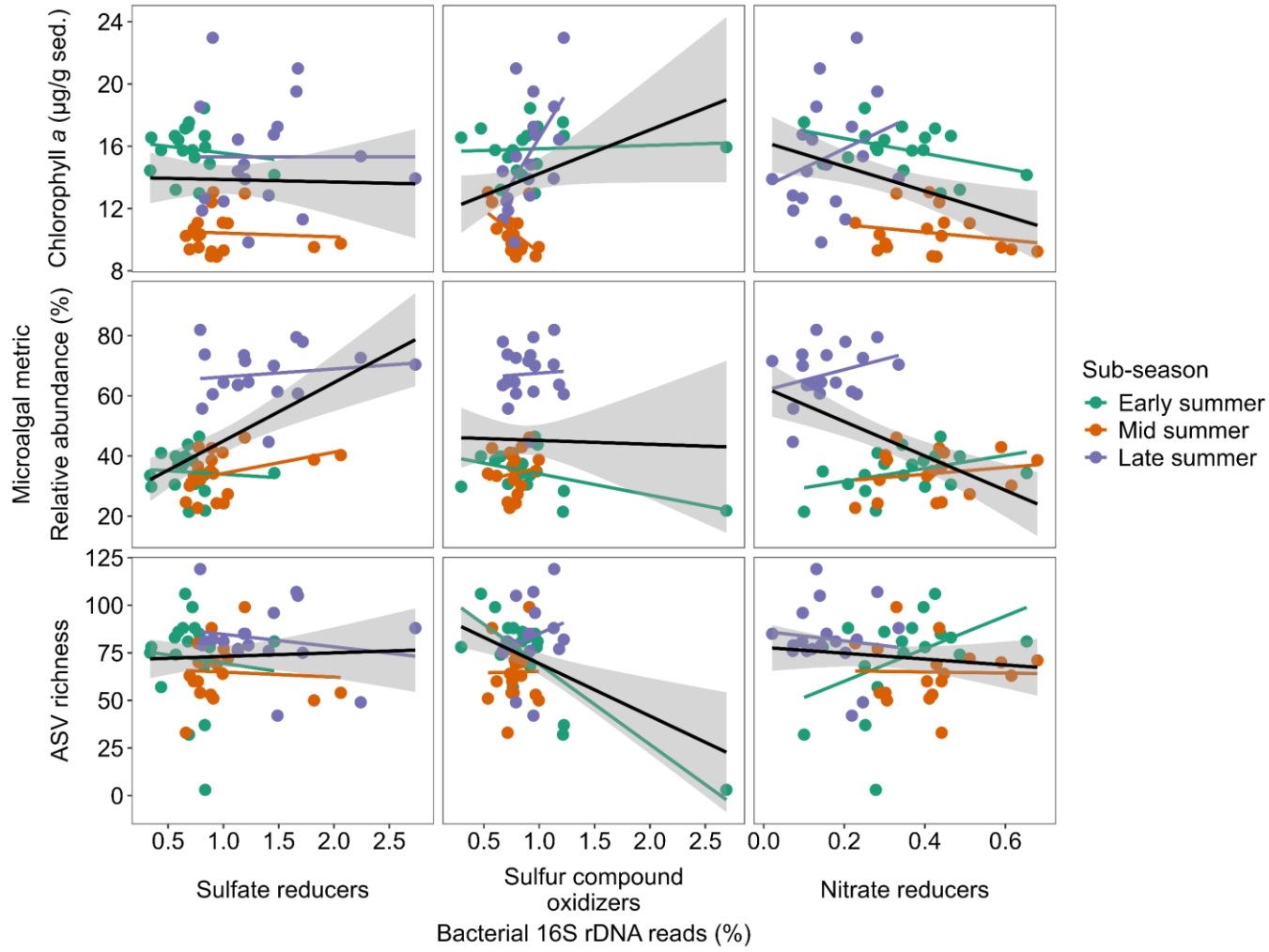


Figure S2F. Relationship between microalgae and bacteria. See figure caption above.

Curriculum Vitae

Candidate's full name:

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University attended (with date and degree obtained):

Bachelor of Science (Honours) in Biology, University of New Brunswick, Fredericton, NB, Canada, 2020

Publication:

Kalu EI, Reyes-Prieto A & Barbeau MA (2023). Community dynamics of microbial eukaryotes in intertidal mudflats in the hypertidal Bay of Fundy. *ISME Communications*. 3:21. https://doi.org/10.1038/s43705-023-00226-8

Conference presentations:

IX European Congress of Protistology & Annual Congress of the International Society of Protistologists (Joint Meeting). 2023, 9–14 July. Vienna, Austria. Insights into protist activity, spatio-seasonal dynamics and multi-trophic linkages in a Canadian hypertidal mudflat. Kalu EI, Barbeau MA & Reyes-Prieto A (oral presentation).

Atlantic Canada Coastal & Estuarine Science Society (20th Annual Conference). 2023, May 23-26. Shippigan, NB, Canada. Differential bottom-up and top-down controls on benthic fauna and microbiota in a hypertidal mudflat. Kalu EI, Vanbaelinghem L, Reyes-Prieto A, Hamilton DJ & Barbeau MA (oral presentation).

Atlantic Canada Coastal & Estuarine Science Society - Bay of Fundy Ecosystem Partnership (Joint Meeting). 2022, May 18-21. Truro, NS, Canada. Effects of nutrient

enrichment and foraging by invertebrates and shorebirds on microbial communities in a hypertidal mudflat. Kalu EI, Reyes-Prieto A, Hamilton DJ & Barbeau MA (oral presentation).

Coastal & Estuarine Research Federation (26th Biennial Conference). 2021, November 7-11. Virtual. Spatiotemporal dynamics of microbial eukaryotes in megatidal mudflats in Atlantic Canada. Kalu EI, Reyes-Prieto A & Barbeau MA (oral presentation).

International Society of Protistologists (Annual Meeting). 2021, July 26-30. Virtual. Community dynamics of microbial eukaryotes in intertidal mudflats in the megatidal Bay of Fundy, Canada. Kalu EI, Reyes-Prieto A & Barbeau MA (oral presentation).

Atlantic Canada Coastal & Estuarine Science Society (Annual Conference). 2021, May 13-14. Virtual. Spatial and temporal dynamics of microbial eukaryotes in muddy intertidal sediments of the Bay of Fundy. Kalu EI, Reyes-Prieto A & Barbeau MA (oral presentation).

University of New Brunswick 27th Annual Graduate Research Conference. 2021, May 6-7. Virtual. Spatial and temporal dynamics of microbial eukaryotes in muddy intertidal sediments of the Bay of Fundy. Kalu EI, Reyes-Prieto A & Barbeau MA (oral presentation).